NOTES ON ALGEBRAIC NUMBER THEORY 3

by

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Abstract. — These are lecture notes for Algebraic Number Theory 3, given by Prof. Dr. Vytautas Paškūnas at Universität Duisburg-Essen during the Winter Semester 25-26. The order of this note is almost strictly based on the schedule of the curriculum.

Contents

These notes follow one main thread: use cohomology to understand local fields. We begin with profinite groups and continuous cohomology, because absolute Galois groups are naturally profinite and their modules are discrete. After the formal construction, we compute the first basic examples in Galois cohomology: additive Hilbert 90, multiplicative Hilbert 90, cyclic cohomology, and Kummer theory.

The first major arithmetic target is the Brauer group of a non-archimedean local field. We compute $H^2(G_K, (K^{\mathrm{sep}})^\times)$ through the maximal unramified extension and the valuation sequence, obtaining the invariant map $\mathrm{inv}_K : \mathrm{Br}(K) \longrightarrow \mathbf{Q}/\mathbf{Z}$.. We then compare this cohomological Brauer group with the algebraic Brauer group defined by central simple algebras and central division algebras, using crossed products and factor sets.

The second part of the notes studies the cohomological dimension of local Galois groups and their inertia subgroups. The structure of unramified, tame, and wild ramification gives the input for the dimension estimates. These estimates prepare the local duality theorems: local Tate duality identifies cohomology groups of finite G_K-modules with the dual cohomology of their Cartier duals, and the Euler–Poincare characteristic formula measures the alternating size of H^0, H^1, and H^2.

The final part uses Tate cohomology to formulate local class field theory. For a finite Galois extension L/K, the fundamental class gives the finite-level reciprocity isomorphism

$$K^\times/N_{L/K}L^\times \simeq \mathrm{Gal}(L/K)^{\mathrm{ab}}.$$

Passing over all finite abelian extensions gives the local reciprocity map

$$\theta_K : K^\times \longrightarrow \mathrm{Gal}(K^{\mathrm{ab}}/K),$$

and the existence theorem identifies the norm subgroups of $K^\times$ with the finite-index subgroups. Thus the final goal is the local class field theory dictionary between finite abelian extensions of K and finite-index subgroups of $K^\times$.

We would like to express our special gratitude to Prof. Vytautas Paškūnas for his lecture. All possible errors are our own. As there are finitely many results in our notes, fortunately there should be only finitely many errors we made. We hope it is a readable and helpful note, although all results here are well-known. Please write to any of us if you find any errors.

Finally, we would like to thank ESAGA (Essen Seminar for Algebraic Geometry and Arithmetic) for its kindness and professionalism, and for offering us the best opportunities to study mathematics.

Lecture 1

This lecture introduces projective limits and profinite groups. The main examples to keep in mind are inverse limits of finite quotients and, later, absolute Galois groups; the point is to understand the topology before using continuous cohomology.

1.1 Projective Limits and Profinite Groups.

— In the first two lectures, we introduce profinite groups and study their basic topology. Infinite Galois groups will be our main examples, and we will recall the infinite Galois correspondence in next lecture.

Definition 1.1. — A directed set $I = (I, \geq)$ is a partially ordered set such that for every $i, j \in I$, there exists $k \in I$ with $k \geq i$ and $k \geq j$.

Definition 1.2. — Let I be a directed set. A projective system

$$P = (I, (G_i)_{i\in I}, (\phi_i^j)_{i\geq j})$$

in a category $\mathscr{C}$ consists of objects

$$G_i \in \mathrm{Ob}(\mathscr{C}), \quad i \in I,$$

and morphisms

$$\phi_i^j : G_i \longrightarrow G_j, \qquad i \geq j,$$

such that $\phi_i^i = \mathrm{id}_{G_i}$ and

$$\phi_j^k \circ \phi_i^j = \phi_i^k \qquad (i \geq j \geq k).$$

Remark 1.3. — A projective system is equivalently a functor from the category whose objects are the elements of I and which has a unique morphism $i \to j$ when $i \geq j$. For a fixed index set I, morphisms of projective systems are natural transformations of the corresponding functors.

Definition 1.4. — A projective limit $G = \varprojlim_i G_i$ of a projective system $P = (I, (G_i)_{i\in I}, (\phi_i^j)_{i\geq j})$ is a pair $(G, (\pi_i)_{i\in I})$ consisting of an object $G \in \mathrm{Ob}(\mathscr{C})$ and morphisms $\pi_i : G \to G_i$ such that

$$\phi_i^j \circ \pi_i = \pi_j \qquad (i \geq j),$$

with the following universal property: if $G' \in \mathrm{Ob}(\mathscr{C})$ and $\pi_i' : G' \to G_i$ are morphisms satisfying $\phi_i^j \circ \pi_i' = \pi_j'$ for all $i \geq j$, then there is a unique morphism $\psi : G' \to G$ such that $\pi_i' = \pi_i \circ \psi$ for all i.

Theorem 1.5. — *Let $\mathscr{C}$ be the category of compact Hausdorff topological groups with continuous group homomorphisms. The projective limit exists in $\mathscr{C}$. By the universal property, it is unique up to unique isomorphism.*

In particular, it can be constructed as follows:

Let $P = (I, (G_i)_{i\in I}, (\phi_i^j)_{i\geq j})$ be a projective system of compact groups over an indexing set I. Then a projective limit of the system is given explicitly by the group

$$G = \left\{ (g_i)_{i\in I} \in \prod_{i\in I} G_i \mid \phi_i^j(g_i) = g_j \textit{ for all } i \geq j \right\} \tag{1.1}$$

and the homomorphisms $\pi_i : G \to G_i$ obtained by inclusion $G \subseteq \prod_i G_i$ followed by projection to G_i.[1]

First we explain the topology induced from the projective limit.

Definition 1.6. — Let (I, G_i, ϕ_i^j) be a projective system of topological groups over an indexing set I, with continuous transition maps. Then the projective limit topology on the projective limit G of construction 1.1 is the subspace topology for the product topology on $\prod_{i\in I} G_i$.[2]

Proof of Theorem 1.5. — The product $\prod_i G_i$ is compact Hausdorff by Tychonoff's theorem. The subset defined in (1.1) is a subgroup. It is also closed: for each pair $i \geq j$, the condition $\phi_i^j(g_i) = g_j$ is the equalizer of the two continuous maps

$$\prod_i G_i \longrightarrow G_j, \qquad (g_i) \mapsto \phi_i^j(g_i), \qquad (g_i) \mapsto g_j,$$

and equalizers are closed in a Hausdorff space. Hence G is a closed subgroup of a compact Hausdorff group, so it is compact Hausdorff. The projection maps are continuous, and the universal property follows directly from the universal property of the product. □

Definition 1.7. — A *profinite group* is a projective limit of a system of finite groups, endowed with the projective limit topology for the discrete topology on the finite groups.

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(1) This construction works in many familiar categories, for example groups, topological groups, and rings, with the evident morphisms.

(2) Equivalently, it is the weakest topology for which all projections $G \to G_i$ are continuous.

Example 1.8. — Let $\mathbf{F}_p[[t]] = \varprojlim_n \mathbf{F}_p[t]/(t^n)$. The projective limit topology is the t-adic topology.

A pro-p group is a profinite group G such that for every open normal subgroup $N \trianglelefteq G$, the quotient G/N is a finite p-group. The group $\mathbf{Z}_p$ is a pro-p group. More generally, the maximal pro-p quotient of a profinite group is obtained as the inverse limit of its finite p-group quotients.

Let F be a field. Set $G_F := \mathrm{Aut}_F(F^{\mathrm{sep}}) = \mathrm{Gal}(F^{\mathrm{sep}}/F)$, with its profinite topology. Let I_F be the set of finite Galois extensions L/F inside F^{sep}, ordered by inclusion. Then

$$G_F \cong \varprojlim_{L \in I_F} \mathrm{Gal}(L/F).$$

Let $F' \subseteq F^{\mathrm{sep}}$ be a normal extension of F. Then $G_{F'} \leq G_F$ is closed and normal. We may thus write

$$\mathrm{Gal}(F'/F) \cong G_F/G_{F'} = \varprojlim_{\substack{L \in I_F \\ L \subseteq F'}} \mathrm{Gal}(L/F).$$

Let $\mathbf{N}$ denote the natural numbers, ordered by divisibility. Then

$$\widehat{\mathbf{Z}} = \varprojlim_{n} \mathbf{Z}/n\mathbf{Z} = \prod_{p} \mathbf{Z}_p,$$

where the last step is the Chinese remainder theorem. We sometimes need a slight modification:

$$\widehat{\mathbf{Z}}^{(p)} = \varprojlim_{p \nmid n} \mathbf{Z}/n\mathbf{Z} = \prod_{\ell \text{ prime}, \ell \neq p} \mathbf{Z}_\ell.$$

1.2 Topology of Profinite Groups.

— We study the topology of profinite groups. In fact, we have:

Theorem 1.9. — *The following are equivalent:*

1. *G is profinite.*
2. *G is compact, Hausdorff, and totally disconnected.*
3. *G is compact Hausdorff and admits a neighborhood basis of the identity by open normal subgroups.*

Lemma 1.10. — *Let G be a profinite group. If $X \subset G$ is a subset, then the closure of X in G is*

$$\overline{X} = \cap_{i \in I}(X \ker \varphi_i),$$

where $\varphi_i : G \twoheadrightarrow G_i$ is the i-th component surjection.

Lemma 1.11. — *Let G be a profinite group, U is an open subgroup. Then U is closed of finite index, and there exists an open normal subgroup $N \subset U$.*

Lecture 2

This lecture recalls infinite Galois theory. We pass from the finite Galois correspondence to the profinite one: closed subgroups correspond to intermediate fields, open subgroups correspond to finite subextensions, and field towers give exact sequences of Galois groups.

2.1 Infinite Galois Groups. — In this section, we recall infinite Galois theory. Let K/k be a possibly infinite Galois extension.

Definition 2.1. — The *Galois group* $\mathrm{Gal}(K/k)$ is the group of all automorphisms of K leaving the elements of k fixed. We define a topology on $\mathrm{Gal}(K/k)$ via a neighborhood basis $\mathscr{U}(K/k)$ at the identity element:

$$\mathscr{U}(K/k) = \{\mathrm{Gal}(K/N) \mid N \in \mathscr{N}\},$$

where $\mathscr{N}$ is the set of all finite Galois subextensions N/k inside K.

Theorem 2.2. — *The group* $\mathrm{Gal}(K/k)$ *is profinite. More exactly,* $\mathrm{Gal}(K/k)$ *is isomorphic to the projective limit of the system* $\{\mathrm{Gal}(N/k) \mid N \in \mathscr{N}\}$.

Proof. — The restriction $\mathrm{Gal}(K/k) \to \mathrm{Gal}(N/k)$ induces a homomorphism

$$\varphi : \mathrm{Gal}(K/k) \to \varprojlim_{N\in\mathscr{N}} \mathrm{Gal}(N/k).$$

For every $\alpha \in K$ there is an $M \in \mathscr{N}$ such that $\alpha \in M$. We put

$$\psi\left((g_N)_{N\in\mathscr{N}}\right)(\alpha) = g_M(\alpha) \quad \text{for} \quad (g_N)_{N\in\mathscr{N}} \in \varprojlim_{N\in\mathscr{N}} \mathrm{Gal}(N/k).$$

This definition does not depend on the choice of M. It is easy to see that ψ is the inverse mapping of φ. Therefore, φ is an isomorphism of abstract groups. Since φ maps $\{\mathrm{Gal}(K/N) \mid N \in \mathscr{N}\}$, a neighborhood basis at $1 \in \mathrm{Gal}(K/k)$, to the corresponding neighborhood basis in the projective limit $\varprojlim_{N\in\mathscr{N}} \mathrm{Gal}(N/k)$, φ is even an isomorphism of topological groups. □

Theorem 2.3. — *Let* $\{K_i \mid i \in I\}$ *be a directed family of Galois extensions of* k *and put* $K = \bigcup_{i\in I} K_i$. *Then* $\{\mathrm{Gal}(K_i/k) \mid i \in I\}$ *is a projective system with limit* $\varprojlim_i \mathrm{Gal}(K_i/k) \simeq \mathrm{Gal}(K/k)$.

Example 2.4. — Let p be a prime and put $k = \mathbf{Q}(\zeta_p)$ and $K = \bigcup_{\nu=1}^{\infty} \mathbf{Q}(\zeta_{p^\nu})$, where ζ_{p^ν} is a primitive p^ν-th root of unity. Then $\mathrm{Gal}(K/k) \simeq \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}_2$ if $p = 2$, and $\mathrm{Gal}(K/k) \simeq \mathbf{Z}_p$ if $p > 2$.

***Theorem 2.5* (Lifts of Isomorphisms).** — *Let* K/k *be a Galois extension with subextensions* K_1/k *and* K_2/k, *and let* $\gamma : K_1 \to K_2$ *be an isomorphism leaving the elements of* k *fixed. Then* γ *can be lifted to an automorphism of* K/k.

Proof. — For each finite Galois subextension M/k of K/k, finite Galois theory gives lifts to M of the restriction of γ to $K_1 \cap M$, after enlarging M inside K if necessary. The possible compatible lifts form a nonempty closed subset of the compact product $\prod_{N\in\mathscr{N}} \mathrm{Gal}(N/k)$. These closed subsets have the finite intersection property. Hence their intersection is nonempty, and an element of the intersection gives a compatible system of automorphisms on all finite Galois subextensions. By the projective-limit description of $\mathrm{Gal}(K/k)$, this compatible system is an automorphism of K extending γ. $\square$

Theorem 2.6. — *If K/k is a Galois extension and M/k is a finite subextension, then*

$$[\mathrm{Gal}(K/k) : \mathrm{Gal}(K/M)] = [M : k].$$

Proof. — Let $[M : k] = n$, and let $\gamma_1, \ldots, \gamma_n$ denote the k-embeddings $M \to K$. By the lifting theorem, each γ_ν can be lifted to an element $\bar{\gamma}_\nu \in \mathrm{Gal}(K/k)$. Then $\{\bar{\gamma}_1, \ldots, \bar{\gamma}_n\}$ is a set of representatives for the cosets of $\mathrm{Gal}(K/M)$ in $\mathrm{Gal}(K/k)$. $\square$

Theorem 2.7. — *Let K/k be a Galois extension and let $\{K_i\}_{i\in I}$ be a cofinal family of finite subextensions of K/k, ordered by inclusion. Then $\{\mathrm{Gal}(K/K_i) \mid i \in I\}$ is a system of open and closed neighborhoods of $1 \in \mathrm{Gal}(K/k)$.*

Proof. — Let $i \in I$, and assume that N/k is a finite normal subextension of K/k containing K_i. Then we know that $\mathrm{Gal}(K/N)$ has finite index in $\mathrm{Gal}(K/K_i)$. Thus if $\mathrm{Gal}(K/N)$ is open and closed, then so is $\mathrm{Gal}(K/K_i)$. If, on the other hand, N/k is an arbitrary finite normal subextension of K/k, then the existence of primitive elements implies that there is an $i \in I$ such that $N \subseteq K_i$, i.e., $\mathrm{Gal}(K/K_i) \subseteq \mathrm{Gal}(K/N)$. But then $\{\mathrm{Gal}(K/K_i) \mid i \in I\}$ is a neighborhood basis at $1 \in \mathrm{Gal}(K/k)$. $\square$

Theorem 2.8. — *Let K/k be a Galois extension with subextension M/k. The topology of $\mathrm{Gal}(K/M)$ is induced by the topology of $\mathrm{Gal}(K/k)$, and $\mathrm{Gal}(K/M)$ is closed in $\mathrm{Gal}(K/k)$.*

Moreover, $\mathrm{Gal}(K/M)$ is open if and only if M/k is finite.

Proof. — For two subextensions M/k and N/k of K/k, we have

$$\mathrm{Gal}(K/M) \cap \mathrm{Gal}(K/N) = \mathrm{Gal}(K/MN).$$

Therefore, $\{\mathrm{Gal}(K/MN) \mid N/k \text{ finite and Galois}\}$ is a neighborhood basis at $1 \in \mathrm{Gal}(K/M)$ for the topology induced by $\mathrm{Gal}(K/k)$, and also for the Krull topology on $\mathrm{Gal}(K/M)$. The subgroup $\mathrm{Gal}(K/M)$ is closed because it is the intersection of the closed stabilizers of the elements of M. If M/k is finite, then the preceding theorem gives

$$[\mathrm{Gal}(K/k) : \mathrm{Gal}(K/M)] = [M : k],$$

so $\mathrm{Gal}(K/M)$ is open. Conversely, if $\mathrm{Gal}(K/M)$ is open, then it contains $\mathrm{Gal}(K/N)$ for some finite Galois subextension N/k of K/k. Hence $M \subseteq N$, and therefore M/k is finite. $\square$

2.2 Galois Correspondence and Exact Sequences. —

Theorem 2.9. — *[Galois Correspondence] Let K/k be a Galois extension. Then $\Phi(M) = \mathrm{Gal}(K/M)$ defines a bijection Φ between the set of all intermediate fields M of K/k and the set of all closed subgroups of $\mathrm{Gal}(K/k)$.*

The inverse map of Φ assigns to each closed subgroup U of $\mathrm{Gal}(K/k)$ its fixed field K^U.

Let us formulate some theorems that follow immediately from classical Galois theory and the preceding results.

Theorem 2.10. — *Let K/k be a Galois extension and let N/k be a Galois subextension. Then the natural sequence*

$$1 \longrightarrow \mathrm{Gal}(K/N) \longrightarrow \mathrm{Gal}(K/k) \longrightarrow \mathrm{Gal}(N/k) \longrightarrow 1$$

is an exact sequence of profinite groups.

Theorem 2.11. — *Let Ω/k be a Galois extension with subextensions K_1/k and K_2/k. If K_1/k is Galois, then so is K_1K_2/K_2, and restriction gives a natural isomorphism*

$$\mathrm{Gal}(K_1K_2/K_2) \longrightarrow \mathrm{Gal}(K_1/K_1 \cap K_2).$$

If K_2/k is also Galois, then there is a natural isomorphism

$$\mathrm{Gal}(K_1K_2/k) \longrightarrow \mathrm{Gal}(K_1/k) \times_G \mathrm{Gal}(K_2/k),$$

where we have put $G = \mathrm{Gal}(K_1 \cap K_2/k)$.

Lecture 3

This lecture defines discrete modules over a profinite group and sets up the category in which continuous cohomology will live. The key point is that continuity of the action forces stabilizers to be open, which is what makes later cochain constructions behave correctly.

Definition 3.1. — Let G be a profinite group. A G-module is an abelian group A together with an action of G on A by group automorphisms.

Let Mod_G be the category of all G-modules. This is the same as the category of modules over the abstract group ring $\mathbf{Z}[G]$.

Proposition 3.2. — *For a G-module A, the following are equivalent:*

1. *the action map $G \times A \to A,\ (g,a) \mapsto g \cdot a$ is continuous when A has the discrete topology;*
2. *for every $a \in A$, the stabilizer $G_a = \{g \in G \mid g \cdot a = a\}$ is open in G;*
3. *$A = \bigcup_{U \subset G \text{ open}} A^U$, where A^U denotes the subgroup of A fixed by U.*

Such modules are called discrete G-modules. *They form a full abelian subcategory of Mod_G, denoted by $\mathscr{C}_G$.*

Example 3.3. — 1. If K/k is Galois, $G = \mathrm{Gal}(K/k)$, and $A = (K, +)$, then A is a discrete G-module, since

$$K = \bigcup_{F \subset K,\ F/k \text{ finite Galois}} F.$$

2. Similarly, for K/k Galois and $G = \mathrm{Gal}(K/k)$, the multiplicative group $A = K^{\times}$ is a discrete G-module.

3. Non-example: for G a profinite group and A a nonzero discrete abelian group, consider the set of all functions

$$\mathscr{F}(G, A) = \{f : G \to A\}$$

with the G-action by right translation. It is an abstract G-module. However, $\mathscr{F}(G, A)$ is discrete if and only if G is finite. Indeed, choose $0 \neq a \in A$ and let δ_1 be the function supported at 1 with value a. Its stabilizer is $\{1\}$, which is open if and only if G is discrete; a compact discrete group is finite.

Lemma 3.4. — *Let G be a profinite group and let A have the discrete topology. For a function $f : G \to A$, the following are equivalent:*

1. *f is continuous;*
2. *f is locally constant;*
3. *there exists an open normal subgroup $U \trianglelefteq G$, with $g_1, ..., g_n \in G$, $a_1, \ldots, a_n \in A$, such that*

$$G = \cup_{i=1}^{n} g_i U, \quad f(x) = a_i, \quad \textit{for all } x \in g_i U.$$

Proof. — **(1) ⇔ (2):** If f is continuous and A is discrete, then for each $x \in G$, $\{f(x)\}$ is open in A, so $f^{-1}(\{f(x)\})$ is an open neighborhood of x on which f is constant. Hence f is locally constant.

Conversely, if f is locally constant, then for any open $V \subset A$, $f^{-1}(V)$ is a union of open sets on which f is constant (hence either contained in or disjoint from $f^{-1}(V)$), so $f^{-1}(V)$ is open. Thus f is continuous.

(2) ⇒ (3): Assume f is locally constant. For each $x \in G$, there exists an open normal subgroup $N_x \trianglelefteq G$ such that $xN_x \subset f^{-1}(f(x))$. The cosets $\{xN_x : x \in G\}$ cover G. By compactness of G, finitely many suffice: $G = \bigcup_{i=1}^{n} g_i N_{g_i}$.

Let $U = \bigcap_{i=1}^{n} N_{g_i}$. Then U is an open normal subgroup of G. For each i, $g_i U \subset g_i N_{g_i}$, and f is constant on $g_i N_{g_i}$ (say $f = a_i$ there). Since $U \subset N_{g_i}$, the cosets $g_i U$ cover G:

$$G = \bigcup_{i=1}^{n} g_i U, \quad f(x) = a_i \text{ for all } x \in g_i U.$$

(3) ⇒ (2): Assume (3) holds. Each coset $g_i U$ is open (since U is open and translation is a homeomorphism). On each $g_i U$, f is constant (equal to a_i). Thus f is locally constant. □

Corollary 3.5. — $C(G,A) = \{f \in \mathscr{F}(G,A) | f \text{ is continuous}\}$[3] with G action by right translation is a discrete G-module.

Moreover,

$$C(G,A) = \bigcup_{U \trianglelefteq G \ open} C(G,A)^U = \bigcup_{U \trianglelefteq G \ open} \mathscr{F}(G/U,A) = \bigcup_{U \trianglelefteq G \ open} \mathscr{F}(G,A)^U.$$

Proof. — For $f \in C(G,A)$, by the lemma above we write $f = \sum_{i=1}^n a_i 1_{g_i U}$ for some $U \trianglelefteq G$ open and appropriate a_i. Then the stabilizer of f contains U, hence open. □

Definition 3.6. — Let H be a closed subgroup of G, and let A be a discrete H-module. We define

$$\mathrm{Ind}_H^G A = \{f \in C(G,A) | f(hg) = hf(g), \quad \forall h \in H, g \in G\},$$

with G action by right translation. This defines a discrete G-module.

Proposition 3.7 **(Frobenius Reciprocity).** — *For B a discrete G-module and A a discrete H-module, we have*

$$\mathrm{Hom}_G(B, \mathrm{Ind}_H^G A) \cong \mathrm{Hom}_H(\mathrm{Res}_H^G B, A).$$

Proof. — We verify that Φ and Ψ are well-defined, G-equivariant (resp. H-equivariant), and mutual inverses.

Forward map Φ: given $F \in \mathrm{Hom}_G(B, \mathrm{Ind}_H^G A)$, define $\Phi(F) \in \mathrm{Hom}_H(\mathrm{Res}_H^G B, A)$ by

$$\Phi(F)(b) = F(b)(1).$$

Check H-linearity: for $h \in H$,

$$\begin{aligned}\Phi(F)(hb) &= F(hb)(1) = (h \cdot F(b))(1) \quad (\text{since } F \text{ is } G\text{-linear}) \\ &= F(b)(1 \cdot h) = F(b)(h) = h \cdot (F(b)(1)) \quad (\text{by definition of } \mathrm{Ind}_H^G A) \\ &= h \cdot \Phi(F)(b).\end{aligned}$$

So $\Phi(F)$ is H-linear.

Backward map Ψ: given $\varphi \in \mathrm{Hom}_H(\mathrm{Res}_H^G B, A)$, define $\Psi(\varphi) \in \mathrm{Hom}_G(B, \mathrm{Ind}_H^G A)$ by

$$[\Psi(\varphi)(b)](g) = \varphi(g \cdot b).$$

Check $\Psi(\varphi)(b) \in \mathrm{Ind}_H^G A$: for $h \in H$, $g \in G$,

$$[\Psi(\varphi)(b)](hg) = \varphi((hg) \cdot b) = \varphi(h \cdot (g \cdot b)) = h \cdot \varphi(g \cdot b) = h \cdot [\Psi(\varphi)(b)](g).$$

[3] In continuous cohomology of a profinite group with discrete coefficients, the cochains must be continuous; since the target A is discrete, this is the same as being locally constant. The larger space $\mathscr{F}(G,A)$ consists of all set-theoretic functions $G \to A$, and for infinite profinite G it is usually not a discrete G-module: for instance, the stabilizer of the function supported at 1 is $\{1\}$, which is not open unless G is finite. If G is finite, then every function is continuous, so $C(G,A) = \mathscr{F}(G,A)$.

Check G-linearity of $\Psi(\varphi)$: for $g' \in G$,

$$[\Psi(\varphi)(g' \cdot b)](g) = \varphi(g \cdot (g' \cdot b)) = \varphi((gg') \cdot b) = [\Psi(\varphi)(b)](gg') = [g' \cdot \Psi(\varphi)(b)](g).$$

So $\Psi(\varphi)$ is G-linear.

Inverses:

1. $\Phi(\Psi(\varphi))(b) = [\Psi(\varphi)(b)](1) = \varphi(1 \cdot b) = \varphi(b)$, so $\Phi \circ \Psi = \mathrm{id}$.
2. $\Psi(\Phi(F))(b)$ evaluated at g is:

$$[\Psi(\Phi(F))(b)](g) = \Phi(F)(g \cdot b) = F(g \cdot b)(1) = (g \cdot F(b))(1) = F(b)(g),$$

so $\Psi(\Phi(F))(b) = F(b)$ for all b, hence $\Psi \circ \Phi = \mathrm{id}$.

Thus Φ and Ψ are mutually inverse isomorphisms. $\square$

Corollary 3.8. — If $H = \{1\}$, then $\mathrm{Hom}_G(B, C(G,A)) \cong \mathrm{Hom}_{\mathbf{Ab}}(B, A)$. Here $\mathbf{Ab}$ denotes the category of abelian groups.

Lemma 3.9. — *The map $A \to C(G,A), \quad a \mapsto [g \mapsto g \cdot a]$ is injective and G-equivariant.*

Moreover, this is the image of the $\mathrm{id}_A \in \mathrm{Hom}_{\mathbf{Ab}}(A, A)$ *under the above isomorphism.*

3.1 Cohomology. — Consider the category $\mathscr{C}_G$ of discrete G-modules, with morphisms the G-equivariant homomorphisms of abelian groups. We have the following operations:

1. It is a full subcategory of the category of $\mathbf{Z}[G]$-modules. Notice that it is closed under extensions, subquotients, and quotients; hence it is an abelian category.
2. We have the adjunctions:

$$\mathrm{Hom}_{\mathbf{Z}[G]}(\iota(M), N) \cong \mathrm{Hom}_{\mathscr{C}_G}(M, N^{sm}),$$

where $N^{sm} := \cup_{U \trianglelefteq G,\ open} N^U$.

For any $U \trianglelefteq G$, *open*, $(N^{sm})^U = N^U$, hence N^{sm} is a discrete G-module.

3. $\mathscr{C}_G$ has enough injectives.

Proof. — Let $M \in \mathscr{C}_G$. Choose an injection of abelian groups $M \hookrightarrow I$ with I injective in $\mathbf{Ab}$. Then $C(G, I)$ is injective in $\mathscr{C}_G$: by Frobenius reciprocity,

$$\mathrm{Hom}_G(M, C(G, I)) = \mathrm{Hom}_{\mathbf{Ab}}(M, I).$$

Since $\mathrm{Hom}_{\mathbf{Ab}}(-, I)$ is exact, so is $\mathrm{Hom}_G(-, C(G, I))$. Finally, the composition

$$M \hookrightarrow C(G, M) \hookrightarrow C(G, I), \quad m \mapsto [g \mapsto \iota(g \cdot m)]$$

is injective. Hence every object of $\mathscr{C}_G$ embeds in an injective object. $\square$

Definition 3.10. — For $n \geq 0$, let $H^n(G, -) : \mathscr{C}_G \to \mathbf{Ab}$ be the n-th right derived functor of the left exact functor $A \mapsto A^G = H^0(G, A)$.

The following are some facts:

1. The resulting cohomology groups do not depend on the choice of injective resolutions.

2. A short exact sequence $0 \to A \to B \to C \to 0$ in $\mathscr{C}_G$ induces a long exact sequence in cohomology.
3. If I is injective in $\mathscr{C}_G$, then $H^n(G, I) = 0$ for any $n > 0$.
4. If $0 \to A \to C^\bullet$ is an injective resolution of A in $\mathscr{C}_G$, then $H^n(G, A)$ is the n-th cohomology of the complex $(C^\bullet)^G$.
5. For A any abelian group, $C(G, A)$ is acyclic.

Proof of (5). — Let us choose an injective I in $\mathbf{Ab}$ and the injection $A \hookrightarrow I$. The exact sequence $0 \to A \to I \to I/A \to 0$ in $\mathbf{Ab}$ with I injective induces an exact sequence in $\mathscr{C}_G$:

$$0 \to C(G, A) \to C(G, I) \to C(G, I/A) \to 0.$$

By taking G-invariants, we obtain a long exact sequence in cohomology:

$$0 \to C(G, A)^G \to C(G, I)^G \to C(G, I/A)^G \to H^1(G, C(G, A)) \to H^1(G, C(G, I)) \to \cdots$$

Since $C(G, I)$ is an injective object in $\mathscr{C}_G$ (as I is injective in $\mathbf{Ab}$ and coinduction preserves injectives), we have $H^n(G, C(G, I)) = 0$ for all $n > 0$ by property (3). Similarly, $C(G, I/A)$ is also injective, so its higher cohomology vanishes. The long exact sequence then forces $H^n(G, C(G, A)) = 0$ for all $n > 0$, proving that $C(G, A)$ is acyclic.

Moreover, $C(G, A)^G \cong A$. Indeed, a function $f : G \to A$ is G-invariant for the right-translation action if and only if it is constant. Thus evaluation at 1 gives an isomorphism $C(G, A)^G \to A$, with inverse $a \mapsto (g \mapsto a)$. This completes the proof. □

Lecture 4

This lecture constructs continuous cohomology using continuous cochains and injective resolutions. We also record the direct-limit properties that let us reduce many computations to finite submodules or finite quotients.

4.1 Continuous Cochain Complexes. — We have seen that:

1. Let $\mathscr{C}_G$ be the category of discrete G-modules with G-equivariant homomorphisms of abelian groups as morphisms. This category is a Serre subcategory of $\mathrm{Mod}_{\mathbf{Z}[G]}$ and it follows that $\mathscr{C}_G$ is an abelian category ([StacksProj, 02MP]).

Moreover, the forgetful functor $\mathscr{C}_G \to \mathrm{Mod}_{\mathbf{Z}[G]}$ is left adjoint to ${}^{sm} : A \mapsto A^{sm}$, where

$$A^{sm} := \bigcup_{N \trianglelefteq G \text{ open}} A^N = \{a \in A \mid G_a \text{ is open in } G\}.$$

2. The category $\mathscr{C}_G$ has enough injectives and the functor $A \mapsto A^G =: H^0(G, A)$ is an additive left exact functor (additivity follows from left-exactness in the abelian category) from $\mathscr{C}_G$ to $\mathbf{Ab}$. Therefore, it admits an initial effaceable δ-functor $(H^i(G, A), \delta^i)$ extending $H^0(G, A)$, which does not depend on the choice of injective resolutions.

On the other hand, for $A \in \mathscr{C}_G$ and a resolution $0 \to A \to C^\bullet$ in $\mathscr{C}_G$ such that each C^i is acyclic (i.e. $H^n(G, C^i) = 0$ for all $i \geq 0$ and $n > 0$), the groups $H^n(G, A)$ are computed by the cohomology of the complex $(C^\bullet)^G$. In particular, injective objects and the modules $C(G, A)$ are acyclic.

Definition 4.1. — Let $A \in \mathscr{C}_G$, and let $i \geq 0$.

1. The group of i-cochains of G with coefficients in A is the set of continuous maps (i.e. locally constant maps) from G^i to A:

$$C^i(G, A) = \{f : G^i \to A \mid f \text{ is continuous}\}.$$

2. The ith differential $d^i = d^i_A : C^i(G, A) \to C^{i+1}(G, A)$ is the map

$$\begin{aligned} d^i(f)(g_0, g_1, \ldots, g_i) &= g_0 \cdot f(g_1, \ldots, g_i) \\ &\quad + \sum_{j=1}^{i} (-1)^j f(g_0, \ldots, g_{j-2}, g_{j-1}g_j, g_{j+1}, \ldots, g_i) + (-1)^{i+1} f(g_0, \ldots, g_{i-1}). \end{aligned}$$

We then define cohomology $H^i(G, A)$ as the cohomology groups of the complex $(C^i(G, A))$.

Proposition 4.2. — *(Computation of H^1). Let $A \in \mathscr{C}_G$.*

1. Let the group of continuous 1-cocycles be

$$Z^1(G, A) := \{f \in C(G, A) \mid f(gh) = g \cdot f(h) + f(g) \textit{ for all } g, h \in G\}$$

and the group of 1-coboundaries be

$$B^1(G, A) := \{f \in C(G, A) \mid \exists a \in A \textit{ s.t. } f(g) = ga - a \textit{ for all } g \in G\}$$

Then there exists an isomorphism $H^1(G, A) \cong Z^1(G, A)/B^1(G, A)$ natural in A.

Proof. — By definition, a class in $H^1(G, A)$ is a 1-cochain $f : G \to A$ satisfying $d^1 f = 0$, modulo cochains of the form $d^0 a$ with $a \in C^0(G, A) = A$. Using the displayed formula for the differential, the condition $d^1 f(g, h) = 0$ is exactly

$$gf(h) - f(gh) + f(g) = 0,$$

or equivalently

$$f(gh) = gf(h) + f(g).$$

Thus the 1-cocycles are precisely the crossed homomorphisms $Z^1(G, A)$ written in the statement. For $a \in A$, the 0-coboundary is

$$(d^0 a)(g) = ga - a,$$

so the subgroup of 1-coboundaries is exactly $B^1(G, A)$. Therefore

$$H^1(G, A) = Z^1(G, A)/B^1(G, A).$$

The construction is functorial in A. □

Example 4.3. — If G acts on A trivially, then $H^1(G,A) = \mathrm{Hom}_{\mathrm{group}}(G,A) \cap C(G,A)$.

In particular, if $A = \mathbf{Z}$ with trivial action, then $H^1(G,\mathbf{Z}) = 0$ and $H^2(G,\mathbf{Z})[n] = H^1(G,\mathbf{Z}/n\mathbf{Z})$.

4.2 Finite Quotients and Direct Limits. — We then study the relationship between the cohomology of profinite groups and the cohomology of finite groups.

Proposition 4.4. — *Let (G_i) be a projective system of profinite groups with surjective transition maps. Let (A_i) be an inductive system of discrete G_i-modules, with transition maps compatible with those of the G_i. Let $G = \varprojlim G_i$ and $A = \varinjlim A_i$. Then for all $q \in \mathbf{N}$:*

$$H^q(G,A) = \varinjlim H^q(G_i, A_i).$$

The proof is given in Appendix, §28.3.

We immediately deduce the following corollaries (the first of which can be taken as a definition of the groups $H^q(G,A)$).

Corollary 4.5. — Let A be a discrete G-module. Then

$$H^q(G,A) = \varinjlim_U H^q(G/U, A^U),$$

where U runs through the set of open normal subgroups of G.

Proof. — We indeed have $G = \varprojlim_U (G/U)$ and $A = \varinjlim_U A^U$. □

Corollary 4.6. — Let A be a discrete G-module. Then

$$H^q(G,A) = \varinjlim H^q(G,B),$$

where B runs through the set of finite type sub-G-modules of A.

Proof. — This follows from the fact that A is the union, hence the inductive limit, of such B's. Indeed, every element of a discrete G-module has open stabilizer, hence finite orbit, so the G-submodule generated by finitely many elements is finitely generated as an abelian group. □

Corollary 4.7. — For $q \geq 1$, the groups $H^q(G,A)$ are torsion.

Proof. — By Corollary 4.5, every class in $H^q(G,A)$ comes from $H^q(G/U, A^U)$ for some open normal subgroup U. The finite group G/U kills its positive-degree cohomology by the restriction–corestriction argument recalled in Appendix, Theorem 28.22 and Corollary 28.24. Hence every class in $H^q(G,A)$ is torsion. □

Lecture 5

We continue our calculation on cohomology theory for profinite groups.

For $i \geq 0$, let G^{i+1} denote the direct product of $i+1$ copies of G. We view $\mathbf{Z}[G^{i+1}]$ as a G-module via the left action

$$g \cdot (g_0, g_1, \ldots, g_i) = (gg_0, gg_1, \ldots, gg_i).$$

We first introduce the standard resolution.

Definition 5.1. — The *(augmented) standard resolution* of $\mathbf{Z}$ by G-modules is the sequence of G-module homomorphisms

$$\cdots \to \mathbf{Z}[G^{i+1}] \xrightarrow{d_i} \mathbf{Z}[G^i] \to \cdots \to \mathbf{Z}[G] \xrightarrow{\varepsilon} \mathbf{Z},$$

where

$$d_i(g_0, \ldots, g_i) = \sum_{j=0}^{i} (-1)^j (g_0, \ldots, g_{j-1}, g_{j+1}, \ldots, g_i)$$

for each $i \geq 1$, and ε is the augmentation map. At times, we may use $(g_0, \ldots, \hat{g}_j, \ldots, g_i) \in G^i$ to denote the i-tuple excluding g_j. To see that this definition is actually reasonable, we need the following lemma.

Proposition 5.2. — *The augmented standard resolution is exact.*

Proof. — As a complex of abelian groups, it has a contracting homotopy. Define

$$s_{-1} : \mathbf{Z} \to \mathbf{Z}[G], \qquad 1 \mapsto (1),$$

and for $i \geq 0$,

$$s_i : \mathbf{Z}[G^{i+1}] \to \mathbf{Z}[G^{i+2}], \qquad (g_0, \ldots, g_i) \mapsto (1, g_0, \ldots, g_i).$$

A direct calculation gives

$$d_{i+1}s_i + s_{i-1}d_i = \mathrm{id}$$

with the evident interpretation at the augmented end. Hence the augmented complex is exact. Since the differentials are G-equivariant, this is an exact sequence of G-modules. □

For a G-module A, we wish to consider the following complex

$$0 \to \mathrm{Hom}_{\mathbf{Z}[G]}(\mathbf{Z}[G], A) \to \cdots \to \mathrm{Hom}_{\mathbf{Z}[G]}(\mathbf{Z}[G^{i+1}], A) \xrightarrow{D^i} \mathrm{Hom}_{\mathbf{Z}[G]}(\mathbf{Z}[G^{i+2}], A) \to \cdots$$

Here, we define $D^i = D^i_A$ by $D^i(\varphi) = \varphi \circ d_{i+1}$. We compare this with the complex of cochains for G.

Theorem 5.3. — *The maps*

$$\psi^i : \mathrm{Hom}_{\mathbf{Z}[G]}(\mathbf{Z}[G^{i+1}], A) \to C^i(G, A)$$

defined by

$$\psi^i(\varphi)(g_1, \ldots, g_i) = \varphi(1, g_1, g_1g_2, \ldots, g_1g_2 \cdots g_i)$$

are isomorphisms for all $i \geq 0$. This provides isomorphisms of complexes in the sense that $\psi^{i+1} \circ D^i = d^i \circ \psi^i$ for all $i \geq 0$. Moreover, these isomorphisms are natural in the G-module A.

Proof. — If $\psi^i(\varphi) = 0$, then $\varphi(1, g_1, g_1g_2, \dots, g_1g_2\cdots g_i) = 0$ for all $g_1, \dots, g_i \in G$. Let $h_0, \dots, h_i \in G$, and define $g_j = h_{j-1}^{-1}h_j$ for all $1 \leq j \leq i$. We then have

$$\varphi(h_0, h_1, \dots, h_i) = h_0\varphi(1, h_0^{-1}h_1, \dots, h_0^{-1}h_i) = h_0\varphi(1, g_1, \dots, g_1\cdots g_i) = 0.$$

Therefore, ψ^i is injective. On the other hand, if $f \in C^i(G, A)$, then defining

$$\varphi(h_0, h_1, \dots, h_i) = h_0 f(h_0^{-1}h_1, \dots, h_{i-1}^{-1}h_i),$$

we have

$$\varphi(gh_0, gh_1, \dots, gh_i) = gh_0 f((gh_0)^{-1}gh_1, \dots, (gh_{i-1})^{-1}gh_i) = g\varphi(h_0, h_1, \dots, h_i)$$

and $\psi^i(\varphi) = f$. Therefore, ψ^i is an isomorphism of groups.

That ψ^i forms a map of complexes is shown in the following computation:

$$\begin{aligned}
\psi^{i+1}(D^i(\varphi))(g_1, \dots, g_{i+1}) &= D^i(\varphi)(1, g_1, \dots, g_1\cdots g_{i+1}) \\
&= \varphi \circ d_{i+1}(1, g_1, \dots, g_1\cdots g_{i+1}) \\
&= \sum_{j=0}^{i+1} (-1)^j \varphi(1, g_1, \dots, \widehat{g_1\cdots g_j}, \dots, g_1\cdots g_{i+1}).
\end{aligned}$$

The latter term equals

$$g_1\psi^i(\varphi)(g_2, \dots, g_{i+1}) + \sum_{j=1}^{i} (-1)^j \psi^i(\varphi)(g_1, \dots, g_{j-1}, g_jg_{j+1}, g_{j+2}, \dots, g_{i+1})$$
$$+ (-1)^{i+1}\psi^i(\varphi)(g_1, \dots, g_i),$$

which is $d^i(\psi^i(\varphi))$.

Finally, suppose that $\alpha : A \to B$ is a G-module homomorphism. We then have

$$\alpha \circ \psi^i(\varphi)(g_1, \dots, g_i) = \alpha \circ \varphi(1, g_1, \dots, g_1\cdots g_i) = \psi^i(\alpha \circ \varphi)(g_1, \dots, g_i),$$

hence the desired naturality. $\square$

Corollary 5.4. — The ith cohomology group of the complex $(\mathrm{Hom}_{\mathbf{Z}[G]}(\mathbf{Z}[G^{i+1}], A), D_A^i)$ is naturally isomorphic to $H^i(G, A)$.

5.1 $H^n(G, A)$ is torsion for $n \geq 1$. — We leave the proof in the Appendix 28.2. It uses corestriction and restriction, and the dimension shifting.

Corollary 5.5. — If $A \in \mathscr{C}_G$ is a $\mathbf{Q}$ vector space, then $H^n(G, A) = 0$, for any $n \geq 1$.

Proof. — The cochain groups $C^i(G, A)$ are $\mathbf{Q}$-vector spaces, hence so is $H^n(G, A)$. By the previous torsion result, $H^n(G, A)$ is also torsion for $n \geq 1$. A torsion $\mathbf{Q}$-vector space is zero. $\square$

Definition 5.6. — Let p be a prime. A *pro-p group* is a profinite group G such that $G \cong \varprojlim G_i$, $|G_i| = p^{l_i}$, $l_i \geq 0$.

Corollary 5.7. — If G is a pro-p group and $A \in \mathscr{C}_G$ is a discrete G-module in which p is invertible (i.e., multiplication by p is an automorphism of A), then $H^n(G, A) = 0$ for all $n \geq 1$.

Proof. — Write $G = \varprojlim G_i$ with G_i finite p-groups. For any open normal subgroup U_i such that $G/U_i \cong G_i$, we have $A^{U_i} \subset A$, and p remains invertible in A^{U_i}. Then $|G_i| \cdot H^n(G_i, A^{U_i}) = 0$. Since $|G_i|$ is a power of p, and multiplication by p is an isomorphism on A^{U_i}, the only element of $H^n(G_i, A^{U_i})$ annihilated by a power of p is zero. Thus $H^n(G_i, A^{U_i}) = 0$ for all i. By Corollary 4.5, $H^n(G, A) = \varinjlim H^n(G_i, A^{U_i}) = 0$. □

Lecture 6

This lecture proves the first basic computations in Galois cohomology: additive Hilbert 90, multiplicative Hilbert 90, cyclic cohomology, and the Kummer exact sequence. The goal is to connect low-degree cohomology with concrete arithmetic information.

6.1 Additive Theory. — Let K/k be a Galois extension. Then $\mathrm{Gal}(K/k)$ acts on the additive group $(K, +)$, so K is a discrete $\mathrm{Gal}(K/k)$-module.

Proposition 6.1. —

$$H^n(\mathrm{Gal}(K/k), K) = 0, \quad \forall n \geq 1.$$

Proof. — By Corollary 4.5,

$$H^n(\mathrm{Gal}(K/k), K) \cong \varinjlim_{F \in \mathscr{M}} H^n(\mathrm{Gal}(F/k), F),$$

where $\mathscr{M} = \{F \mid k \subseteq F \subseteq K, \text{ with } F/k \text{ finite Galois}\}$. It is therefore enough to treat the case where K/k is finite Galois. Put $G = \mathrm{Gal}(K/k)$. By the normal basis theorem, there is an element $\alpha \in K$ such that

$$\{\sigma(\alpha) \mid \sigma \in G\}$$

is a k-basis of K. Define a k-linear map

$$\Phi : C(G, k) \longrightarrow K, \qquad f \longmapsto \sum_{g \in G} f(g)\, g^{-1}(\alpha),$$

where $C(G, k)$ is the k-vector space of continuous functions $G \to k$. Since G is finite here, this is simply the space of all functions on G. The group G acts on $C(G, k)$ by right translation in this convention: $(\tau f)(g) = f(g\tau)$. Then

$$\Phi(\tau f) = \sum_{g \in G} f(g\tau) g^{-1}(\alpha) = \sum_{h \in G} f(h)(h\tau^{-1})^{-1}(\alpha) = \tau \left(\sum_{h \in G} f(h) h^{-1}(\alpha) \right) = \tau \Phi(f),$$

so Φ is G-equivariant. Since the functions δ_g supported at g map to $g^{-1}(\alpha)$, the image contains a normal basis of K; hence Φ is surjective. Both sides have k-dimension $|G|$, so Φ is an isomorphism. Finally $C(G,k)$ is the coinduced module $\mathrm{Coind}_{\{1\}}^G(k)$; since G is finite, it is also the induced module $\mathrm{Ind}_{\{1\}}^G(k)$. Shapiro's lemma gives

$$H^n(G,K) \cong H^n(G, C(G,k)) \cong H^n(\{1\}, k) = 0 \qquad (n \geq 1).$$

□

6.2 Multiplicative Theory. —

Theorem 6.2 **("generalized" Hilbert 90).** — *Let K/k be a Galois extension of fields. Then $H^1(\mathrm{Gal}(K/k), K^\times) = 0$.*

First we recall the linear independence of the characters:

Lemma 6.3. — *Let G be a group, $\chi_1, \ldots, \chi_n : G \to L^\times$ be different group homomorphisms. Then χ_i are linearly independent in $\mathscr{F}(G,L)$.*

Proof. — Suppose that there exists $\lambda_1, \ldots, \lambda_n \in L$ and not all $\lambda_i = 0$, such that $\sum \lambda_i \chi_i = 0$ in $\mathscr{F}(G,L)$, i.e. for any $g \in G$, $\sum \lambda_i \chi_i(g) = 0$.

Without loss of generality, we assume that $n \geq 2$, and $\lambda_1 \neq 0$. Since $\chi_1 \neq \chi_2$, there exists $h \in G$ such that $\chi_1(h) \neq \chi_2(h)$. For any $g \in G$, we have the equation

$$\sum_{i=1}^n \lambda_i \chi_i(g) = 0. \tag{1}$$

Replacing g by hg in (1), we obtain

$$\sum_{i=1}^n \lambda_i \chi_i(hg) = \sum_{i=1}^n \lambda_i \chi_i(h) \chi_i(g) = 0. \tag{2}$$

Now multiply (1) by $\chi_1(h)$ and subtract (2):

$$\sum_{i=1}^n \lambda_i (\chi_1(h) - \chi_i(h)) \chi_i(g) = 0.$$

Since $\chi_1(h) - \chi_1(h) = 0$, the term for $i = 1$ vanishes. For $i = 2$, we have $\chi_1(h) - \chi_2(h) \neq 0$. Thus we obtain a shorter nontrivial linear relation among the χ_i (with $\lambda_2' = \lambda_2(\chi_1(h) - \chi_2(h)) \neq 0$). By induction, we eventually reach a contradiction. Hence all λ_i must be zero, proving linear independence. □

proof of "generalized" Hilbert 90. — Let $\mathscr{E}$ denote the set of finite Galois extensions of k in K. Then

$$H^1(\mathrm{Gal}(K/k), K^\times) = \varinjlim_{E \in \mathscr{E}} H^1(\mathrm{Gal}(E/k), E^\times),$$

which reduces us to the case that K/k is finite Galois. Let $G = \mathrm{Gal}(K/k)$, and let $f : G \to K^\times$ be a 1-cocycle. We may view the elements $\sigma \in G$ as distinct multiplicative characters $K^\times \to K^\times$ via $\alpha \mapsto \sigma(\alpha)$. These characters are linearly independent. Hence the map $\sum_{\sigma\in G} f(\sigma)\sigma : K^\times \to K$ is not zero. Choose $\alpha \in K^\times$ such that $z = \sum_{\sigma\in G} f(\sigma)\sigma(\alpha) \neq 0$. For any $\tau \in G$, we have

$$\begin{aligned}\tau^{-1}(z) &= \sum_{\sigma\in G} \tau^{-1}(f(\sigma)) \cdot \tau^{-1}\sigma(\alpha)\\ &= \sum_{\sigma\in G} \tau^{-1}(f(\tau\sigma))\sigma(\alpha)\\ &= \sum_{\sigma\in G} \tau^{-1}(f(\tau)\cdot \tau f(\sigma))\sigma(\alpha)\\ &= \tau^{-1}(f(\tau)) \sum_{\sigma\in G} f(\sigma)\sigma(\alpha)\\ &= \tau^{-1}(f(\tau))z.\end{aligned}$$

Thus, $f(\tau) = \frac{z}{\tau(z)}$, so f is the 1-coboundary of z^{-1}. Hence every 1-cocycle is a 1-coboundary, and $H^1(G, K^\times) = 0$. ☐

6.3 Digression: Cohomology of finite cyclic groups. — Let G be a cyclic group of finite order. We prove that the Tate cohomology[4] groups with coefficients in a module A are *periodic* in degree of period 2, up to isomorphisms determined by a choice of generator g of G.

The first thing that we will observe is that for such a group G, there is an even nicer projective resolution of $\mathbf{Z}$ than the standard one: i.e., consider the sequence

$$\cdots \to \mathbf{Z}[G] \xrightarrow{N_G} \mathbf{Z}[G] \xrightarrow{g-1} \mathbf{Z}[G] \xrightarrow{N_G} \mathbf{Z}[G] \xrightarrow{g-1} \mathbf{Z}[G] \xrightarrow{\varepsilon} \mathbf{Z} \to 0,$$

where the boundary maps are multiplication N_G in even degree and by $g-1$ in odd degree.

Proposition 6.4. — *The Tate cohomology groups of A are the cohomology groups of the two-periodic complex*

$$\cdots \to A \xrightarrow{g-1} A \xrightarrow{N_G} A \xrightarrow{g-1} A \to \cdots,$$

with a map $g-1$ following the term in degree 0.

Proof. — Note first that for $i \in \{-1, 0\}$, the group $\hat{H}^i(G, A)$ is by definition isomorphic to the ith cohomology group of the complex in question. Let $C_\bullet$ denote the projective resolution of $\mathbf{Z}$ as before. The complex $C_\bullet \otimes_{\mathbf{Z}[G]} A$ that ends

$$\cdots \to A \xrightarrow{g-1} A \xrightarrow{N_G} A \xrightarrow{g-1} A \to 0$$

computes $H_i(G, A)$, yielding the result for $i \leq -2$.

[4] Tate cohomology is introduced later in Definition 23.5. Until then, one can read the positive-degree statements as ordinary group cohomology. The Tate version is obtained by splicing a projective resolution with its dual, which is why it also includes degrees $0, -1, -2, \ldots$.

Multiplication by g induces the endomorphism

$$\mathrm{Hom}_{\mathbf{Z}[G]}(\mathbf{Z}[G], A) \to \mathrm{Hom}_{\mathbf{Z}[G]}(\mathbf{Z}[G], A), \quad \varphi \mapsto (x \mapsto \varphi(gx) = g\varphi(x)).$$

Via the isomorphism of G-modules $\mathrm{Hom}_{\mathbf{Z}[G]}(\mathbf{Z}[G], A) \cong A$ given by evaluation at 1, the latter endomorphism is identified with multiplication by g on A. The complex $\mathrm{Hom}_{\mathbf{Z}[G]}(C_\bullet, A)$ that computes $H^i(G, A)$ is therefore isomorphic to

$$0 \to A \xrightarrow{g-1} A \xrightarrow{N_G} A \xrightarrow{g-1} A \to \cdots,$$

providing the result for $i \geq 1$. □

The next two corollaries use the Tate groups $\widehat{H}^{-1}$ and $\widehat{H}^0$ for a finite cyclic group. For the formal definition and the long exact sequence, see Definition 23.5 and Theorem 23.6.

Corollary 6.5. — Let L/K be a *finite cyclic* extension of fields, and let $N_{L/K} : L^\times \to K^\times$ be the norm map. Then

$$\ker N_{L/K} = \left\{ \alpha \in L^\times \mid \alpha = \frac{\sigma(\beta)}{\beta} \text{ for some } \beta \in L^\times \right\},$$

where σ is a generator of $\mathrm{Gal}(L/K)$.

Proof. — Since σ generates $G = \mathrm{Gal}(L/K)$, the element $\sigma - 1 \in \mathbf{Z}[G]$ generates $I_G = \ker(\varepsilon)$, and so the statement at hand is $\ker N_{L/K} = I_G L^\times$, which is to say $\hat{H}^{-1}(G, L^\times) = 0$. Since G is cyclic, we have $\hat{H}^{-1}(G, L^\times) \cong H^1(G, L^\times)$, by Proposition 6.4. Thus, the result follows from Theorem 6.2. □

Proposition 6.6. — *Assume that K/k is an extension of finite fields. Then $H^r(\mathrm{Gal}(K/k), K^\times) = 0$, for any $r \geq 1$.*

Proof. — Let $k = \mathbf{F}_q$, $K = \mathbf{F}_{q^m}$, and put $G = \mathrm{Gal}(K/k)$. We have $H^1(G, K^\times) = 0$ by Hilbert's Theorem 90 (Theorem 6.2). Since G is cyclic, the positive-degree cohomology is two-periodic. Thus all odd-degree groups vanish. In even positive degree,

$$H^r(G, K^\times) \cong \widehat{H}^0(G, K^\times) = k^\times / N_{K/k}(K^\times).$$

The group $K^\times$ is cyclic. If α is a generator of order $q^m - 1$, then

$$N_{K/k}(\alpha) = \alpha^{1+q+\cdots+q^{m-1}}$$

has order $q-1$, hence generates $k^\times$. Thus the norm map is surjective and $\widehat{H}^0(G, K^\times) = 0$. Therefore $H^r(G, K^\times) = 0$ for all $r \geq 1$. □

6.4 Kummer Theory. —

Proposition 6.7. — *Let K be a field of characteristic not dividing $n \geq 1$, and let μ_n be the group of roots of unity in a separable closure K^{sep} of K. Let $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$ be the absolute Galois group. Then there are canonical isomorphisms*

$$K^\times / K^{\times n} \xrightarrow{\sim} H^1(G_K, \mu_n) \quad \textit{and} \quad H^2(G_K, \mu_n) \xrightarrow{\sim} \mathrm{Br}(K)[n],$$

where, in this cohomological formulation, $\mathrm{Br}(K) = H^2(G_K, (K^{\mathrm{sep}})^\times)$ *and* $\mathrm{Br}(K)[n]$ *denotes its* n*-torsion subgroup.*

Proof. — Since K^{sep} is separably closed, we have an exact sequence

$$1 \to \mu_n \to (K^{\mathrm{sep}})^\times \xrightarrow{\times n} (K^{\mathrm{sep}})^\times \to 1$$

of discrete G_K-modules, where the map $\times n$ is raising to the n-th power. This sequence is exact because every element of $(K^{\mathrm{sep}})^\times$ has an n-th root in the separably closed field K^{sep}, and the kernel of raising to the n-th power is precisely μ_n.

Taking the long exact cohomology sequence associated to this short exact sequence, we obtain:

$$\begin{aligned} 0 \to H^0(G_K, \mu_n) \to H^0(G_K, (K^{\mathrm{sep}})^\times) &\xrightarrow{\times n} H^0(G_K, (K^{\mathrm{sep}})^\times) \\ \to H^1(G_K, \mu_n) \to H^1(G_K, (K^{\mathrm{sep}})^\times) &\xrightarrow{\times n} H^1(G_K, (K^{\mathrm{sep}})^\times) \\ \to H^2(G_K, \mu_n) \to H^2(G_K, (K^{\mathrm{sep}})^\times) &\xrightarrow{\times n} H^2(G_K, (K^{\mathrm{sep}})^\times) \to \cdots \end{aligned}$$

Now $H^0(G_K, (K^{\mathrm{sep}})^\times) = K^\times$ by definition of Galois invariants. By Hilbert's Theorem 90 (Theorem 6.2), we have $H^1(G_K, (K^{\mathrm{sep}})^\times) = 0$. Moreover, by definition of the Brauer group, $\mathrm{Br}(K) = H^2(G_K, (K^{\mathrm{sep}})^\times)$.

The long exact sequence therefore breaks into the following exact sequences:

$$K^\times \xrightarrow{\times n} K^\times \to H^1(G_K, \mu_n) \to 0, \quad 0 \to H^2(G_K, \mu_n) \to \mathrm{Br}(K) \xrightarrow{\times n} \mathrm{Br}(K).$$

The first exact sequence gives the isomorphism $K^\times / K^{\times n} \cong H^1(G_K, \mu_n)$. The second exact sequence shows that $H^2(G_K, \mu_n)$ is isomorphic to the kernel of multiplication by n on $\mathrm{Br}(K)$, which is precisely the n-torsion subgroup $\mathrm{Br}(K)[n]$. This completes the proof. □

Remark 6.8. — The image of $b \in K^\times$ in $H^1(G_K, \mu_n)$ is represented explicitly as follows. Choose an n-th root $\sqrt[n]{b}$ in K^{sep} and send

$$g \longmapsto \frac{g(\sqrt[n]{b})}{\sqrt[n]{b}}.$$

A different choice of n-th root of b differs by an element $\xi \in \mu_n$, and gives a cohomologous cocycle.

This shows that the choice of the cocycle is independent of any choices.

In the special case $\mu_n \subseteq K$, the action of G_K on μ_n is trivial, and hence

$$K^\times / K^{\times n} \cong H^1(G_K, \mu_n) \cong \mathrm{Hom}_{\mathrm{cont}}(G_K, \mu_n),$$

which is Kummer duality.

6.5 Local Fields and Units. — We denote by K a non-archimedean local field with absolute value $|\cdot|$ (i.e. $\mathbf{F}_p((t))$ or $\mathbf{Q}_p$). We denote by $\mathscr{O}_K$ the valuation ring of K, and $\mathfrak{m}_K = \{x \in K : |x| < 1\}$ the maximal ideal of $\mathscr{O}_K$. $\mathfrak{m}_K$ is generated by the uniformizer ϖ_K.

Then every element $x \in K^\times$ can be written uniquely as $x = u \cdot \varpi_K^n$ with u a unit and $n \in \mathbf{Z}$ ($n := \mathrm{val}(x)$ is the valuation of x). This decomposition gives an isomorphism

$$K^\times \cong \mathscr{O}_K^\times \times \mathbf{Z},$$

where $\mathscr{O}_K^\times$ denotes the group of units in the valuation ring. The topology on $K^\times$ is such that $\mathscr{O}_K^\times$ is an open subgroup, and the quotient $\mathscr{O}_K^\times/(1+\mathfrak{m}_K^n)$ is finite for each $n \geq 1$.

In particular, for a local field, we have the following exact sequence:

$$1 \to \mathscr{O}_K^\times \to K^\times \xrightarrow{\mathrm{val}} \mathbf{Z} \to 0.$$

Lecture 7

This lecture studies the cohomology of the maximal unramified extension of a non-archimedean local field. The valuation sequence reduces the computation of H^2 to the unramified Galois group and leads to the invariant map inv_K.

First we need a technical lemma.

Lemma 7.1. — *Let G be a finite group and let M be a G-module together with a decreasing filtration by G-submodules*

$$\cdots \subset M^2 \subset M^1 \subset M^0 = M,$$

such that $M \cong \varprojlim_i M/M^i$. If, for some $n \geq 1$, one has $H^n(G, M^i/M^{i+1}) = 0$ for every $i \geq 0$, then $H^n(G, M) = 0$.

Proof. — Let $f \in Z^n(G, M)$. For each $r \geq 1$, put $M_r := M/M^r$. We first show by induction on r that the image of f in $Z^n(G, M_r)$ is a coboundary.

For $r = 1$, the module is M/M^1, and the hypothesis for $i = 0$ gives $H^n(G, M/M^1) = 0$, so there is a cochain $\psi_1 \in C^{n-1}(G, M_1)$ with $d\psi_1 = f$ in $C^n(G, M_1)$. Assume now that $\psi_r \in C^{n-1}(G, M_r)$ satisfies $d\psi_r = f$ modulo M^r. Lift ψ_r arbitrarily to a cochain $\widetilde{\psi}_r \in C^{n-1}(G, M_{r+1})$. Then $f - d\widetilde{\psi}_r$ takes values in M^r/M^{r+1} and is an n-cocycle there. By the hypothesis $H^n(G, M^r/M^{r+1}) = 0$, there exists

$$\eta_r \in C^{n-1}(G, M^r/M^{r+1})$$

such that $d\eta_r = f - d\widetilde{\psi}_r$ in $C^n(G, M^r/M^{r+1})$. Set $\psi_{r+1} = \widetilde{\psi}_r + \eta_r$. Then $d\psi_{r+1} = f$ modulo M^{r+1}, and ψ_{r+1} reduces to ψ_r modulo M^r.

Thus the ψ_r form a compatible system in

$$C^{n-1}(G, M/M^r).$$

Since G is finite and $M \simeq \varprojlim_r M/M^r$, we also have

$$C^{n-1}(G, M) \simeq \varprojlim_r C^{n-1}(G, M/M^r).$$

Let $\psi \in C^{n-1}(G, M)$ be the inverse-limit cochain. The equality $d\psi_r = f$ modulo M^r for every r implies $d\psi = f$ in $C^n(G, M)$. Therefore every n-cocycle is a coboundary, and $H^n(G, M) = 0$. □

Aim: for a local field K, let K^{sep} be a separable closure. We want to calculate $H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times) \cong \mathbf{Q}/\mathbf{Z}$.

We next introduce Teichmüller lifts. They give canonical representatives of residue-field elements inside the valuation ring, and they are useful for splitting the unit group into residue-field units and principal units.

***Theorem 7.2* (Hensel's Lemma).** — *Let $(K, |\cdot|)$ be a complete non-archimedean field, $f(x) \in \mathscr{O}_K[x]$ such that $0 \neq \overline{f}(x) \in k[x]$ (such polynomial $f(x)$ is called primitive). Suppose we have*

$$\overline{f}(x) = \overline{u}(x)\overline{v}(x)$$

such that $(\overline{u}(x), \overline{v}(x)) = 1$. Then $f(x)$ admits a factorization

$$f(x) = u(x)v(x)$$

where $u(x), v(x) \in \mathscr{O}_K[x]$, such that $\deg u(x) = \deg \overline{u}(x)$, $u(x) \equiv \overline{u}(x) \pmod{\mathfrak{m}_K}$, and $v(x) \equiv \overline{v}(x) \pmod{\mathfrak{m}_K}$.

Corollary 7.3. — Let $f(x) \in \mathscr{O}_K[x]$, $a_0 \in \mathscr{O}_K$ such that $f(a_0) \equiv 0 \pmod{\mathfrak{m}_K}$ and $f'(a_0) \not\equiv 0 \pmod{\mathfrak{m}_K}$. Then there exists a unique $a \in \mathscr{O}_K$ such that $f(a) = 0$ and $a \equiv a_0 \pmod{\mathfrak{m}_K}$.

In particular, if the residue field is $k = \mathbf{F}_q$, then for every $\lambda \in k$ we may choose a lift $a_0 \in \mathscr{O}_K$ of λ and apply the result to $f(x) = x^q - x$. The unique root congruent to a_0 modulo $\mathfrak{m}_K$ is denoted $[\lambda]$ and is called the Teichmüller lift of λ.

Lemma 7.4. — *For any $i \geq 1$, we have isomorphisms*

$$\frac{1 + \mathfrak{m}_K^i}{1 + \mathfrak{m}_K^{i+1}} \cong \frac{\mathfrak{m}_K^i}{\mathfrak{m}_K^{i+1}} \cong k,$$

where the first map is given by

$$1 + x \longmapsto x \pmod{\mathfrak{m}_K^{i+1}}, \qquad x \in \mathfrak{m}_K^i,$$

and the second by $\varpi_K^i u \mapsto \bar{u} \in k$. For $i = 0$, one instead has $\mathscr{O}_K^\times/(1 + \mathfrak{m}_K) \cong k^\times$.

Proof. — The first map is well-defined because every element of $1 + \mathfrak{m}_K^i$ has the form $1 + x$ with $x \in \mathfrak{m}_K^i$, and changing x by an element of $\mathfrak{m}_K^{i+1}$ gives the same class modulo $1 + \mathfrak{m}_K^{i+1}$. It is a group homomorphism since, for $x, y \in \mathfrak{m}_K^i$,

$$(1 + x)(1 + y) = 1 + x + y + xy,$$

and $xy \in \mathfrak{m}_K^{2i} \subseteq \mathfrak{m}_K^{i+1}$ for $i \geq 1$. Hence the product maps to $x+y$ modulo $\mathfrak{m}_K^{i+1}$. Its kernel consists exactly of those $1+x$ with $x \in \mathfrak{m}_K^{i+1}$, i.e. $1+\mathfrak{m}_K^{i+1}$, so the first map is an isomorphism.

For the second isomorphism, choose a uniformizer ϖ_K. Every element of $\mathfrak{m}_K^i$ can be written as $\varpi_K^i u$ with $u \in \mathscr{O}_K$, and its class modulo $\mathfrak{m}_K^{i+1}$ is sent to $\bar{u} \in k$. The kernel is precisely $\mathfrak{m}_K^{i+1}$, and every residue class $\bar{u} \in k$ occurs. Thus $\mathfrak{m}_K^i/\mathfrak{m}_K^{i+1} \cong k$ as additive groups. $\square$

Lemma 7.5. — *For any $i \geq 1$, we have*

$$1+\mathfrak{m}_K^i \cong \varprojlim_{j \geq i} \frac{1+\mathfrak{m}_K^i}{1+\mathfrak{m}_K^j}.$$

In particular, $1+\mathfrak{m}_K^i$ is a pro-p-group.

Proof. — The first assertion is just completeness of $\mathscr{O}_K$ for the $\mathfrak{m}_K$-adic topology, restricted to the subgroup $1+\mathfrak{m}_K^i$. The subgroups $1+\mathfrak{m}_K^j$ form a neighborhood basis of 1, and

$$\bigcap_{j \geq i}(1+\mathfrak{m}_K^j) = \{1\}.$$

Thus an element of the inverse limit is exactly a compatible system of approximations modulo $1+\mathfrak{m}_K^j$, which converges uniquely to an element of $1+\mathfrak{m}_K^i$.

For each $j > i$, the finite quotient $(1+\mathfrak{m}_K^i)/(1+\mathfrak{m}_K^j)$ has a filtration whose successive quotients are

$$(1+\mathfrak{m}_K^r)/(1+\mathfrak{m}_K^{r+1}) \cong k \qquad (i \leq r < j).$$

Since k is a finite field of characteristic p, each successive quotient is a finite p-group. Hence every finite quotient in the inverse system is a finite p-group, and $1+\mathfrak{m}_K^i$ is pro-p. $\square$

7.1 Unramified extension.

— Recall: extension of the valuation:

Theorem 7.6. — *Let $(K, |\cdot|)$ be a complete non-archimedean field. Let L/K be a finite extension of degree n. Then there exists a unique extension $|\cdot|_L$ of $|\cdot|$ to a non-archimedean valuation on L. Moreover, for any $x \in L$, $|x|_L = |N_{L/K}(x)|^{1/n}$ and L is complete for $|\cdot|_L$.*

Hence $\mathscr{O}_K, \varpi_K, \ldots$ are stable under $\mathrm{Gal}(L/K)$. We consider the unramified extension.

Theorem 7.7. — *Let k'/k be a finite separable extension.*

(1) There exists an unramified extension K'/K with residue field k'. This extension is unique up to isomorphism and K'/K is Galois if and only if k'/k is Galois.

(2) For any finite extension L/K with residue field k_L. There is a natural bijection between the set of K-embeddings of K' in L and the set of k-embeddings of k' in k_L. In particular, if k'/k is Galois, then $\mathrm{Gal}(K'/K) \cong \mathrm{Gal}(k'/k)$.

We have the following exact sequence of $\mathrm{Gal}(L/K)$-modules arising from the valuation on L:

$$0 \to \mathscr{O}_L^\times \to L^\times \xrightarrow{v_L} \mathbf{Z} \to 0,$$

where the Galois action on $\mathbf{Z}$ is trivial. This sequence induces a long exact sequence in cohomology.

Our primary interest lies in the cohomology group $H^2(\mathrm{Gal}(L/K), L^\times)$.

Proposition 7.8. — *If L/K is a finite unramified extension, then*

$$H^n(\mathrm{Gal}(L/K), \mathscr{O}_L^\times) = 0 \quad \textit{for all } n \geq 1.$$

Proof. — Let $G = \mathrm{Gal}(L/K)$, which is isomorphic to $\mathrm{Gal}(k_L/k)$ and is therefore finite. The ring of integers $\mathscr{O}_L$ has a natural decreasing filtration by the higher unit groups, which are G-submodules:

$$\cdots \subset U^{(2)} \subset U^{(1)} \subset U^{(0)} = \mathscr{O}_L^\times,$$

where $U^{(i)} = 1 + \mathfrak{m}_L^i$ for $i \geq 1$, and we set $U^{(0)} = \mathscr{O}_L^\times$ for convenience. This filtration satisfies $\mathscr{O}_L^\times \cong \varprojlim_i \mathscr{O}_L^\times / U^{(i)}$, as $\mathscr{O}_L^\times$ is complete with respect to the $\mathfrak{m}_L$-adic topology.(5)

Motivated by the lemma, we study the cohomology of quotients. In fact, the successive quotients have trivial cohomology:

1. For $i = 0$, $U^{(0)}/U^{(1)} \cong k_L^\times$. Since L/K is unramified, $G \simeq \mathrm{Gal}(k_L/k)$ is cyclic. we get
$$H^n(G, k_L^\times) = 0 \qquad (n \geq 1),$$
by proposition 6.6, using Hilbert 90 and two-periodicity of cyclic cohomology.
2. For $i \geq 1$, $U^{(i)}/U^{(i+1)} \cong k_L$, which is the additive group of the finite field k_L. Hence
$$H^n(G, k_L) = 0 \qquad (n \geq 1)$$
by additive Hilbert 90, Proposition 6.1.

In any case, for the purpose of applying technical lemma, we need $H^n(G, U^{(i)}/U^{(i+1)}) = 0$ for all $i \geq 0$ and all $n \geq 1$. This holds by the above arguments. Thus $H^n(G, \mathscr{O}_L^\times) = 0$ for all $n \geq 1$. $\square$

Definition 7.9. — Let K^{ur} be the composition of all finite unramified extensions L/K in $\bar{K}$. Then $K^{\mathrm{ur}} \subset K^{\mathrm{sep}}$ and K^{ur}/K is Galois.

We have $\mathrm{Gal}(K^{\mathrm{ur}}/K) \cong \mathrm{Gal}(\bar{k}/k) \cong \widehat{\mathbf{Z}}$.

Theorem 7.10. —

$$H^2(\mathrm{Gal}(K^{\mathrm{ur}}/K), (K^{\mathrm{ur}})^\times) \cong \mathbf{Q}/\mathbf{Z}.$$

(5)This is why we work first at finite unramified levels. The union $\mathscr{O}_{K^{\mathrm{ur}}} = \bigcup_L \mathscr{O}_L$ is not complete for the π-adic topology in general; its completion is $\mathscr{O}_{\widehat{K^{\mathrm{ur}}}}$. Thus the argument for units uses finite L/K and then passes to K^{ur} by direct limits in cohomology.

Proof. — 1. **Cohomology of $\widehat{\mathbf{Z}}$ with rational coefficients.** Consider the short exact sequence of trivial $\widehat{\mathbf{Z}}$-modules:

$$0 \to \mathbf{Z} \to \mathbf{Q} \to \mathbf{Q}/\mathbf{Z} \to 0.$$

Since $\mathbf{Q}$ is a $\mathbf{Q}$-vector space and $\widehat{\mathbf{Z}}$ is a profinite group, we have $H^n(\widehat{\mathbf{Z}}, \mathbf{Q}) = 0$ for all $n \geq 1$. Thus

$$H^2(\widehat{\mathbf{Z}}, \mathbf{Z}) \cong H^1(\widehat{\mathbf{Z}}, \mathbf{Q}/\mathbf{Z}) \cong \mathrm{Hom}_{\mathrm{cont}}(\widehat{\mathbf{Z}}, \mathbf{Q}/\mathbf{Z}) \cong \mathbf{Q}/\mathbf{Z}.$$

The last isomorphism follows because continuous characters $\widehat{\mathbf{Z}} \to \mathbf{Q}/\mathbf{Z}$ correspond to elements of $\mathbf{Q}/\mathbf{Z}$ via $f \mapsto f(1)$.

2. **The valuation exact sequence for K^{ur}.** For the maximal unramified extension K^{ur}, the valuation ring is

$$\mathscr{O}_{K^{\mathrm{ur}}} = \{x \in K^{\mathrm{ur}} : v(x) \geq 0\} = \bigcup_{L/K \text{ finite unramified}} \mathscr{O}_L,$$

The valuation $v : (K^{\mathrm{ur}})^\times \to \mathbf{Z}$ is surjective and $\mathrm{Gal}(K^{\mathrm{ur}}/K)$-equivariant (with trivial action on $\mathbf{Z}$), giving a short exact sequence of discrete $\mathrm{Gal}(K^{\mathrm{ur}}/K)$-modules:

$$0 \to \mathscr{O}_{K^{\mathrm{ur}}}^\times \to (K^{\mathrm{ur}})^\times \xrightarrow{v} \mathbf{Z} \to 0.$$

By taking the direct limit over all such L, we obtain

$$H^n(\mathrm{Gal}(K^{\mathrm{ur}}/K), \mathscr{O}_{K^{\mathrm{ur}}}^\times) = 0 \quad \text{for all } n \geq 1.$$

Applying Galois cohomology to the short exact sequence:

$$\cdots \to H^2(G_{\mathrm{ur}}, \mathscr{O}_{K^{\mathrm{ur}}}^\times) \to H^2(G_{\mathrm{ur}}, (K^{\mathrm{ur}})^\times) \to H^2(G_{\mathrm{ur}}, \mathbf{Z}) \to H^3(G_{\mathrm{ur}}, \mathscr{O}_{K^{\mathrm{ur}}}^\times) \to \cdots$$

where $G_{\mathrm{ur}} = \mathrm{Gal}(K^{\mathrm{ur}}/K) \cong \widehat{\mathbf{Z}}$.

Since $H^2(G_{\mathrm{ur}}, \mathscr{O}_{K^{\mathrm{ur}}}^\times) = 0$ and $H^3(G_{\mathrm{ur}}, \mathscr{O}_{K^{\mathrm{ur}}}^\times) = 0$, the long exact sequence gives an isomorphism

$$H^2(G_{\mathrm{ur}}, (K^{\mathrm{ur}})^\times) \cong H^2(G_{\mathrm{ur}}, \mathbf{Z}).$$

3. Thus we have $H^2(\widehat{\mathbf{Z}}, \mathbf{Z}) \cong \mathbf{Q}/\mathbf{Z}$. Therefore,

$$H^2(\mathrm{Gal}(K^{\mathrm{ur}}/K), (K^{\mathrm{ur}})^\times) \cong \mathbf{Q}/\mathbf{Z}.$$

□

Lecture 8

This lecture records restriction, inflation, and inflation–restriction. The main goal is to compare the cohomology of a profinite group G, a closed normal subgroup N, and the quotient G/N, first in low degree and then in the higher-degree form needed later. As a corollary, we compute $H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times)$, and we will see that for every finite Galois extension L/K of degree n, $H^2(\mathrm{Gal}(L/K), L^\times)$ is cyclic of order n.

8.1 Restriction and inflation-restriction exact sequence. — Let G be a profinite group and let H be a closed subgroup of G. If I is an injective object in $\mathscr{C}_G$, then its restriction I_H is injective in $\mathscr{C}_H$. This induces a map of complexes $(I^\bullet)^G \to (I^\bullet)^H$, and consequently a map in cohomology.

Remark 8.1. — If H is normal in G, then G/H acts on $(I^n)^H$ and the transition maps are G/H-equivariant. Hence $H^n(H, A)$ inherits a natural G/H-action.

Proposition 8.2. — *Let N be a closed normal subgroup of G, and let $A \in \mathscr{C}_G$. If $H^i(N, A) = 0$ for $1 \le i \le n-1$, then there exists an exact sequence:*

$$0 \to H^n(G/N, A^N) \to H^n(G, A) \to H^n(N, A)^{G/N}.$$

The first map is the inflation map, and the second is the restriction map (followed by taking G/N-invariants).

Lemma 8.3. — *If I is injective in $\mathscr{C}_G$, then I^N is injective in $\mathscr{C}_{G/N}$.*

Proof. — This follows from the natural isomorphism:

$$\mathrm{Hom}_G(M, I) = \mathrm{Hom}_{G/N}(M, I^N),$$

which holds for any G/N-module M viewed as a G-module via the quotient map $G \to G/N$. □

Lemma 8.4 **(Dimension shifting).** — *Let $Q = G/N$, let $A \in \mathscr{C}_G$, and let $A \to I^\bullet$ be an injective resolution in $\mathscr{C}_G$. Put*

$$C^\bullet = (I^\bullet)^N, \qquad B^r = \mathrm{im}(C^{r-1} \to C^r) \quad (r \ge 1).$$

If $H^i(N, A) = 0$ for $1 \le i \le n-1$, then

$$0 \to A^N \to C^0 \to C^1 \to \cdots \to C^{n-1} \to B^n \to 0$$

is exact, and there is a natural identification

$$H^n(Q, A^N) \cong \mathrm{coker}\big((C^{n-1})^Q \to (B^n)^Q\big).$$

Proof. — The cohomology of $C^\bullet$ is $H^\bullet(N, A)$. Hence the vanishing hypothesis gives the displayed exact sequence. By the preceding lemma each C^j is injective in $\mathscr{C}_Q$. Repeatedly applying the long exact cohomology sequence to

$$0 \to B^r \to C^r \to B^{r+1} \to 0 \qquad (0 \le r \le n-1),$$

where $B^0 = A^N$, gives the usual dimension-shifting identifications. The last step is the exact sequence

$$(C^{n-1})^Q \to (B^n)^Q \to H^1(Q, B^{n-1}) \to 0,$$

after shifting $n-1$ times. Thus

$$H^n(Q, A^N) \cong H^1(Q, B^{n-1}) \cong \mathrm{coker}\big((C^{n-1})^Q \to (B^n)^Q\big).$$

□

Proof of Proposition 8.2. — Let $Q = G/N$ and choose an injective resolution $A \to I^\bullet$ in $\mathscr{C}_G$. Put $C^\bullet = (I^\bullet)^N$ and

$$Z^n = \ker(C^n \to C^{n+1}), \qquad B^n = \operatorname{im}(C^{n-1} \to C^n).$$

By Lemma 8.4,

$$H^n(Q, A^N) \cong (B^n)^Q / d\big((C^{n-1})^Q\big).$$

Since $(C^\bullet)^Q = (I^\bullet)^G$, we also have

$$H^n(G, A) = H^n((I^\bullet)^G) = (Z^n)^Q / d\big((C^{n-1})^Q\big).$$

The inflation map is induced by the inclusion $(B^n)^Q \subset (Z^n)^Q$. The restriction map is induced by

$$(Z^n)^Q \longrightarrow (Z^n/B^n)^Q = H^n(N, A)^Q.$$

Because taking Q-invariants is left exact, the kernel of this last map is exactly $(B^n)^Q$. Therefore the kernel of restriction on $H^n(G, A)$ is precisely the image of inflation from $H^n(Q, A^N)$. This proves the exact sequence

$$0 \to H^n(G/N, A^N) \xrightarrow{\text{Inf}} H^n(G, A) \xrightarrow{\text{Res}} H^n(N, A)^{G/N}.$$

□

Remark 8.5. — Equivalently, this is the edge exact sequence of the Hochschild–Serre spectral sequence

$$E_2^{p,q} = H^p(G/N, H^q(N, A)) \Rightarrow H^{p+q}(G, A).$$

Under the hypothesis $H^q(N, A) = 0$ for $1 \leq q \leq n-1$, the intermediate rows in total degree n vanish. Thus $E_2^{n,0} = H^n(G/N, A^N)$ survives to E_∞, and the two edge maps are inflation and restriction [**7**, Chap. I, §2.6]; [**4**, Theorem 1.44, pp. 28–29], see also in the appendix section.

Corollary 8.6. — Suppose that $H^n(N, A) = 0$ for all $n \geq 1$. Then there exists a natural isomorphism

$$H^n(G/N, A^N) \cong H^n(G, A) \quad \text{for all } n \geq 0.$$

Proof. — Consider an injective resolution $A \to I^\bullet$ in $\mathscr{C}_G$. Restricting to N, the complex $I^\bullet$ remains an injective resolution in $\mathscr{C}_N$ (since restriction preserves injectives). By hypothesis,

$$H^n((I^\bullet)^N) \cong H^n(N, A) = 0,$$

so the complex of N-invariants $(I^\bullet)^N$ is exact. Therefore, it is an injective resolution of A^N in $\mathscr{C}_{G/N}$. Hence,

$$H^n(G/N, A^N) = H^n(((I^\bullet)^N)^{G/N}) = H^n((I^\bullet)^G) \cong H^n(G, A),$$

where the last equality follows from the definition of group cohomology via injective resolutions. This completes the proof. □

Let K be a local field and let $I_K := \mathrm{Gal}(K^{\mathrm{sep}}/K^{\mathrm{ur}})$ be the inertia subgroup. Then:

$$H^i(I_K, (K^{\mathrm{sep}})^\times) = 0 \quad \text{for any } i \geq 2.$$

In fact, as will be proved later, for any torsion module M, we have $H^i(I_K, M) = 0$ for all $i \geq 2$. Since we already know $H^1(I_K, (K^{\mathrm{sep}})^\times) = 0$ by Hilbert 90:

Corollary 8.7. —

$$H^2(\mathrm{Gal}(K^{\mathrm{ur}}/K), (K^{\mathrm{ur}})^\times) \cong H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times).$$

Proof. — Apply Proposition 8.2 to

$$1 \to I_K \to \mathrm{Gal}(K^{\mathrm{sep}}/K) \to \mathrm{Gal}(K^{\mathrm{ur}}/K) \to 1$$

with coefficients $(K^{\mathrm{sep}})^\times$. We have $H^1(I_K, (K^{\mathrm{sep}})^\times) = 0$ by Hilbert 90 and $H^2(I_K, (K^{\mathrm{sep}})^\times) = 0$ by the vanishing stated above. Since $((K^{\mathrm{sep}})^\times)^{I_K} = (K^{\mathrm{ur}})^\times$, the exact sequence for $n = 2$ gives the desired isomorphism. □

Corollary 8.8. — Let L/K be a finite Galois extension. Then we have an exact sequence

$$0 \to H^2(\mathrm{Gal}(L/K), L^\times) \to H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times) \xrightarrow{\mathrm{Res}} H^2(\mathrm{Gal}(K^{\mathrm{sep}}/L), (K^{\mathrm{sep}})^\times).$$

Proof. — Consider the tower of fields $K \subset L \subset K^{\mathrm{sep}}$. Let $G = \mathrm{Gal}(K^{\mathrm{sep}}/K)$ and $N = \mathrm{Gal}(K^{\mathrm{sep}}/L)$, so that $G/N \cong \mathrm{Gal}(L/K)$. We apply the inflation-restriction exact sequence (Proposition 8.2) with $n = 2$. To do this, we need to check the condition $H^1(N, (K^{\mathrm{sep}})^\times) = 0$. Here

$$N = \mathrm{Gal}(K^{\mathrm{sep}}/L) = G_L, \qquad K^{\mathrm{sep}} = L_{\mathrm{sep}},$$

after choosing the same separable closure. Thus

$$H^1(N, (K^{\mathrm{sep}})^\times) = H^1(\mathrm{Gal}(L_{\mathrm{sep}}/L), L_{\mathrm{sep}}^\times).$$

Hilbert's Theorem 90, in its infinite Galois form proved in Theorem 6.2, says that this group is zero. Therefore, we obtain the exact sequence

$$0 \to H^2(\mathrm{Gal}(L/K), ((K^{\mathrm{sep}})^\times)^N) \to H^2(G, (K^{\mathrm{sep}})^\times) \to H^2(N, (K^{\mathrm{sep}})^\times).$$

Since $((K^{\mathrm{sep}})^\times)^N = L^\times$, we obtain exactly the desired sequence. □

Proposition 8.9. — *The restriction map* Res : $H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times) \to H^2(\mathrm{Gal}(K^{\mathrm{sep}}/L), (K^{\mathrm{sep}})^\times)$ *is compatible with the invariant map in the sense that the following diagram commutes:*

$$\begin{tikzcd}
H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times) \arrow[r, "\mathrm{Res}"] \arrow[d, "\mathrm{inv}_K"'] & H^2(\mathrm{Gal}(K^{\mathrm{sep}}/L), (K^{\mathrm{sep}})^\times) \arrow[d, "\mathrm{inv}_L"] \\
\mathbf{Q}/\mathbf{Z} \arrow[r, "{[L:K]}"] & \mathbf{Q}/\mathbf{Z}
\end{tikzcd}$$

where the bottom map is multiplication by the degree $[L : K]$.

Proof. — We first recall what the invariant map means in this notation. Corollary 8.8 gives an isomorphism

$$H^2(\mathrm{Gal}(K^{\mathrm{ur}}/K), (K^{\mathrm{ur}})^\times) \xrightarrow{\sim} H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times),$$

because the inertia subgroup has no contribution in degrees 1 and 2 for the coefficient module $(K^{\mathrm{sep}})^\times$. Thus, to compute $\mathrm{inv}_K(\alpha)$ for $\alpha \in H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times)$, we first pull α back through this isomorphism to the unramified group, and then use the valuation sequence

$$1 \to \mathscr{O}_{K^{\mathrm{ur}}}^\times \to (K^{\mathrm{ur}})^\times \xrightarrow{v_K} \mathbf{Z} \to 0.$$

Concretely, this gives

$$H^2(\mathrm{Gal}(K^{\mathrm{ur}}/K), (K^{\mathrm{ur}})^\times) \xrightarrow{v_K} H^2(\mathrm{Gal}(K^{\mathrm{ur}}/K), \mathbf{Z}) \xrightarrow{\delta^{-1}} H^1(\mathrm{Gal}(K^{\mathrm{ur}}/K), \mathbf{Q}/\mathbf{Z}) \xrightarrow{\chi \mapsto \chi(\mathrm{Frob}_K)} \mathbf{Q}/\mathbf{Z}.$$

Now let $e = e(L/K)$ and $f = f(L/K)$. The restriction map is compatible with each arrow in this construction. On the valuation part, the normalized valuations satisfy

$$v_L(x) = e\, v_K(x) \qquad (x \in (K^{\mathrm{ur}})^\times \subset (L^{\mathrm{ur}})^\times),$$

so the valuation arrow contributes the factor e. On the last arrow, the Frobenius of L maps to the f-th power of the Frobenius of K, so evaluating a restricted character contributes the factor f. The connecting isomorphism δ^{-1} is functorial, so it contributes no extra factor. Therefore the full composite invariant satisfies

$$\mathrm{inv}_L(\mathrm{Res}_{L/K}\, \alpha) = ef\, \mathrm{inv}_K(\alpha) = [L : K]\, \mathrm{inv}_K(\alpha).$$

This proves the commutativity of the displayed diagram. □

Corollary 8.10. — Let L/K be a finite Galois extension of degree n. Then $H^2(\mathrm{Gal}(L/K), L^\times)$ is a cyclic group of order n, generated by the unique element $u_{L/K} \in H^2(\mathrm{Gal}(L/K), L^\times)$ such that

$$\mathrm{inv}_K(\mathrm{Inf}(u_{L/K})) = \frac{1}{n} \in \mathbf{Q}/\mathbf{Z}.$$

This element is called the fundamental class of L/K.

Proof. — By the preceding inflation–restriction corollary, inflation identifies $H^2(\mathrm{Gal}(L/K), L^\times)$ with the kernel of

$$\mathrm{Res} : H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times) \longrightarrow H^2(\mathrm{Gal}(K^{\mathrm{sep}}/L), (K^{\mathrm{sep}})^\times).$$

Proposition 8.9 identifies this restriction map, after applying the invariant isomorphisms, with multiplication by n on $\mathbf{Q}/\mathbf{Z}$. Hence

$$H^2(\mathrm{Gal}(L/K), L^\times) \cong \ker\big(\mathbf{Q}/\mathbf{Z} \xrightarrow{\times n} \mathbf{Q}/\mathbf{Z}\big) = \frac{1}{n}\mathbf{Z}/\mathbf{Z}.$$

This group is cyclic of order n. The element corresponding to $1/n$ is the unique class $u_{L/K}$ whose inflation has invariant $1/n$. □

Remark 8.11. — Serre's proof gives a useful way to read Proposition 8.9 and the preceding corollary without first isolating the cohomology of the inertia group I_K. The important bridge is the passage from finite Galois extensions to the absolute group. Since

$$K^{\mathrm{sep}} = \bigcup_{L/K \text{ finite Galois}} L,$$

one has

$$H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times) = \varinjlim_{L/K \text{ finite Galois}} H^2(\mathrm{Gal}(L/K), L^\times),$$

where the transition maps are inflation. Equivalently,

$$H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times) = \bigcup_{L/K \text{ finite Galois}} \mathrm{Inf}_{L/K} H^2(\mathrm{Gal}(L/K), L^\times).$$

Now fix such an L/K and put $L^{\mathrm{ur}} = LK^{\mathrm{ur}}$. Serre considers the unramified kernel

$$H^2(L/K)_{\mathrm{nr}} := \ker\Big(H^2(\mathrm{Gal}(K^{\mathrm{ur}}/K), (K^{\mathrm{ur}})^\times) \xrightarrow{\mathrm{Res}} H^2(\mathrm{Gal}(L^{\mathrm{ur}}/L), (L^{\mathrm{ur}})^\times)\Big).$$

Corollary 1.9 in Serre says that $H^2(L/K)_{\mathrm{nr}}$ is cyclic of order $n = [L : K]$ and is generated by an element $u^{\mathrm{ur}}_{L/K}$ satisfying

$$\mathrm{inv}_K(u^{\mathrm{ur}}_{L/K}) = \frac{1}{n}.$$

Proposition 1.15 in Serre then identifies the finite relative group with this unramified kernel after mapping both into the absolute group:

$$\mathrm{Inf}_{L/K} H^2(\mathrm{Gal}(L/K), L^\times) = \mathrm{Inf}_{K^{\mathrm{ur}}/K} H^2(L/K)_{\mathrm{nr}} \subset H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times).$$

Thus, explicitly,

$$H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times) = \bigcup_{L/K \text{ finite Galois}} \mathrm{Inf}_{K^{\mathrm{ur}}/K} H^2(L/K)_{\mathrm{nr}}.$$

In this approach, Proposition 8.9 is proved first on the unramified side by comparing Frobenius and valuations, and the fundamental-class corollary follows immediately from the restriction formula

$$\mathrm{inv}_L(\mathrm{Res}_{L/K}\,\alpha) = [L : K]\,\mathrm{inv}_K(\alpha).$$

This is the content of Serre's Proposition 1.8 and Proposition 1.15 in the chapter "Local class field theory" [**2**, Serre, Local class field theory, §1.3 and §1.6, pp. 7–11].

Lecture 9

This lecture begins the algebraic side of the Brauer group. We introduce simple algebras, central simple algebras, and the structural tools needed to compare Brauer classes with Galois cohomology.

9.1 Simple and Central Algebras. —

Definition 9.1. — Let K be a field. A K**-algebra** is a finite-dimensional algebra A (possibly non-commutative) with unit together with an algebra homomorphism $K \to Z(A) \subset A$, where $Z(A)$ is the center of A.

We say a K-algebra is **simple** if it has no proper non-zero two-sided ideals.

We say a K-algebra is **central** over K if $Z(A) = K$.

Example 9.2. — (1) If L/K is a finite field extension, then L is a simple K-algebra. It is central iff $L = K$.

(2) A **division algebra** (or **skew field**) over K is a K-algebra where every non-zero element has a two-sided inverse. Clearly, a division algebra has no non-zero proper ideals, hence is simple.

(3) Example of a skew field: the Hamilton quaternions $\mathbb{H}$ over $\mathbf{R}$.

(4) For any K-algebra A, the matrix algebra $M_n(A)$ is a K-algebra. If D is a division algebra over K, then $M_n(D)$ is a simple K-algebra.

Proof. — Let $I \subseteq M_n(D)$ be a non-zero two-sided ideal. We will show that $I = M_n(D)$: let $E_{ij} \in M_n(D)$ denote the standard matrix units, i.e., the matrix with 1_D in the (i, j)-position and 0 elsewhere. These satisfy:

1. $E_{ij}E_{kl} = \delta_{jk}E_{il}$, where δ_{jk} is the Kronecker delta.
2. For any $X = (x_{ij}) \in M_n(D)$, we have $E_{pq}XE_{rs} = x_{qr}E_{ps}$.

Take a non-zero matrix $X = (x_{ij}) \in I$. Since $X \neq 0$, there exist indices p, q such that $x_{pq} \neq 0$ in D. Using the matrix units and the fact that I is a two-sided ideal, we have:

$$E_{pp}XE_{qq} = x_{pq}E_{pq} \in I.$$

Since D is a division algebra and $x_{pq} \neq 0$, x_{pq} has a multiplicative inverse $x_{pq}^{-1} \in D$. Multiply $x_{pq}E_{pq}$ on the left by $x_{pq}^{-1}E_{pp}$:

$$(x_{pq}^{-1}E_{pp})(x_{pq}E_{pq}) = x_{pq}^{-1}x_{pq}E_{pq} = 1_D E_{pq} = E_{pq} \in I.$$

Now that $E_{pq} \in I$, we can obtain all matrix units:

$$E_{ij} = E_{ip}E_{pq}E_{qj} \in I \quad \text{for all } i, j = 1, \ldots, n.$$

Any matrix $Y = (y_{ij}) \in M_n(D)$ can be written as a linear combination:

$$Y = \sum_{i,j=1}^{n} y_{ij}E_{ij}.$$

Since each $E_{ij} \in I$ and I is an ideal (hence closed under left multiplication by elements of D and under addition), we have $Y \in I$. Therefore $I = M_n(D)$.

Since I was an arbitrary non-zero two-sided ideal, $M_n(D)$ has no proper non-zero two-sided ideals, i.e., it is simple. □

9.2 Modules over Simple Algebras. — We study modules over simple K-algebras.

Proposition 9.3. — *Let A be a finite-dimensional simple K-algebra. Let M be a minimal left ideal of A (i.e., M contains no proper non-zero A-submodule). Such an M exists because A is finite-dimensional over K.*

Then ${}_AA \cong M^{\oplus n}$ for some $n \geq 1$.

Proof. — Consider the two-sided ideal $\sum_{a \in A} M \cdot a$. Since A is simple, this sum equals A. Because A is finite-dimensional, there exist $a_1, \dots, a_n \in A$ such that $Ma_1 + \cdots + Ma_n = A$. Choose $m_i \in M$ such that $1 = \sum_{i=1}^n m_i a_i$, and assume n is minimal. Minimality implies $Ma_i \neq 0$ for all i.

One checks that the map $M \to Ma_i$ given by $m \mapsto ma_i$ is an isomorphism of left A-modules. Moreover, the sum $\bigoplus_{i=1}^n Ma_i$ is direct; otherwise, a linear dependence would contradict the minimality of n. Hence ${}_AA \cong M^{\oplus n}$. □

Corollary 9.4. — Every left A-module is semisimple (a direct sum of simple modules). In particular, if N_1 and N_2 are finite-dimensional A-modules, then $N_1 \cong N_2$ iff $\dim_K N_1 = \dim_K N_2$.

***Theorem 9.5* (Schur's Lemma).** — *Let A be a finite-dimensional K-algebra and M an irreducible (simple) A-module. Then M is finite-dimensional over K and $\mathrm{End}_A(M)$ is a division algebra over K.*

***Theorem 9.6* (Wedderburn).** — *Every simple K-algebra A is isomorphic to $M_n(D)$ for a unique $n \geq 1$ and a unique (up to isomorphism) division algebra D over K.*

Sketch. — Let M be a minimal left ideal of A. By the proposition, ${}_AA \cong M^{\oplus n}$.

Let A^{op} be the opposite K-algebra: it is the same K-vector space as A, with multiplication $a * b := ba$. Then $\mathrm{End}_A({}_AA) \cong A^{\mathrm{op}}$ via right multiplication, and $(A^{\mathrm{op}})^{\mathrm{op}} \cong A$. Thus

$$A^{\mathrm{op}} \cong \mathrm{End}_A({}_AA) \cong \mathrm{End}_A(M^{\oplus n}) \cong M_n(\mathrm{End}_A(M)).$$

By Schur's lemma, $\mathrm{End}_A(M)$ is a division algebra. Set $D := \mathrm{End}_A(M)^{\mathrm{op}}$, which is also a division algebra. Then $A \cong (M_n(D^{\mathrm{op}}))^{\mathrm{op}} \cong M_n(D)$. Uniqueness follows from the uniqueness of the division algebra in the Wedderburn decomposition. □

Definition 9.7. — Let A and B be K-algebras. The **tensor product** $A \otimes_K B$ is the tensor product of A and B as K-vector spaces, with multiplication defined on simple tensors by

$$(a \otimes b)(a' \otimes b') = aa' \otimes bb', \quad \forall a, a' \in A,\ b, b' \in B,$$

and extended bilinearly. This makes $A \otimes_K B$ into a K-algebra.

Remark 9.8. — There are natural embeddings $A \hookrightarrow A \otimes_K B$ and $B \hookrightarrow A \otimes_K B$ given by $a \mapsto a \otimes 1$ and $b \mapsto 1 \otimes b$, which are K-algebra homomorphisms. If $\{e_i\}$ is a K-basis of A and $\{f_j\}$ is a K-basis of B, then $\{e_i \otimes f_j\}$ is a K-basis of $A \otimes_K B$, so $\dim_K(A \otimes_K B) = (\dim_K A)(\dim_K B)$.

The next lemma shows that the center of a tensor product is the tensor product of the centers.

Lemma 9.9. —

$$Z(A \otimes_K B) = Z(A) \otimes_K Z(B).$$

Proof. — ($\supseteq$): If $a \in Z(A)$ and $b \in Z(B)$, then for any $a' \in A$, $b' \in B$,

$$(a \otimes b)(a' \otimes b') = aa' \otimes bb' = a'a \otimes b'b = (a' \otimes b')(a \otimes b).$$

Thus $a \otimes b$ commutes with all simple tensors, hence with all elements of $A \otimes B$. By linearity, $Z(A) \otimes Z(B) \subseteq Z(A \otimes B)$.

($\subseteq$): Let $z = \sum_{i=1}^{n} a_i \otimes b_i \in Z(A \otimes B)$ with n minimal. Minimality implies $\{a_i\}$ and $\{b_i\}$ are linearly independent over K.

For any $a \in A$, since z commutes with $a \otimes 1$:

$$(a \otimes 1)z = \sum_i (aa_i) \otimes b_i = z(a \otimes 1) = \sum_i (a_i a) \otimes b_i.$$

Thus $\sum_i (aa_i - a_i a) \otimes b_i = 0$. Since $\{b_i\}$ is linearly independent, we have $aa_i - a_i a = 0$ for all i, so $a_i \in Z(A)$.

Similarly, for any $b \in B$, commuting with $1 \otimes b$ gives $\sum_i a_i \otimes (bb_i - b_i b) = 0$. Since $\{a_i\}$ is linearly independent, we get $bb_i - b_i b = 0$ for all i, so $b_i \in Z(B)$.

Hence $z \in Z(A) \otimes Z(B)$. □

Theorem 9.10. — *Let A be a central simple K-algebra and B any K-algebra. Then any two-sided ideal $\mathfrak{a}$ of $A \otimes_K B$ is of the form $A \otimes_K \mathfrak{b}$ for some two-sided ideal $\mathfrak{b}$ of B.*

Proof. — **Claim 1:** $\mathfrak{a} \cap B \neq 0$ (where we identify B with $1 \otimes B \subset A \otimes B$).

Take a non-zero $x = \sum_{i=1}^{n} a_i \otimes b_i \in \mathfrak{a}$ with n minimal. Minimality implies $\{a_1, \ldots, a_n\}$ are linearly independent over K and $\{b_1, \ldots, b_n\}$ are linearly independent over K. In particular, $a_1 \neq 0$.

Since A is simple, $Aa_1A = A$. Thus there exist $c_j, d_j \in A$ such that $1 = \sum_{j=1}^{l} c_j a_1 d_j$. Then

$$\tilde{x} := \sum_{j=1}^{l} (c_j \otimes 1)x(d_j \otimes 1) = 1 \otimes b_1 + \sum_{i=2}^{n} \left(\sum_j c_j a_i d_j \right) \otimes b_i \in \mathfrak{a}.$$

If $n > 1$, note that $a_2 \notin Z(A) = K$ (otherwise $\{a_i\}$ would be linearly dependent). Hence there exists $a \in A$ such that $aa_2 - a_2 a \neq 0$. Consider

$$y := (a \otimes 1)\tilde{x} - \tilde{x}(a \otimes 1) = \sum_{i=2}^{n} (aa_i - a_i a) \otimes b_i \in \mathfrak{a}.$$

This y is non-zero (since the coefficient of b_2 is non-zero) and has at most $n-1$ terms (the b_1 term cancels because $a \cdot 1 - 1 \cdot a = 0$), contradicting the minimality of n. Therefore $n = 1$, so $\tilde{x} = 1 \otimes b_1 \in \mathfrak{a}$, and thus $b_1 \in \mathfrak{a} \cap B$, proving the claim.

Let $\mathfrak{b} := \mathfrak{a} \cap B$, which is a two-sided ideal of B. Clearly $A \otimes \mathfrak{b} \subseteq \mathfrak{a}$.

Claim 2: $\mathfrak{a} = A \otimes \mathfrak{b}$.

Consider the quotient map $\varphi : A \otimes B \to A \otimes (B/\mathfrak{b})$. Its kernel is $A \otimes \mathfrak{b}$. If $\mathfrak{a} \neq A \otimes \mathfrak{b}$, then $\varphi(\mathfrak{a})$ is a non-zero two-sided ideal of $A \otimes (B/\mathfrak{b})$. By Claim 1 applied to this quotient, $\varphi(\mathfrak{a}) \cap (B/\mathfrak{b}) \neq 0$. But this means there exists $b \in B \setminus \mathfrak{b}$ such that $1 \otimes b \in \mathfrak{a} + (A \otimes \mathfrak{b})$, i.e., $1 \otimes b \in \mathfrak{a}$ modulo $A \otimes \mathfrak{b}$. This implies $b \in \mathfrak{a} \cap B = \mathfrak{b}$, a contradiction. Hence $\mathfrak{a} = A \otimes \mathfrak{b}$. □

Corollary 9.11. — Let A, B be central simple algebras over K. Then $A \otimes_K B$ is central simple over K.
Proof. — Simplicity follows from Theorem 9.10. For centrality, we use

$$Z(A \otimes_K B) = Z(A) \otimes_K Z(B).$$

Since A and B are central, $Z(A) = Z(B) = K$, hence $Z(A \otimes_K B) = K$. □

Corollary 9.12. — Let L/K be a finite field extension and A a central simple K-algebra. Then

$$A_L := A \otimes_K L$$

is a central simple algebra over L.
Proof. — Simplicity follows from Theorem 9.10, applied with $B = L$, because L has only the two ideals 0 and L.

Moreover,

$$Z(A \otimes_K L) = Z(A) \otimes_K L = K \otimes_K L = L,$$

so it is central over L. □

Lemma 9.13. — *If K is algebraically closed and D is a finite-dimensional division algebra over K, then $D = K$.*
Proof. — Let $a \in D \setminus K$. Then $K(a) \subset D$ is a commutative subfield, hence a finite extension of K. Since K is algebraically closed, we must have $K(a) = K$, a contradiction. Thus $D = K$. □

Corollary 9.14. — Let A be a finite-dimensional simple algebra over an algebraically closed field K. Then

$$A \cong M_n(K).$$

Proof. — By Wedderburn's theorem, $A \cong M_n(D)$ for some division algebra D over K. By the previous lemma, $D = K$, hence $A \cong M_n(K)$. □

Corollary 9.15. — Let A be a central simple K-algebra. Then $\dim_K A$ is a perfect square.
Proof. — Let $\overline{K}$ be an algebraic closure of K. Then

$$A \otimes_K \overline{K} \cong M_n(\overline{K})$$

for some n. Hence

$$\dim_K A = \dim_{\overline{K}}(A \otimes_K \overline{K}) = n^2.$$

□

Theorem 9.16. — *Let A be a central simple K-algebra with $\dim_K A = n^2$. Then the map*

$$A \otimes_K A^{\mathrm{op}} \longrightarrow \mathrm{End}_K(A), \qquad \sum_i a_i \otimes b_i \mapsto \left(x \mapsto \sum_i a_i x b_i \right)$$

is an isomorphism of K-algebras. In particular,

$$A \otimes_K A^{\mathrm{op}} \cong \mathrm{End}_K(A) \cong M_{n^2}(K).$$

Proof. — Let F denote the map. First, F is a K-algebra homomorphism since left and right multiplications commute. Since A is central simple, $A \otimes_K A^{\mathrm{op}}$ is simple (Theorem 9.10). Thus $\ker F$ is either 0 or the whole algebra. Clearly $F \neq 0$, so $\ker F = 0$. Finally,

$$\dim_K (A \otimes_K A^{\mathrm{op}}) = (n^2)^2 = \dim_K \mathrm{End}_K(A),$$

so F is an isomorphism. □

9.3 Tensor Products and Brauer Classes. —

Definition 9.17. — The *Brauer group* of a field K is defined as

$$\mathrm{Br}(K) := \{\text{finite-dimensional central simple } K\text{-algebras}\}/ \sim,$$

where $A \sim B$ if there exist integers $m, n \geq 1$ such that

$$A \otimes_K M_m(K) \cong B \otimes_K M_n(K).$$

Equivalently, $A \sim B$ if and only if in their Wedderburn decompositions

$$A \cong M_r(D), \quad B \cong M_s(D'),$$

the division algebras D and D' are isomorphic as K-algebras.

We denote by $[A]$ the equivalence class of A.

The group operation on $\mathrm{Br}(K)$ is defined by

$$[A] \cdot [B] := [A \otimes_K B].$$

Proposition 9.18. — *With the above operation,* $\mathrm{Br}(K)$ *is an abelian group.*

Proof. — First we check that the operation is well-defined. Let $A \sim A'$ and $B \sim B'$. Then there exist integers $r, s, t, u \geq 1$ such that

$$A \otimes_K M_r(K) \cong A' \otimes_K M_s(K), \qquad B \otimes_K M_t(K) \cong B' \otimes_K M_u(K).$$

Tensoring these two isomorphisms and using

$$M_r(K) \otimes_K M_t(K) \cong M_{rt}(K), \qquad M_s(K) \otimes_K M_u(K) \cong M_{su}(K),$$

we obtain

$$(A \otimes_K B) \otimes_K M_{rt}(K) \cong (A' \otimes_K B') \otimes_K M_{su}(K).$$

Hence $A \otimes_K B \sim A' \otimes_K B'$, so $[A] \cdot [B]$ depends only on the classes $[A]$ and $[B]$. Also, if A and B are central simple over K, then $A \otimes_K B$ is central simple over K by the previous corollary, so the operation is closed on $\mathrm{Br}(K)$.

1. The identity element is $[K]$, since
$$A \otimes_K K \cong A.$$
2. The inverse of $[A]$ is $[A^{\mathrm{op}}]$, because
$$A \otimes_K A^{\mathrm{op}} \cong \mathrm{End}_K(A) \cong M_{n^2}(K),$$
so $[A][A^{\mathrm{op}}] = [K]$.
3. Associativity follows from the canonical K-algebra isomorphism
$$(A \otimes_K B) \otimes_K C \cong A \otimes_K (B \otimes_K C).$$
4. Commutativity follows from the flip isomorphism
$$A \otimes_K B \longrightarrow B \otimes_K A, \qquad a \otimes b \longmapsto b \otimes a.$$
This is a K-algebra isomorphism, since
$$(a \otimes b)(a' \otimes b') = aa' \otimes bb'$$
is sent to $bb' \otimes aa'$, which is exactly
$$(b \otimes a)(b' \otimes a')$$
in $B \otimes_K A$.

Therefore the tensor product operation makes $\mathrm{Br}(K)$ into an abelian group. □

Remark 9.19. — Every central simple K-algebra A admits a decomposition $A \cong M_n(D)$ for a unique (up to isomorphism) central division K-algebra D. Thus, each class $[A] \in \mathrm{Br}(K)$ is uniquely determined by the isomorphism class of D. In this sense, the Brauer group classifies central division algebras over K up to isomorphism. Equivalently,
$$\mathrm{Br}(K) \;\cong\; \{\text{central division } K\text{-algebras}\}/\cong .$$
The group law corresponds to tensor product followed by passing to the associated division algebra (i.e. removing the matrix factor via Wedderburn's theorem 9.6).

Remark 9.20. — If K is algebraically closed, then $\mathrm{Br}(K) = 0$. Examples involving $\mathbf{R}$, $\mathbf{Q}_p$, and $\mathbf{Q}$ (with the local-global principle) can be found in Liang Xiao's note.

Lecture 10

This lecture continues the algebra of central simple algebras. The main tools are Skolem–Noether and double centralizer result. We will construct a separable extension of K for every given central division algebra D/K. This leads us to the connection with Brauer group and H^2.

10.1 Skolem–Noether and Double Centralizers.

— Recall what we proved before:

1. For any K-algebra R, there is a natural isomorphism $R \otimes_K M_n(K) \cong M_n(R)$: define $R \otimes_K M_n(K) \to M_n(R), \quad r \otimes (a_{ij}) \mapsto (ra_{ij})$; this is an isomorphism of K-algebras with inverse $(r_{ij}) \mapsto \sum_{i,j} r_{ij} \otimes e_{ij}$.
2. As a corollary, $M_m(K) \otimes_K M_n(K) \cong M_{mn}(K)$.
3. For central simple K-algebras A_1, A_2, $A_1 \sim A_2$ is equivalent to the existence of $m_1, m_2 \geq 1$ such that $A_1 \otimes_K M_{m_1}(K) \cong A_2 \otimes_K M_{m_2}(K)$, thus it is an equivalent relation.
4. The previous proposition proves that tensor product gives a well-defined abelian group structure on $\mathrm{Br}(K)$.

Our goal today is to relate the Brauer group to Galois cohomology:

$$\mathrm{Br}(K) \cong H^2\big(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times\big).$$

This works for any field.

Theorem 10.1 **(Skolem-Noether).** — *Let B be a finite-dimensional central simple K-algebra and A a simple K-algebra. If*

$$f, g : A \to B$$

are K-algebra homomorphisms, then there exists $b \in B^\times$ such that

$$g(a) = bf(a)b^{-1} \quad \textit{for all } a \in A.$$

Proof. — Consider $C = A \otimes_K B^{\mathrm{op}}$. Since B is central, B^{op} is also central, Hence C is simple.

Associated with every homomorphism $f : A \to B$, we define a C-module structure on B by

$$(a \otimes b)_f \cdot x = f(a)\, x\, b.$$

We denote this module by B_f. Similarly we have B_g. For the two homomorphisms $f, g : A \to B$, we have

$$\dim_K B_f = \dim_K B_g,$$

Since C is simple, we have $B_f \simeq B_g$ as C-modules. Let $\varphi : B_f \to B_g$ be a C-module isomorphism and set $b = \varphi(1)$, then for any $x \in B$,

$$\varphi(x) = \varphi\big((1 \otimes x)_f \cdot 1\big) = (1 \otimes x)_g \cdot \varphi(1) = bx.$$

Applying the same argument to $\psi = \varphi^{-1} : B_g \to B_f$, we obtain

$$\psi(x) = b'x, \quad \text{where } b' = \psi(1).$$

Then

$$x = (\varphi \circ \psi)(x) = bb'x,$$

so $bb' = 1$. Similarly, $b'b = 1$, hence $b \in B^\times$.

Finally, for any $a \in A$, we have

$$bf(a) = \varphi(f(a)) = \varphi\big((a \otimes 1)_f \cdot 1\big) = (a \otimes 1)_g \cdot \varphi(1) = g(a)b.$$

□

***Theorem 10.2* (Double Centralizer).** — *Let A be a central simple K-algebra with $\dim_K A = n$, and let $B \subseteq A$ be a simple K-subalgebra with $\dim_K B = m$.*

Define the centralizer

$$Z_A(B) := \{x \in A \mid xb = bx \text{ for all } b \in B\}.$$

Then:

1. There is an isomorphism of K-algebras

$$Z_A(B) \otimes_K M_m(K) \cong A \otimes_K B^{\mathrm{op}}.$$

2. $Z_A(B)$ is a simple K-algebra and

$$\dim_K Z_A(B) = \frac{n}{m}.$$

3. $Z_A(Z_A(B)) = B$.

Proof. — The proof relies on two observations.

(1) For K-algebras A, B and subalgebras $A' \subset A$, $B' \subset B$, one has

$$Z_{A\otimes_K B}(A' \otimes_K B') = Z_A(A') \otimes_K Z_B(B').$$

(2) In $\mathrm{End}_K(A)$, the left and right regular representations satisfy

$$Z_{\mathrm{End}_K(A)}(L(A)) = R(A^{\mathrm{op}}).$$

Now we prove the theorem: consider the embeddings

$$f, g : B \to A \otimes_K \mathrm{End}_K(B) \cong A \otimes_K M_m(K)$$

defined by

$$f(b) = b \otimes \mathrm{id}_B, \qquad g(b) = 1 \otimes \lambda_b,$$

where $\lambda_b(x) = bx$. Since $Z(A \otimes_K M_m(K)) = K$, the algebra $A \otimes_K M_m(K)$ is central. Hence by Skolem–Noether 10.1, f and g are conjugate. Thus there exists $x \in (A \otimes_K \mathrm{End}_K(B))^\times$ such that

$$f(b) = xg(b)x^{-1}.$$

Taking centralizers,

$$Z_{A\otimes_K \mathrm{End}_K(B)}(f(B)) \cong Z_{A\otimes_K \mathrm{End}_K(B)}(g(B)).$$

Now compute both sides:

$$Z_{A\otimes_K \mathrm{End}_K(B)}(B \otimes_K K) = Z_A(B) \otimes_K \mathrm{End}_K(B).$$

$$Z_{A\otimes_K \mathrm{End}_K(B)}(K \otimes_K \lambda(B)) = A \otimes_K Z_{\mathrm{End}_K(B)}(\lambda(B)) = A \otimes_K R(B^{\mathrm{op}}).$$

Thus

$$Z_A(B) \otimes_K \operatorname{End}_K(B) \cong A \otimes_K B^{\mathrm{op}}.$$

Since $\operatorname{End}_K(B) \cong M_m(K)$, this proves (1).

(2) Since $A \otimes_K B^{\mathrm{op}}$ is simple, it follows that

$$Z_A(B) \otimes_K \operatorname{End}_K(B)$$

is simple, hence $Z_A(B)$ is simple. Taking dimensions,

$$\dim_K Z_A(B) \cdot m^2 = (\dim_K A)(\dim_K B) = nm,$$

Thus we have

$$\dim_K Z_A(B) = \frac{n}{m}.$$

(3) Clearly $B \subset Z_A(Z_A(B))$. Using (2) to $Z_A(B)$, we obtain

$$\dim_K Z_A(Z_A(B)) = \frac{n}{\dim_K Z_A(B)} = m.$$

So $B = Z_A(Z_A(B))$ by dimension considerations. □

10.2 Splitting Fields and Maximal Subfields. —

Corollary 10.3. — Let A be a central simple K-algebra with $\dim_K A = d^2$. If L/K is a field extension of degree ℓ together with a K-embedding $L \hookrightarrow A$, then $\ell \mid d$ and $Z_A(L)$ is a central simple algebra over L with

$$\dim_L Z_A(L) = \left(\frac{d}{\ell}\right)^2.$$

In particular, if $\ell = d$, then $Z_A(L) = L$, so L is a maximal subfield of A.

Proof. — We use the chosen embedding to regard L as a commutative subfield of A. Since L is commutative, $L \subseteq Z_A(L)$. From the theorem 10.2,

$$\dim_K Z_A(L) = \frac{d^2}{\ell}.$$

Thus

$$\dim_L Z_A(L) = \frac{d^2/\ell}{\ell} = \left(\frac{d}{\ell}\right)^2.$$

Moreover,

$$L \subseteq Z(Z_A(L)) \subseteq Z_A(Z_A(L)) = L,$$

so $Z_A(L)$ is central over L. Since $Z_A(L)$ is central simple over L, its L-dimension is a square. Hence d/ℓ is an integer, and therefore $\ell \mid d$. □

Remark 10.4. — The hypothesis $L \hookrightarrow A$ is essential. The corollary is not saying that every finite extension of K of degree ℓ can be embedded in A. For example, take $K = \mathbf{Q}$ and $A = M_2(\mathbf{Q})$, so $d = 2$. A quadratic field such as $\mathbf{Q}(i)$ can be embedded in $M_2(\mathbf{Q})$ by letting it act on itself as a two-dimensional $\mathbf{Q}$-vector space, and here $\ell = 2 \mid d$. But a cubic field cannot embed in $M_2(\mathbf{Q})$, since the corollary would force $3 \mid 2$. Similarly over $K = \mathbf{Q}_p$, a quadratic field extension embeds in $M_2(\mathbf{Q}_p)$ by the regular representation, while a cubic extension cannot embed in $M_2(\mathbf{Q}_p)$. Thus the divisibility $\ell \mid d$ is a necessary condition for an embedding $L \subset A$, not a statement about arbitrary finite extensions of K.

Corollary 10.5. — Let D be a central division algebra over K with $\dim_K D = d^2$. If $P \subseteq D$ is a maximal subfield, then $[P : K] = d$.

Proof. — Notice that every maximal subfield $P \subset D$ necessarily contains K, since otherwise the subring generated by P and K would be a subfield of D strictly containing P.

Furthermore, since D is finite-dimensional, maximal subfields exist. Let $P \subset D$ be a maximal subfield. Then $P = Z_D(P)$. Indeed, if $a \in Z_D(P) \setminus P$, then $P[a]$ would be a subfield strictly containing P, a contradiction.

Applying the previous corollary, we obtain $\dim_K P = d$. □

Corollary 10.6. — If $L \subseteq D$ is a maximal subfield, then

$$Z_D(L) = L, \quad \text{and} \quad D \otimes_K L \cong M_d(L).$$

Proof. — The equality $Z_D(L) = L$ is the defining property of a maximal subfield. The previous corollary gives $[L : K] = d$. Regard D as a right L-vector space and define

$$D \otimes_K L \longrightarrow \mathrm{End}_L(D), \qquad a \otimes b \longmapsto (x \mapsto axb).$$

The source is simple by Theorem 9.10, applied with $B = L$, so this nonzero homomorphism is injective. Both sides have L-dimension d^2, hence it is an isomorphism. Since $\mathrm{End}_L(D) \cong M_d(L)$, we get $D \otimes_K L \cong M_d(L)$. □

Remark 10.7. — There is a well-defined homomorphism

$$\mathrm{Br}(K) \to \mathrm{Br}(L), \quad [A] \mapsto [A \otimes_K L].$$

If L is a maximal subfield of a division algebra D, then

$$D \otimes_K L \cong M_d(L),$$

so $[D]$ lies in the kernel.

Proposition 10.8. — *Let D be a central division algebra over a field K. Then D contains a maximal subfield P which is a separable extension of K.*

Proof. — There is nothing to prove if $\operatorname{char} K = 0$ or if K is finite. So we may assume that K is an infinite field of characteristic $p > 0$.

It is enough to show that there exists an element $a \in D \setminus K$ which is separable over K. Indeed, given such an element, we can argue by induction on $\dim_K D = d^2$.

Let $\ell = [K(a) : K] > 1$. Then by Corollary, the centralizer $Z_D(K(a))$ is a central division algebra over $K(a)$ with $\dim_{K(a)} Z_D(K(a)) = \left(\frac{d}{\ell}\right)^2 < d^2$. By the induction hypothesis, $Z_D(K(a))$ contains a maximal subfield P which is separable over $K(a)$. Then P is separable over K, and by Corollary, $[P : K(a)] = \frac{d}{\ell}$, so $[P : K] = d$, thus P is a maximal subfield of D.

Now, we show the existence of such an element a. Let P be any maximal subfield of D. If d is not a power of p, then P/K cannot be purely inseparable, so it contains a separable element, and we are done.

Thus, assume $d = p^\alpha$. Suppose, for contradiction, that every element of $D \setminus K$ is purely inseparable over K. Then for all $a \in D, a^{p^\alpha} \in K$. Choose a basis $e_1 = 1, e_2, \ldots, e_{d^2}$ of D over K, and let $t_1, \ldots, t_{d^2}$ be variables. Then there exist polynomials $f_1, \ldots, f_{d^2} \in K[t_1, \ldots, t_{d^2}]$ such that

$$(t_1 e_1 + \cdots + t_{d^2} e_{d^2})^{p^\alpha} = f_1(t_1, \ldots, t_{d^2}) e_1 + \cdots + f_{d^2}(t_1, \ldots, t_{d^2}) e_{d^2}.$$

Since $a^{p^\alpha} \in K$ for all $a \in D$, we must have

$$f_2(a_1, \ldots, a_{d^2}) = \cdots = f_{d^2}(a_1, \ldots, a_{d^2}) = 0$$

for all $(a_1, \ldots, a_{d^2}) \in K^{d^2}$. Because K is infinite, this implies

$$f_2 = \cdots = f_{d^2} = 0.$$

Therefore, for all $(a_1, \ldots, a_{d^2}) \in \overline{K}^{d^2}$, $a^{p^\alpha} \in \overline{K}$.

But $D \otimes_K \overline{K} \cong M_d(\overline{K})$, and this is false in $M_d(\overline{K})$ (for example, for a matrix unit e_{11}, $e_{11}^{p^\alpha} = e_{11} \notin \overline{K}$). This contradiction proves the existence of a separable element. $\square$

Lecture 11

This lecture starts the comparison between relative Brauer groups and degree-two Galois cohomology. We represent a class in $\mathrm{Br}(L/K)$ by a central simple algebra split by L, choose descent data, and attach to it a factor set in $Z^2(\mathrm{Gal}(L/K), L^\times)$.

We have shown that if $L \subset D$ is a maximal subfield of a central division algebra D over K, then $(\dim_K L)^2 = \dim_K D$.

Definition 11.1. — Let A be a finite-dimensional central simple K-algebra, and let L/K be a field extension. We say that A *splits over* L (or that L is a *splitting field* for A) if

$$A \otimes_K L \cong M_n(L) \text{ as } L\text{-algebras}.$$

Define the *relative Brauer group*

$$\mathrm{Br}(L/K) := \ker\big(\mathrm{Br}(K) \to \mathrm{Br}(L)\big), \quad [A] \mapsto [A \otimes_K L].$$

Proposition 11.2. —
$$\mathrm{Br}(K) = \bigcup_{L/K \text{ finite Galois}} \mathrm{Br}(L/K).$$

Proof. — We prove $\subseteq$. Let $[A] \in \mathrm{Br}(K)$, and write $A \cong M_r(D)$ with D a central division algebra over K. Let $K' \subset D$ be a maximal subfield. By Proposition 10.8, K'/K is finite and separable. By the Double Centralizer Theorem 10.2,
$$Z_D(K') \otimes_K M_m(K) \cong D \otimes_K (K')^{\mathrm{op}}.$$
Since K' is maximal, we have $Z_D(K') = K'$, and $[K' : K]^2 = \dim_K D$. Thus $m = [K' : K]$, and $D \otimes_K K' \cong M_m(K')$. It follows that
$$A \otimes_K K' \cong M_r(D \otimes_K K') \cong M_{rm}(K').$$

Let L/K be a finite Galois extension containing K'. Then
$$A \otimes_K L \cong (A \otimes_K K') \otimes_{K'} L \cong M_{rm}(L),$$
so $[A] \in \mathrm{Br}(L/K)$. □

Remark 11.3. — There is an identification
$$H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K), (K^{\mathrm{sep}})^\times) \cong \varinjlim_{L/K \text{ finite Galois}} H^2(\mathrm{Gal}(L/K), L^\times).$$

To obtain the isomorphism
$$\mathrm{Br}(L/K) \cong H^2(\mathrm{Gal}(L/K), L^\times),$$
one must construct the correspondence and verify compatibility with the transition maps.

Theorem 11.4. — *Let L/K be a finite extension of degree n.*

1. *If A is a central simple K-algebra with $\dim_K A = n^2$ and $L \subset A$, then*
$$A \otimes_K L \cong M_n(L).$$
2. *If A is a finite-dimensional central simple K-algebra that splits over L, then there exists a central simple K-algebra A' such that*
$$A \sim A', \quad \dim_K A' = n^2, \quad L \subset A'.$$

Proof. — (1) Regard A as a right L-vector space. Since $\dim_K A = n^2$ and $[L : K] = n$, we have $\dim_L A = n$. For $a \in A$, left multiplication
$$\lambda_a : A \longrightarrow A, \qquad x \longmapsto ax,$$
is L-linear, hence gives a K-algebra homomorphism
$$f : A \longrightarrow \mathrm{End}_L(A) \cong M_n(L).$$
The given embedding $L \subset A$ also gives a K-algebra homomorphism
$$g : L \longrightarrow \mathrm{End}_L(A)$$

by right multiplication. The images of $f(A)$ and $g(L)$ commute, so by the universal property of tensor products there is an L-algebra homomorphism

$$A \otimes_K L \longrightarrow \mathrm{End}_L(A), \qquad a \otimes b \longmapsto f(a)g(b).$$

It is injective because $A \otimes_K L$ is simple, and both sides have L-dimension n^2. Therefore it is an isomorphism, so $A \otimes_K L \cong M_n(L)$.

(2) Write $A \cong M_r(D)$ with D a central division algebra over K. Since A splits over L, the division algebra D also splits over L: indeed, if $D \otimes_K L \cong M_s(\Delta)$ with Δ a central division algebra over L, then

$$A \otimes_K L \cong M_r(D \otimes_K L) \cong M_{rs}(\Delta),$$

and comparison with a full matrix algebra over L forces $\Delta = L$.

Put $m^2 = \dim_K D$, so $D \otimes_K L \cong M_m(L)$ and also $D^{\mathrm{op}} \otimes_K L \cong M_m(L)$. Let $V = L^m$. Via this splitting, V is a left D^{op}-module. Since L commutes with D^{op} inside $D^{\mathrm{op}} \otimes_K L$, multiplication by elements of L gives a K-embedding

$$L \hookrightarrow \mathrm{End}_{D^{\mathrm{op}}}(V).$$

Set

$$A' = \mathrm{End}_{D^{\mathrm{op}}}(V).$$

Then $A' \cong M_t(D)$ for some t, so A' is central simple over K and is similar to A. Moreover

$$\dim_K V = mn = t \dim_K D = t\, m^2,$$

hence $t = n/m$, and therefore

$$\dim_K A' = t^2 \dim_K D = \left(\frac{n}{m}\right)^2 m^2 = n^2.$$

Thus A' is similar to A, contains an isomorphic copy of L, and has the desired dimension. □

Corollary 11.5. — Let L/K be a finite extension with $[L : K] = n$. Then every class $c \in \mathrm{Br}(L/K)$ can be represented by a central simple K-algebra A such that

$$\dim_K A = n^2, \quad L \subset A.$$

Moreover, A is unique up to isomorphism as a K-algebra.

Proof. — Existence follows from the theorem. For uniqueness, suppose A_1, A_2 are two such algebras. Then $A_1 \sim A_2$, i.e. $A_1 \cong M_r(D), \quad A_2 \cong M_s(D)$ for the same division algebra D.

Since $\dim_K A_1 = \dim_K A_2 = n^2$, we must have $r = s = 1$, hence $A_1 \cong A_2$. □

11.1 Construction of the Map $\beta_{L/K}$. — Let L/K be a finite Galois extension with group $G = \mathrm{Gal}(L/K)$ and $|G| = n$. Let A be a central simple K-algebra with $\dim_K A = n^2$ and suppose we are given an embedding $\iota : L \hookrightarrow A$ (by Theorem 11.4). For each $\sigma \in G$, the maps ι and $\iota \circ \sigma$ are K-algebra embeddings of L into A. By the Skolem–Noether theorem 10.1, there exists $x_\sigma \in A^\times$ such that

$$x_\sigma a x_\sigma^{-1} = \sigma(a), \quad \forall a \in L.$$

Lemma 11.6. —

$$\{x_\sigma \mid \sigma \in G\} \textit{ is an } L\textit{-basis of } A.$$

Proof. — Since $\dim_K A = n^2$ and $[L : K] = n$, we have $\dim_L A = n = |G|$. It suffices to show that $\{x_\sigma\}$ is linearly independent over L. Assume a minimal linear relation

$$a_1 x_{\sigma_1} + \cdots + a_r x_{\sigma_r} = 0, \quad a_i \in L,\ a_i \neq 0.$$

Then $r > 1$. Choose $\alpha \in L$ such that $L = K(\alpha)$. Using $x_\sigma a = \sigma(a) x_\sigma$, we compute:

$$0 = \sigma_r(\alpha) \sum_{i=1}^{r} a_i x_{\sigma_i} - \left(\sum_{i=1}^{r} a_i x_{\sigma_i} \right) \alpha = \sum_{i=1}^{r-1} a_i (\sigma_r(\alpha) - \sigma_i(\alpha)) x_{\sigma_i}.$$

This is a shorter nontrivial relation, a contradiction. Hence the x_σ are linearly independent and form a basis. □

Define

$$a_{\sigma,\tau} := x_\sigma x_\tau x_{\sigma\tau}^{-1}.$$

Lemma 11.7. —

$$a_{\sigma,\tau} \in L^\times.$$

Proof. — For any $a \in L$, we compute

$$(x_\sigma x_\tau) a (x_\sigma x_\tau)^{-1} = \sigma(\tau(a)) = (\sigma\tau)(a) = x_{\sigma\tau} a x_{\sigma\tau}^{-1}.$$

Thus

$$c_{\sigma,\tau} := x_{\sigma\tau}^{-1} x_\sigma x_\tau \in Z_A(L).$$

Since L is a maximal subfield of A, we have $Z_A(L) = L$, hence $c_{\sigma,\tau} \in L^\times$. Therefore

$$a_{\sigma,\tau} = x_\sigma x_\tau x_{\sigma\tau}^{-1} = x_{\sigma\tau} c_{\sigma,\tau} x_{\sigma\tau}^{-1} = (\sigma\tau)(c_{\sigma,\tau}) \in L^\times.$$

□

This factor appears in studying the following structure: by linear independence, we have $A = \oplus_{\sigma \in G} L x_\sigma$. Notice that for any $a_\sigma, a_\tau \in L$ we have

$$(a_\sigma x_\sigma)(a_\tau x_\tau) = (a_\sigma x_\sigma a_\tau x_\sigma^{-1}) x_\sigma x_\tau = (a_\sigma \sigma(a_\tau)) x_\sigma x_\tau.$$

Thus to understand multiplication in A, it is enough to describe the products $x_\sigma x_\tau$ for all $\sigma, \tau \in G$. For this, we compute the action of these products on L. For any $a \in L$, we have

$$(x_\sigma x_\tau) a (x_\sigma x_\tau)^{-1} = x_\sigma (x_\tau a x_\tau^{-1}) x_\sigma^{-1} = \sigma(\tau(a)) = (\sigma\tau)(a) = x_{\sigma\tau} a x_{\sigma\tau}^{-1}.$$

It follows that $c_{\sigma,\tau} := x_{\sigma\tau}^{-1} x_\sigma x_\tau$ centralizes L, and therefore $c_{\sigma,\tau} \in L^\times$ by the preceding lemma. Now we can write

$$x_\sigma x_\tau = x_{\sigma\tau} c_{\sigma,\tau} = a_{\sigma,\tau} x_{\sigma\tau} \quad \text{with} \quad a_{\sigma,\tau} = x_{\sigma\tau} c_{\sigma,\tau} x_{\sigma\tau}^{-1} = (\sigma\tau)(c_{\sigma,\tau}) \in L^\times.$$

Thus, multiplication in A is *completely determined* by specifying the elements $a_{\sigma,\tau} \in L^\times$ for all $\sigma, \tau \in G$. The collection $\{a_{\sigma,\tau}\}$ is called a *factor set* of A relative to L.

It is often convenient to view factor sets as functions $G \times G \to L^\times$. These functions satisfy a system of relations derived from the associative law in A: take any $\rho, \sigma, \tau \in G$, then

$$(x_\rho x_\sigma) x_\tau = (a_{\rho,\sigma} x_{\rho\sigma}) x_\tau = a_{\rho,\sigma}(x_{\rho\sigma} x_\tau) = a_{\rho,\sigma} a_{\rho\sigma,\tau} x_{(\rho\sigma)\tau}$$

and

$$x_\rho(x_\sigma x_\tau) = x_\rho(a_{\sigma,\tau} x_{\sigma\tau}) = (x_\rho a_{\sigma,\tau} x_\rho^{-1})(x_\rho x_{\sigma\tau}) = \rho(a_{\sigma,\tau}) a_{\rho,\sigma\tau} x_{\rho(\sigma\tau)}.$$

Since $x_{(\rho\sigma)\tau} = x_{\rho(\sigma\tau)}$, we obtain that

$$\rho(a_{\sigma,\tau}) a_{\rho,\sigma\tau} = a_{\rho,\sigma} a_{\rho\sigma,\tau}, \quad \text{for all } \rho, \sigma, \tau \in G.$$

Notice that these conditions are exactly the conditions defining 2-cocycles on G with values in $L^\times$. Thus every factor set gives an element of $Z^2(G, L^\times)$. It remains to show that a different choice of $\{x_\sigma\}$ leads to a cohomologous 2-cocycle.

Lemma 11.8. — *If $\{x_\sigma\}$ and $\{x'_\sigma\}$ are two choices, then*

$$a'_{\sigma,\tau} = b_\sigma \sigma(b_\tau) b_{\sigma\tau}^{-1} \cdot a_{\sigma,\tau}$$

for some $b_\sigma \in L^\times$. In particular, we have the well-defined map

$$\beta_{L/K} : \mathrm{Br}(L/K) \to H^2(G, L^\times), \quad [A] \mapsto [a_{\sigma,\tau}].$$

Proof. — Both x_σ and x'_σ implement the same automorphism σ of L. Hence, for every $a \in L$,

$$(x'_\sigma x_\sigma^{-1}) a (x'_\sigma x_\sigma^{-1})^{-1} = a.$$

Thus $x'_\sigma x_\sigma^{-1} \in Z_A(L) = L$, so there is a unique $b_\sigma \in L^\times$ with $x'_\sigma = b_\sigma x_\sigma$. Compute:

$$x'_\sigma x'_\tau = (b_\sigma x_\sigma)(b_\tau x_\tau) = b_\sigma \sigma(b_\tau) x_\sigma x_\tau = b_\sigma \sigma(b_\tau) a_{\sigma,\tau} x_{\sigma\tau}.$$

Since $x'_{\sigma\tau} = b_{\sigma\tau} x_{\sigma\tau}$, we obtain

$$a'_{\sigma,\tau} = b_\sigma \sigma(b_\tau) b_{\sigma\tau}^{-1} a_{\sigma,\tau}.$$

□

In the next step, one proves that $\beta_{L/K}$ is an isomorphism by showing:

- injectivity (cohomologous cocycles give isomorphic algebras),
- surjectivity (every cocycle defines a crossed product),
- compatibility with multiplication.

Lecture 12

This lecture completes the dictionary between degree-two cocycles and relative Brauer classes. We construct crossed-product algebras from factor sets, check compatibility with tensor products, and verify that enlarging the splitting field corresponds to inflation. Thus the crossed-product construction gives the desired group isomorphism

$$H^2(\mathrm{Gal}(L/K), L^\times) \simeq \mathrm{Br}(L/K).$$

12.1 Crossed Products from Cocycles. — Let L/K be finite Galois, put $G = \mathrm{Gal}(L/K)$, and let $a = (a_{\sigma,\tau}) \in Z^2(G, L^\times)$ be normalized. The crossed-product algebra attached to a is

$$A(a) = \bigoplus_{\sigma\in G} Le_\sigma,$$

with multiplication determined by

$$e_\sigma x = \sigma(x)e_\sigma, \qquad e_\sigma e_\tau = a_{\sigma,\tau}e_{\sigma\tau} \quad (x \in L).$$

The cocycle identity is exactly the associativity condition for triple products $e_\rho e_\sigma e_\tau$. The center is K: an element centralizing L has only the e_1-component, and commuting with all e_σ forces that component to lie in $L^G = K$. Simplicity is proved by the standard minimal-support argument: if a nonzero two-sided ideal contains $\sum a_\sigma e_\sigma$ with the smallest possible number of nonzero coefficients, then commutators with elements of L reduce the support unless only one coefficient occurs; in that case a unit multiple of some e_σ lies in the ideal, hence 1 lies in the ideal. Therefore $A(a)$ is central simple over K and split by L [**2**, Chap. IV, Lemma 3.13, p. 136].

If a and a' differ by a coboundary, say

$$a'_{\sigma,\tau} = b_\sigma\sigma(b_\tau)b_{\sigma\tau}^{-1}a_{\sigma,\tau},$$

then $A(a') \to A(a)$, $e'_\sigma \mapsto b_\sigma e_\sigma$, is a K-algebra isomorphism. Conversely, after using Skolem–Noether to make an isomorphism fix L, the two standard bases differ by scalars $b_\sigma \in L^\times$, so the two factor sets are cohomologous. Hence the crossed-product construction is inverse to the factor-set map $\beta_{L/K}$ from Lecture 11 [**2**, Chap. IV, Theorem 3.11, pp. 135–136].

12.2 Compatibility with the Group Law. — We now check that the bijection is a group homomorphism. Since $L^\times$ is written multiplicatively, the sum of two cohomology classes is represented by the pointwise product

$$(aa')_{\sigma,\tau} = a_{\sigma,\tau}a'_{\sigma,\tau}.$$

The standard tensor-product lemma for crossed products says that

$$A(aa') \sim A(a) \otimes_K A(a'),$$

so the class of $A(a)$ in $\mathrm{Br}(K)$ adds exactly as the class of a in $H^2(G, L^\times)$. Here is the idea of the proof. Put $A = A(a)$, $B = A(a')$, and $C = A(aa')$. Regard A and B as left L-vector spaces and form

$$V = A \otimes_L B.$$

There is a right action of $A \otimes_K B$ on V coming from right multiplication on the two factors, and there is a left action of C given on basis elements by

$$(le''_\sigma) \cdot (a_0 \otimes b_0) = l\, e_\sigma a_0 \otimes e'_\sigma b_0.$$

The cocycle relation shows that these two actions are well-defined and commute. Thus one obtains an injective homomorphism

$$(A \otimes_K B)^{\mathrm{op}} \longrightarrow \mathrm{End}_C(V).$$

Both sides have the same K-dimension, hence this is an isomorphism. Since $V \simeq C^n$ as a left C-module, where $n = [L : K]$, this gives $A \otimes_K B \simeq M_n(C)$, and therefore $A(aa') \sim A(a) \otimes_K A(a')$ [**2**, Chap. IV, Lemma 3.15, p. 137].

12.3 Compatibility with Inflation. — Finally let E/K be finite Galois with $L \subset E$. Put $\Gamma = \mathrm{Gal}(E/K)$ and let

$$\rho : \Gamma \longrightarrow G = \mathrm{Gal}(L/K)$$

be the restriction map. If $a \in Z^2(G, L^\times)$, its inflation is the cocycle

$$\widehat{a}_{\gamma,\delta} = a_{\rho(\gamma),\rho(\delta)} \in L^\times \subset E^\times.$$

Let

$$A = A(a) = \bigoplus_{\sigma \in G} Le_\sigma, \qquad B = A_E(\widehat{a}) = \bigoplus_{\gamma \in \Gamma} Ey_\gamma$$

be the corresponding crossed products. Write $n = [L : K]$, $m = [E : K]$, and $h = [E : L]$, so $m = hn$. The compatibility statement is

$$B \cong M_h(A) \qquad \text{as central simple } K\text{-algebras.}$$

This is the algebra form of the commutative diagram

$$\begin{tikzcd}
H^2(\mathrm{Gal}(L/K), L^\times) \arrow[r, "\mathrm{Inf}"] \arrow[d] & H^2(\mathrm{Gal}(E/K), E^\times) \arrow[d] \\
\mathrm{Br}(L/K) \arrow[r] & \mathrm{Br}(E/K).
\end{tikzcd}$$

One way to see the isomorphism is to use the same module method as in Lemma 3.15. The algebra B acts on a bimodule built from A and E; after choosing an L-basis of E, this bimodule is free of rank h over A. The induced map from B to the endomorphism algebra of this free module is injective because B is simple, and the dimensions agree:

$$\dim_K B = m^2 = h^2 n^2 = \dim_K M_h(A).$$

Hence $B \simeq M_h(A)$. Therefore passing from L to the larger splitting field E changes the crossed product only by a matrix algebra, i.e. it does not change the Brauer class. This proves that the crossed-product isomorphisms commute with inflation.

Lecture 13-14

These lectures study reduced norm and reduced trace for central simple algebras. After defining them through a splitting field, we prove independence of the splitting field and use the reduced norm to detect units. We then introduce C_1-fields as a source of vanishing results for Brauer groups; in particular, Tsen's theorem gives the C_1-property for function fields of curves over algebraically closed fields.

13.1 Reduced Norm and Trace. — Today, we introduce the reduced norm and trace. Let K be a field and A be a central simple algebra over K. Let L/K be a finite Galois extension such that A splits over L. Choose an isomorphism $\varphi : A \otimes_K L \cong M_n(L)$ of L-algebras. For $a \in A$, define

$$\mathrm{Nred}_A(a) := \det(\varphi(a \otimes 1)), \qquad \mathrm{Tred}_A(a) := \mathrm{Tr}(\varphi(a \otimes 1)).$$

These are the reduced norm and reduced trace of a.

Since φ is a homomorphism of L-algebras, the reduced norm is multiplicative, and the reduced trace is additive. For the well-definedness, we have to check the following:

1. The reduced norm is independent of choice of isomorphisms φ follows from the Skolem-Noether. In fact, if we have $\varphi, \psi : A \otimes_K L \overset{\cong}{\to} M_n(L)$, then by Skolem-Noether, there exists $x \in \mathrm{GL}_n(L)$ such that $\psi(a \otimes 1) = x\varphi(a \otimes 1)x^{-1}$, which implies that the determinant and the trace are the same.
2. The reduced norm and trace are independent of the choice of splitting fields L. This means that if A splits over two fields L and M, then the reduced norm and trace defined through $A \otimes_K L \cong M_n(L)$ and $A \otimes_K M \cong M_{n'}(M)$ are the same. In fact, consider the compositum LM and we have the embeddings $A \otimes_K L \cong M_n(L) \hookrightarrow M_n(LM)$ and $A \otimes_K M \cong M_n(M) \hookrightarrow M_n(LM)$. More precisely, choose a finite Galois extension E/K containing both L and M and splitting A. After scalar extension to E, the two E-algebra isomorphisms

   $$(A \otimes_K L) \otimes_L E \simeq A \otimes_K E \simeq (A \otimes_K M) \otimes_M E$$

   identify both definitions with the ordinary determinant and trace on $M_n(E)$. Since determinant and trace are invariant under conjugation, and any two splittings over E differ by an inner automorphism by Skolem–Noether, both constructions give the same elements of E. As the values are already fixed by $\mathrm{Gal}(E/K)$, they lie in K. This is the standard independence-of-splitting-field argument for reduced norm and reduced trace [**3**, Chap. 2, §2.6, p. 38].

Proposition 13.1. — We have $\mathrm{Nred}_A(a) \in K$ and $\mathrm{Tred}_A(a) \in K$ for all $a \in A$.

Proof. — An element $\sigma \in \mathrm{Gal}(L/K)$ induces two maps $\sigma_1 : A \otimes_K L \xrightarrow{\cong} A \otimes_K L$, $\sum_i a_i \otimes \lambda_i \mapsto \sum_i a_i \otimes \sigma(\lambda_i)$ and $\sigma_2 : M_n(L) \xrightarrow{\cong} M_n(L)$, $(a_{ij}) \mapsto (\sigma(a_{ij}))$. These are isomorphisms of K-algebras, but not isomorphisms of L-algebras. Let $\psi = \sigma_2 \circ \varphi \circ \sigma_1^{-1} : A \otimes_K L \xrightarrow{\cong} M_n(L)$. This is an isomorphism of L-algebras, since $\sigma_1^{-1}(\lambda x) = \sigma^{-1}(\lambda)\sigma_1^{-1}(x)$ for all $\lambda \in L$ and φ is L-linear and $\sigma_2(\lambda y) = \sigma(\lambda)\sigma_2(y)$ for all $\lambda \in L$. By Skolem-Noether, there exists $x_\sigma \in \mathrm{GL}_n(L)$ such that $\varphi(y) = x_\sigma \psi(y) x_\sigma^{-1}$ for all $y \in A \otimes_K L$, which implies that $\det(\varphi(y)) = \det(\psi(y))$ for all $y \in A \otimes_K L$. If $y = a \otimes 1$ for some $a \in A$, then $\sigma_1^{-1}(a \otimes 1) = a \otimes \sigma^{-1}(1) = a \otimes 1$. This shows that

$$\mathrm{Nred}(a) = \det(\varphi(a\otimes 1)) = \det(\psi(a\otimes 1)) = \det(\sigma_2(\varphi(a\otimes 1))) = \sigma(\det(\varphi(a\otimes 1))) = \sigma(\mathrm{Nred}(a)),$$

where the fourth equality follows from that σ_2 is induced by $\sigma \in \mathrm{Gal}(L/K)$ on each component and the determinant is a polynomial with variables in the matrix, and therefore we have $\mathrm{Nred}(a) \in L^{\mathrm{Gal}(L/K)} = K$. By a similar argument we get the result for the reduced trace. □

Corollary 13.2. — Let $e_1, \cdots, e_{n^2}$ be a K-basis of A. Then there exists a homogeneous polynomial $\Phi(t_1, \cdots, t_{n^2})$ of homogeneous degree n such that $\mathrm{Nred}_A(\lambda_1 e_1 + \cdots + \lambda_{n^2} e_{n^2}) = \Phi(\lambda_1, \cdots, \lambda_{n^2})$ for all $\lambda_i \in K$.

Proof. — Recall that L is a Galois extension of K such that A splits over L. Let $K(t_1, \cdots, t_{n^2})$ be the fraction field of $K[t_1, \cdots, t_{n^2}] := K'$ and similarly for $L(t_1, \cdots, t_{n^2}) := L'$. Consider $A' := A \otimes_K K'$, which is central simple over K' and splits over L', since $A' \otimes_{K'} L' = (A \otimes_K L) \otimes_L L'$.

Consider the reduced norm

$$\mathrm{Nred}_{A'}(t_1 e_1 + \cdots + t_{n^2} e_{n^2}) \in K' \cap L[t_1, \cdots, t_{n^2}] = K[t_1, \cdots, t_{n^2}],$$

which is the determinant of an element in $M_n(L')$ of the form $\left(\sum_{i=1}^{n^2} m_{rs}^{(i)}\, t_i \right)_{r,s}$ and hence it is a homogeneous polynomial in L of degree n over L. Indeed, after choosing a splitting $A \otimes_K L \simeq M_n(L)$, each basis vector e_i maps to a matrix $(m_{rs}^{(i)})_{r,s}$. Therefore $\sum_i t_i e_i$ maps to the matrix whose (r,s)-entry is the linear form $\sum_i m_{rs}^{(i)} t_i$. The determinant is a homogeneous polynomial of degree n in the variables t_i with coefficients in L. By the previous proposition the reduced norm is fixed by $\mathrm{Gal}(L/K)$, hence its coefficients lie in K; equivalently the polynomial belongs to $K[t_1, \ldots, t_{n^2}]$. □

Recall the Cayley-Hamilton theorem.

***Theorem 13.3* (Cayley-Hamilton).** — Let R be a ring and M be a finitely generated R-module with generators $m_1, \cdots, m_n$. For any endomorphism $\varphi \in \mathrm{End}_R(M)$, $\chi_\varphi(\varphi) = 0$ in $\mathrm{End}_R(M)$, where $\chi_\varphi(T) := \det(T \cdot I_n - \varphi)$ is the characteristic polynomial of φ.

In particular, for a matrix $A \in M_n(R)$, it can be viewed as an element in $\mathrm{End}_R(R^n)$ and hence we have $\chi_A(A) = 0$ in $M_n(R)$, which is the Cayley-Hamilton theorem in linear algebra.

Proof. — Let $a_{ij} \in R$ satisfy that $\varphi(m_i) = \sum_{j=1}^n a_{ij} m_j$. Then $\Delta := (\delta_{ij}\varphi - a_{ij})_{ij}$ is a $n \times n$ matrix over the commutative subring $R[\varphi]$ of $\mathrm{End}_R(M)$, where we note that the commutativity follows from that φ is R-linear. Let $\overline{m}$ be $(m_1, \cdots, m_n)^T \in R^n$ and we have $\Delta \cdot \overline{m} = 0$. As $\Delta \in M_n(R[\varphi])$ and $R[\varphi]$ is commutative, we may take the adjoint matrix Γ of Δ and obtain $(\Gamma\Delta) \cdot \overline{m} = \det(\Delta)\overline{m} = 0$. Since $\det(\Delta) \in R[\varphi]$ and by R-linearity, $\det(\Delta)$ annihilates the whole M, which means that $\det(\Delta) = 0$. We conclude by noticing that $\det(\Delta) = \chi_\varphi(\varphi)$, since $\det(\Delta)$ is defined by substituting T in $\chi_\varphi(T)$ by φ. □

Remark 13.4. — The subtlety in the last theorem is that in the characteristic polynomial $\chi_\varphi(T) := \det(T \cdot I_n - \varphi)$, when we evaluate it in $\mathrm{End}_R(M)$ at φ, the element $\varphi \cdot I_n$ is not obtained by the usual composition in $\mathrm{End}_R(M)$, but actually the diagonal matrix $\mathrm{diag}(T, \cdots, T)$.

Proposition 13.5. — We have $a \in A$ is a unit iff $\mathrm{Nred}(a) \neq 0$.

Proof. — If a is invertible, multiplicativity gives

$$\mathrm{Nred}(a)\mathrm{Nred}(a^{-1}) = \mathrm{Nred}(1) = 1,$$

so $\mathrm{Nred}(a) \neq 0$.

Conversely, let

$$\chi_a(T) = T^n + \lambda_{n-1}T^{n-1} + \cdots + \lambda_1 T + \lambda_0 \in K[T], \qquad \lambda_0 = (-1)^n \mathrm{Nred}(a).$$

This is the reduced characteristic polynomial of a, obtained by computing the ordinary characteristic polynomial of $\varphi(a \otimes 1)$ over a splitting field. The coefficients lie in K by the same Galois-invariance argument used for the reduced norm and trace. Moreover, the Cayley–Hamilton identity holds in A: after extending scalars to a splitting field it becomes the ordinary Cayley–Hamilton theorem for matrices, and the map $A \hookrightarrow A \otimes_K L$ is injective.

The Cayley–Hamilton identity gives

$$0 = \chi_a(a) = a^n + \lambda_{n-1}a^{n-1} + \cdots + \lambda_1 a + \lambda_0.$$

If $\mathrm{Nred}(a) \neq 0$, then $\lambda_0 \in K^\times$, and hence

$$a \cdot \big(-\lambda_0^{-1}(a^{n-1} + \lambda_{n-1}a^{n-2} + \cdots + \lambda_1)\big) = 1.$$

The same polynomial expression also gives a right inverse, since it lies in the commutative subalgebra $K[a] \subset A$. Thus $a \in A^\times$. □

Corollary 13.6. — A is a division algebra iff $\mathrm{Nred}_A(t_1e_1 + \cdots + t_{n^2}e_{n^2}) = 0$ does not have a nontrivial solution in K^{n^2}.

Proof. — We have A is a division ring iff $A^\times = A\backslash\{0\}$ iff $\mathrm{Nred}_A(a) \neq 0$ for all $a \neq 0$ in A iff $\mathrm{Nred}_A(t_1e_1 + \cdots + t_{n^2}e_{n^2}) = 0$ has no nontrivial solution in K^{n^2}, where the last equality follows from the linear independence of $e_1, \cdots, e_{n^2}$. □

13.2 C_1-Fields. —

Definition 13.1. — Let k be a field. We say that k has property C_1 if every homogeneous polynomial
$$f \in k[x_1, \dots, x_n]$$
of degree $d < n$ has a nontrivial zero in k^n.

Remark 13.2. — It is obvious that algebraically closed fields are C_1.

Proposition 13.3. — *If k is a C_1-field, then* $\mathrm{Br}(k) = 0$.

Proof. — It suffices to show that every central division algebra D over k is trivial. Let $\dim_k D = n^2$ and let $\{e_1, \dots, e_{n^2}\}$ be a k-basis of D. Consider
$$x = \sum_{i=1}^{n^2} t_i e_i,$$
and its reduced norm $\mathrm{Nrd}_D(x)$. This is a homogeneous polynomial of degree n in n^2 variables.

If $n > 1$, then $n < n^2$, so by the C_1 property there exists a nontrivial solution (t_i) such that $\mathrm{Nrd}_D(x) = 0$. Hence x is noninvertible, so D is not a division algebra. Therefore $D = k$, and $\mathrm{Br}(k) = 0$. □

Remark 13.4. — More generally, for $r \geq 1$, a field k is called C_r if every homogeneous polynomial of degree d in n variables with $d^r < n$ has a nontrivial zero.

Goal. If K is a local field, we aim to show that K^{ur} is C_1. This implies
$$H^2(\mathrm{Gal}(K^{\mathrm{sep}}/K^{\mathrm{ur}}), (K^{\mathrm{sep}})^\times) = 0.$$

Lemma 13.5. — *If K is C_1 and L/K is finite, then L is C_1.*

Proof. — Let $[L : K] = m$ and fix a K-basis $v_1, \dots, v_m$ of L. Let $f \in L[x_1, \dots, x_n]$ be homogeneous of degree $d < n$. Write $x_i = \sum_{j=1}^m x_{ij} v_j$, where x_{ij} are variables over K. Then $f(x_1, \dots, x_n)$ is an element of $L[x_{ij}]$. Taking the field norm
$$\Phi(x_{ij}) := N_{L/K}(f(x_1, \dots, x_n)),$$
we obtain a homogeneous polynomial $\Phi \in K[x_{ij}]$ of degree md.

Since $md < mn$, by the C_1 property of K, there exists a nontrivial solution (x_{ij}) such that $\Phi = 0$. Hence
$$N_{L/K}(f(x_1, \dots, x_n)) = 0,$$
which implies $f(x_1, \dots, x_n) = 0$.

Since the solution is nontrivial, we obtain a nontrivial zero of f in L^n. □

Theorem 13.6 **(Chevalley).** — *Finite fields are C_1. In particular, the Brauer group of a finite field vanishes.*

Proof. — Let $k = \mathbf{F}_q$, $q = p^r$, and let $f \in k[x_1, \dots, x_n]$ be homogeneous of degree $d < n$. We prove the stronger claim that the number $N(f)$ of solutions of $f(x_1, \dots, x_n) = 0$ in k^n is divisible by p. Since
$$1 - f(x)^{q-1}$$

is 1 when $f(x) = 0$ and 0 when $f(x) \neq 0$, we have

$$N(f) = \sum_{x \in k^n} (1 - f(x)^{q-1}) = q^n - \Sigma(f), \qquad \Sigma(f) := \sum_{x \in k^n} f(x)^{q-1}.$$

Thus it suffices to show $\Sigma(f) = 0$ in k.

Expand f^{q-1} as a sum of monomials

$$x_1^{r_1} \cdots x_n^{r_n}, \quad \sum r_i = d(q-1).$$

We show that $\Sigma(x_1^{r_1} \cdots x_n^{r_n}) = 0$ in k for every occurring monomial.

If some $r_i = 0$, then summing over $x_i \in k$ gives a factor $q = 0$ in k. Otherwise all $r_i > 0$. Since $\sum_i r_i = d(q-1) < n(q-1)$, at least one exponent, say r_1, satisfies $0 < r_1 < q-1$. Fixing $(\alpha_2, \dots, \alpha_n) \in k^{n-1}$ and taking a generator ω of the cyclic group $k^\times$, we get

$$\sum_{\alpha \in k} \alpha^{r_1} \alpha_2^{r_2} \cdots \alpha_n^{r_n} = \alpha_2^{r_2} \cdots \alpha_n^{r_n} \sum_{i=0}^{q-2} \omega^{ir_1} = (\alpha_2^{r_2} \cdots \alpha_n^{r_n}) \frac{(\omega^{r_1})^{q-1} - 1}{\omega^{r_1} - 1},$$

which equals 0 in k. We conclude by letting $(\alpha_2, \dots, \alpha_n)$ run over k^{n-1}. Hence $N(f) \equiv 0 \pmod p$. For a homogeneous f, the nonzero scalar multiples of any nonzero solution come in orbits of size $q-1$, while the trivial solution is a single point. Since $N(f)$ is divisible by p, it cannot be equal to 1. Therefore f has a nontrivial zero, and k is C_1. □

***Theorem 13.7* (Tsen).** — Let k be an algebraically closed field. Then $k(t) = \mathrm{Frac}(k[t])$ is C_1.

In particular, with Lemma 13.5, for every function field K of an algebraic curve over k, K is a C_1 field.

Proof. — By Lemma 13.5, it is enough to prove the result for $K = k(t)$, since the function field of an algebraic curve over k is a finite extension of $k(t)$ after choosing a nonconstant rational function on the curve.

Let $f \in k(t)[x_1, \dots, x_n]$ be homogeneous of degree $d < n$. After multiplying by a common denominator, we may assume that

$$f \in k[t][x_1, \dots, x_n].$$

It is enough to find a nontrivial zero in $k[t]^n$. Choose an integer $N > 0$ and look for a solution of the form

$$x_i = \sum_{j=0}^{N} a_{ij} t^j.$$

Here the a_{ij} are indeterminates over k. If r is the maximum of the t-degrees of the coefficients of f, then after substitution we can write

$$f(x_1, \dots, x_n) = \sum_{\ell=0}^{dN+r} f_\ell(a_{10}, \dots, a_{nN}) t^\ell,$$

where each f_ℓ is a homogeneous polynomial of degree d in the $n(N+1)$ variables a_{ij}.

Thus we need a nonzero common solution of

$$f_\ell(a_{ij}) = 0 \qquad (0 \le \ell \le dN + r).$$

For N sufficiently large,

$$dN + r + 1 < n(N+1),$$

because $d < n$. Hence these equations define a closed subset of

$$\mathbf{P}_k^{n(N+1)-1}$$

cut out by at most $n(N+1) - 1$ homogeneous equations. By the standard projective non-emptiness theorem over an algebraically closed field, this closed subset has a k-point. Choosing homogeneous coordinates for this point gives $\lambda_{ij} \in k$, not all zero, such that

$$f_\ell(\lambda_{ij}) = 0 \qquad \text{for all } \ell.$$

Now set

$$x_i = \sum_{j=0}^{N} \lambda_{ij} t^j \in k[t].$$

The tuple $(x_1, \dots, x_n)$ is nonzero and satisfies $f(x_1, \dots, x_n) = 0$. Thus $k(t)$ is C_1. $\square$

Remark 13.7. — Tsen's theorem has the following geometric interpretation. Let C be a smooth projective curve with function field K. The homogeneous polynomial $f \in K[x_1, \dots, x_n]$ defines an $(n-1)$-dimensional projective variety equipped with a surjective morphism $p : X \to C$. A nontrivial solution of

$$f(x_1, \dots, x_n) = 0 \text{ in } K^n$$

defines a section of p, i.e. a morphism $s : C \to X$ with $p \circ s = \mathrm{id}_C$. In particular, $s(C) \subset X$ is a closed subvariety of dimension 1 mapped isomorphically onto C by p.

Lecture 15 (Not completed, Excursion)

This excursion is about deformation theory of Galois representations. We introduce Mazur's finiteness condition and framed deformation functors, compute the tangent space, and explain how lifting across a small extension produces an obstruction class in $H^2(G, \operatorname{ad}\bar\rho)$.

14.1 Mazur's Finiteness Condition and Deformation Functors. — We begin the excursion into the deformation theory of Galois representations.

Let G be a profinite group, and let $\bar\rho : G \to \mathrm{GL}_n(k)$ be a continuous representation, where k is a finite field of characteristic p.

Question: is there a lift of $\bar\rho$ to characteristic zero? More precisely, let $L/\mathbf{Q}_p$ be a finite extension, let $\varpi \in \mathscr{O}_L$ be a uniformizer, and let k be the residue field. Does there exist a finite extension L'/L and a continuous homomorphism $\rho : G \to \mathrm{GL}_n(\mathscr{O}_{L'})$ such that $\rho \equiv \bar{\rho} \pmod{\mathfrak{m}_{L'}}$?

The answer is yes when $G = G_F$ and $F/\mathbf{Q}_p$ is finite: for GL_n this is proved in On local Galois deformation rings, and for general reductive groups in On local Galois deformation rings: generalised reductive groups.

Definition 14.1. — A profinite group G satisfies Φ_p (Mazur's finiteness condition at a prime p) if, for every open normal subgroup $N \triangleleft G$, the set $\mathrm{Hom}_{\mathrm{cts}}(N, \mathbf{F}_p)$ is finite.

Lemma 14.2. — *Let G act continuously on a finite-dimensional $\mathbf{F}_p$-vector space V. If G satisfies Φ_p, then $\dim_{\mathbf{F}_p} H^1(G,V) < \infty$.*

Proof. — Let V be finite-dimensional with basis $v_1, \dots, v_n$. Let $N = \cap_{i=1}^n \mathrm{Stab}_G(v_i)$. It is open and is the kernel of the action, hence normal.

By inflation-restriction:

$$0 \to H^1(G/N, V) \to H^1(G,V) \overset{Res}{\to} H^1(N,V)$$

the first term is the first cohomology of a finite group with coefficients in the finite-dimensional vector space V, hence is finite-dimensional. The third term is isomorphic to

$$H^1(N, \mathbf{F}_p) \otimes_{\mathbf{F}_p} V \cong \mathrm{Hom}_{\mathrm{cts}}(N, \mathbf{F}_p) \otimes_{\mathbf{F}_p} V,$$

which is finite-dimensional by Φ_p. $\square$

Remark 14.3. — If $G = G_F$, then $\mathrm{Hom}_{\mathrm{cts}}(G_F, \mathbf{Z}/p\mathbf{Z})$ is in bijection with cyclic Galois extensions E/F of degree dividing p inside F^{sep}. The nonzero homomorphisms correspond to cyclic extensions of degree exactly p, by sending

$$\varphi \longmapsto (F^{\mathrm{sep}})^{\ker \varphi}.$$

Example 14.4. — Let F be a global field, i.e. a number field or the function field of a smooth projective curve over a finite field. Fix a finite set S of places of F. Then there are only finitely many extensions E/F of a given degree unramified outside S (Hermite–Minkowski). Hence Φ_p holds for $G = \mathrm{Gal}(F^S/F)$, where F^S is the union of all finite extensions E/F unramified outside S.

Another example is when $F/\mathbf{Q}_p$ is finite. Then G_F satisfies Φ_p.

Proof. — Let $N \triangleleft G_F$ be open. Then $N = G_E$ for a finite extension E/F, so it is enough to prove that

$$H^1(G_E, \mathbf{F}_p) = \mathrm{Hom}_{\mathrm{cts}}(G_E, \mathbf{F}_p)$$

is finite for every finite extension E/F. Replace E by $E' = E(\mu_p)$. The inflation–restriction sequence for $1 \to G_{E'} \to G_E \to \mathrm{Gal}(E'/E) \to 1$ shows that finiteness of $H^1(G_{E'}, \mathbf{F}_p)$ implies finiteness of $H^1(G_E, \mathbf{F}_p)$, since $\mathrm{Gal}(E'/E)$ is finite. Thus we may assume $\mu_p \subset E$.

Under this assumption $\mu_p \simeq \mathbf{F}_p$ as a trivial G_E-module. The Kummer exact sequence

$$1 \to \mu_p \to (E^{\mathrm{sep}})^\times \xrightarrow{x \mapsto x^p} (E^{\mathrm{sep}})^\times \to 1$$

and Hilbert 90 give

$$H^1(G_E, \mathbf{F}_p) \cong H^1(G_E, \mu_p) \cong E^\times/E^{\times p}.$$

Finally $E^\times \simeq \langle \varpi_E \rangle \times \mathscr{O}_E^\times$, and $(E^\times)^p$ has finite index in $E^\times$; equivalently, $E^\times/E^{\times p}$ is finite. Hence $H^1(G_E, \mathbf{F}_p)$ is finite for all finite E/F, so G_F satisfies Φ_p. $\square$

Now we formulate the deformation problem.

Let $\mathfrak{a}_{\mathscr{O}}$ be the category of local Artinian $\mathscr{O}$-algebras with residue field k; morphisms are local $\mathscr{O}$-algebra homomorphisms inducing the identity on k.

For an Artinian local ring $(A, \mathfrak{m}_A)$, there exists n such that $\mathfrak{m}_A^n = 0$, and $\dim_k \mathfrak{m}_A/\mathfrak{m}_A^2 < \infty$. Thus A is Noetherian, and every quotient is Noetherian.

Since the residue field k is finite, every object A of $\mathfrak{a}_{\mathscr{O}}$ is finite as a set.

Definition 14.5. — Given a continuous representation $\bar{\rho}: G \to \mathrm{GL}_n(k)$, the framed deformation functor

$$D_{\bar{\rho}}^{\square} : \mathfrak{a}_{\mathscr{O}} \to \mathrm{Set}$$

sends $(A, \mathfrak{m}_A)$ to the set of continuous representations

$$\rho : G \to \mathrm{GL}_n(A)$$

such that $\rho \equiv \bar{\rho} \pmod{\mathfrak{m}_A}$.

We are not taking representations up to isomorphism.

***Theorem 14.6* (Mazur).** — *If G satisfies Φ_p, then $D_{\bar{\rho}}^{\square}$ is pro-representable by a complete local Noetherian $\mathscr{O}$-algebra $(R_{\bar{\rho}}^{\square}, \mathfrak{m}^{\square})$ with residue field k.*

Remark 14.7. — Fact: such rings are always of the form $\mathscr{O}[[x_1, \dots, x_n]]/I$.

Sketch: if $(R, \mathfrak{m})$ is complete local Noetherian with residue field k, then

$$\dim_k \mathfrak{m}/(\mathfrak{m}^2, \varpi) < \infty.$$

Choose $a_1, \dots, a_n \in \mathfrak{m}$ whose images form a basis of this quotient. For every N, the map

$$\mathscr{O}[[x_1, \dots, x_n]] \to R/\mathfrak{m}^N, \qquad x_i \mapsto a_i + \mathfrak{m}^N,$$

is surjective. Passing to the inverse limit gives a surjection

$$\mathscr{O}[[x_1, \dots, x_n]] \twoheadrightarrow R.$$

The minimal number of variables in such a presentation is detected by the tangent space.

Pro-representability means that there exists a continuous universal representation $\rho^{\square} : G \to \mathrm{GL}_n(R^{\square})$ such that $\rho^{\square} \otimes_{R^{\square}} k = \bar{\rho}$, together with a functorial bijection

$$\mathrm{Hom}_{\hat{\mathfrak{a}_{\mathscr{O}}}}(R_{\bar{\rho}}^{\square}, A) := \varinjlim_n \mathrm{Hom}_{\mathfrak{a}_{\mathscr{O}}}(R^{\square}/(\mathfrak{m}^{\square})^n, A) \cong D_{\bar{\rho}}^{\square}(A).$$

It is given by

$$\varphi \mapsto \rho^{\square} \otimes_{R^{\square}, \varphi} A := (G \to \mathrm{GL}_n(R_{\bar{\rho}}^{\square}) \to \mathrm{GL}_n(A)).$$

14.2 Tangent Spaces and Obstructions. —

Exercise 14.8. — Let G be a free profinite group with generators $\gamma_1, \dots, \gamma_m$. Then the framed deformation ring is a formal power series ring

$$R_{\bar\rho}^{\square} = \mathscr{O}[[x_{ij}^{(l)} \mid 1 \leq l \leq m,\ 1 \leq i, j \leq n]].$$

If $G = \widehat{\mathbf{Z}} \oplus \widehat{\mathbf{Z}}$ with generators γ_1, γ_2, then one must impose the relation $\gamma_1\gamma_2 = \gamma_2\gamma_1$. For the trivial residual representation, the deformation ring is obtained from the free case by imposing the matrix commutator equations.

Lemma 14.9. —

$$D_{\bar\rho}^{\square}(k[\varepsilon]) \cong Z^1(G, \mathrm{ad}(\bar\rho)).$$

Lemma 14.10. — *Let V be any finite-dimensional k-vector space with continuous G-action. Then*

$$0 \to H^0(G, V) \to V \to Z^1(G, V) \to H^1(G, V) \to 0$$

is exact, where $v \in V$ maps to the coboundary $g \mapsto gv - v$.

Hence, assuming Φ_p,

$$\dim_k Z^1(G, V) = \dim_k V - \dim_k H^0(G, V) + \dim_k H^1(G, V).$$

Theorem 14.11 **(Mazur).** — *Assume G satisfies Φ_p. Put*

$$r = \dim_k Z^1(G, \mathrm{ad}(\bar\rho)), \qquad s = \dim_k H^2(G, \mathrm{ad}(\bar\rho)).$$

Then $R_{\bar\rho}^{\square}$ has a presentation

$$R_{\bar\rho}^{\square} \cong \frac{\mathscr{O}[[x_1, \dots, x_r]]}{I}$$

where the ideal I can be generated by at most s elements.

Thus, after possibly adding redundant zero relations if fewer than s generators are needed, one often writes this informally as

$$R_{\bar\rho}^{\square} \simeq \mathscr{O}[[x_1, \dots, x_r]]/(f_1, \dots, f_s)$$

[**5**, §1.6–1.7].

Remark 14.12. — Obstruction theory is controlled by $H^2(G, \mathrm{ad}(\bar\rho))$. In the local-field case $G = G_F$, this is written $H^2(G_F, \mathrm{ad}(\bar\rho))$.

Main idea: let $\varphi : A \to B$ be a small extension in $\mathfrak{a}_{\mathscr{O}}$, with $\ker\varphi = (t)$ and $\mathfrak{m}_A \cdot t = 0$.

For example, $\mathscr{O}/\varpi^2 \to k$ is such a surjection.

Given $\rho \in D_{\bar\rho}^{\square}(B)$, can we lift it to some $\tilde\rho \in D_{\bar\rho}^{\square}(A)$? For each $g \in G$, choose a lift $\tilde\rho_g \in \mathrm{GL}_n(A)$ such that $\tilde\rho_g \equiv \rho_g \pmod{(t)}$. Define

$$\kappa(g, h) = \tilde\rho_g \tilde\rho_h \tilde\rho_{gh}^{-1} = I + tM(g, h).$$

We verify that $(g, h) \mapsto M(g, h)$ is a 2-cocycle, and that its class is zero in $H^2(G, \mathrm{ad}(\bar\rho))$ exactly when a lifting exists.

Here is the precise obstruction calculation. Because $\mathfrak{m}_A t = 0$, the ideal (t) is a k-vector space and $1 + tX$ is multiplied by the rule $(1 + tX)(1 + tY) = 1 + t(X + Y)$. The choice of lifts $\tilde{\rho}_g$ gives

$$\tilde{\rho}_g \tilde{\rho}_h = (1 + tM(g, h))\tilde{\rho}_{gh}, \qquad M(g, h) \in \mathrm{ad}(\bar{\rho}).$$

Associativity of the threefold product $\tilde{\rho}_g \tilde{\rho}_h \tilde{\rho}_\ell$ gives

$$g \cdot M(h, \ell) - M(gh, \ell) + M(g, h\ell) - M(g, h) = 0,$$

so M is a continuous 2-cocycle for the G-module $\mathrm{ad}(\bar{\rho})$, where g acts by conjugation through $\bar{\rho}(g)$. If the lifts are changed by $\tilde{\rho}'_g = (1 + tN(g))\tilde{\rho}_g$, then the cocycle changes by the coboundary dN. Hence the class

$$\mathrm{ob}_{A \to B}(\rho) := [M] \in H^2(G, \mathrm{ad}(\bar{\rho}))$$

is independent of all choices. It vanishes exactly when, after modifying the chosen lifts by a 1-cochain N, the new lifts satisfy $\tilde{\rho}'_g \tilde{\rho}'_h = \tilde{\rho}'_{gh}$; equivalently they define a representation $G \to \mathrm{GL}_n(A)$ lifting ρ [**5**, §1.6–1.7].

The Euler–Poincare formula says that for every finite-dimensional k-representation V of G_F,

$$\dim_k H^0(G_F, V) - \dim_k H^1(G_F, V) + \dim_k H^2(G_F, V) = -[F : \mathbf{Q}_p] \dim_k V.$$

For $V = \mathrm{ad}(\bar{\rho})$ of dimension d^2, and

$$r = \dim_k Z^1(G_F, V) = d^2 - \dim_k H^0(G_F, V) + \dim_k H^1(G_F, V), \qquad s = \dim_k H^2(G_F, V),$$

we obtain

$$r - s = d^2 + [F : \mathbf{Q}_p] d^2.$$

Thus obstruction theory predicts that the framed ring has relative dimension $d^2 + d^2[F : \mathbf{Q}_p]$ over $\mathscr{O}$. This prediction is a theorem of Böckle–Iyengar–Paškūnas: if $F/\mathbf{Q}_p$ is finite and $\bar{\rho} : G_F \to \mathrm{GL}_d(k)$ is continuous, then the universal framed deformation ring $R^{\square}_{\bar{\rho}}$ is flat over $\mathscr{O}$ and is a local complete intersection of relative dimension $d^2 + d^2[F : \mathbf{Q}_p]$ [**1**, Theorem 1.1]. Consequently, in this local case, the Krull dimension of the special fibre is

$$\dim R^{\square}_{\bar{\rho}}/(\varpi) = d^2 + d^2[F : \mathbf{Q}_p].$$

Lecture 16 (Not completed, Excursion and mysterious function field)

This lecture has two roles. First we finish the deformation-theoretic excursion by using local Tate duality to interpret obstruction groups. Then we return to C_1-fields and function-field arguments, which provide the vanishing input used later for unramified fields and cohomological dimension.

15.1 Local Tate Duality in Deformation Theory. — Still in the excursion, let $F/\mathbf{Q}_p$ be finite and let V be a finite-dimensional $\mathbf{F}_p$-vector space with continuous G_F-action. Local Tate duality gives a perfect pairing

$$H^i(G_F, V) \times H^{2-i}(G_F, V^*(1)) \longrightarrow H^2(G_F, \mathbf{F}_p(1)) \simeq \mathbf{F}_p,$$

where $V^* = \mathrm{Hom}_{\mathbf{F}_p}(V, \mathbf{F}_p)$ and $V^*(1) = V^* \otimes_{\mathbf{F}_p} \mathbf{F}_p(1)$. If $\omega : G_F \to \mathbf{F}_p^\times$ is the mod-p cyclotomic character, then the action on the underlying vector space V^* is

$$(g \cdot \ell)(v) = \omega(g)\ell(g^{-1}v).$$

This is useful in deformation theory because it replaces the obstruction group H^2 by the dual of an H^0 group, which is usually much easier to compute.

Example 15.1. — Let $\bar{\rho}$ act on a finite-dimensional k-vector space V. Then $\mathrm{ad}(\bar{\rho}) = \mathrm{End}_k(V)$, and the trace pairing identifies $\mathrm{ad}(\bar{\rho})^*$ with $\mathrm{ad}(\bar{\rho})$. Local Tate duality gives

$$H^2(G_F, \mathrm{ad}(\bar{\rho}))^\vee \cong H^0(G_F, \mathrm{ad}(\bar{\rho})(1)) \cong \mathrm{Hom}_{G_F}(\bar{\rho}, \bar{\rho}(1)).$$

Hence, if $\mathrm{Hom}_{G_F}(\bar{\rho}, \bar{\rho}(1)) = 0$, then $H^2(G_F, \mathrm{ad}(\bar{\rho})) = 0$. In that case the framed deformation functor is unobstructed; equivalently, the framed deformation ring is a formal power series ring in $r = \dim_k Z^1(G_F, \mathrm{ad}\,\bar{\rho})$ variables.

15.2 The C_1-Property of Unramified Fields. — We now return to the C_1-property for maximal unramified extensions.

Example 15.2. — Let $K = \mathbf{F}_p((t))$ and $L = K(t^{1/p})$. Then L is a reduced K-algebra, but $L \otimes_K \bar{K}$ is non-reduced. This example explains why purely inseparable phenomena must be kept separate from smoothness arguments.

Definition 15.3. — A finitely generated field extension K/k is separably generated if there exists a transcendence basis $t_1, \ldots, t_d \in K$ such that $K/k(t_1, \ldots, t_d)$ is a finite separable extension.

Proposition 15.4 **(Separating transcendence basis theorem)**

Let F be a field and let K/F be a finitely generated field extension. If K/F is separable, then K is separably generated over F.

Proof. — Since K/F is finitely generated and separable, one can choose algebraically independent elements $t_1, \ldots, t_d \in K$ such that K is finite separable over $F(t_1, \ldots, t_d)$. □

Corollary 15.5. — Let F be a field. If X is geometrically reduced and of finite type over F, then X^{sm} is open and dense in X.

Proof. — The smooth locus is open. It remains to see that it contains the generic point of each irreducible component. Geometric reducedness implies that the function field of each component is a separable finitely generated extension of F. By the proposition, these function fields are separably generated over F, which is equivalent to generic smoothness for varieties over a field. Thus every component has a nonempty smooth open subset, so X^{sm} is dense. □

Corollary 15.6. — If k is a finite field and X is geometrically reduced and of finite type over $k((t))^{\mathrm{ur}}$, then X^{sm} is open and dense in X.
Proof. — We have

$$k((t))^{\mathrm{ur}} = \bigcup_{k'/k \text{ finite}} k'((t)).$$

Since X is of finite type, after enlarging k' if necessary it descends to a geometrically reduced scheme X' of finite type over $k'((t))$. By the previous corollary, $(X')^{\mathrm{sm}}$ is open and dense in X'. Smoothness is stable under base change, so X^{sm} is open and dense in X. □

***Theorem 15.1* (Greenberg approximation).** — Let k be a perfect field and let

$$S = \{f_1, \dots, f_m\} \subset k[[t]][x_1, \dots, x_n]$$

be a finite system of polynomial equations. Then there is an integer $N_0(S) > 0$ such that, for every $N > N_0(S)$, the existence of a common solution of

$$f_i(x_1, \dots, x_n) \equiv 0 \pmod{t^N} \qquad (i = 1, \dots, m)$$

in $k[[t]]^n$ implies the existence of an actual common zero in $k[[t]]^n$ [**3**, Theorem 6.2.12, pp. 144–146].
***Theorem 15.2* (Lang).** — Let k be a perfect field. Then the maximal unramified extension $k((t))^{\mathrm{nr}}$ is a C_1-field. In particular, if k is algebraically closed, then $k((t))$ is a C_1-field [**3**, Theorem 6.2.11, pp. 144–145].
Proof. — We prove the first assertion. Let

$$f \in k((t))^{\mathrm{nr}}[x_1, \dots, x_n]$$

be homogeneous of degree $d < n$. After multiplying by a common denominator, we may assume that the coefficients of f lie in $k'[[t]]$ for some finite extension k'/k, because

$$k((t))^{\mathrm{nr}} = \bigcup_{k'/k \text{ finite}} k'((t)).$$

For each N, choose a polynomial approximation

$$f^{(N)} \in k'[t][x_1, \dots, x_n], \qquad f^{(N)} \equiv f \pmod{t^N}.$$

Let $\overline{k'}$ be an algebraic closure of k'. By Tsen's theorem, $\overline{k'}(t)$ is C_1, so $f^{(N)}$ has a nontrivial zero over $\overline{k'}(t)$. Since the zero involves only finitely many coefficients, after replacing k' by a finite extension inside $\overline{k'}$ we may assume that this zero is defined over $k'(t)$. By homogeneity, after clearing denominators and dividing by a common power of t, we may choose a zero

$$(a_1^{(N)}, \dots, a_n^{(N)}) \in k'[t]^n$$

such that at least one coordinate, say $a_j^{(N)}$, is a unit in $k'[[t]]$.

Reducing modulo t^N, we obtain

$$f(a_1^{(N)}, \dots, a_n^{(N)}) \equiv 0 \pmod{t^N}.$$

Choose $b^{(N)} \in k'[[t]]$ with

$$b^{(N)} a_j^{(N)} \equiv 1 \pmod{t^N}.$$

Apply Greenberg approximation over the perfect field k' to the system

$$f(x_1, \ldots, x_n) = 0, \qquad y x_j - 1 = 0.$$

For N larger than the Greenberg constant of this system, the congruence solution

$$(a_1^{(N)}, \ldots, a_n^{(N)}, b^{(N)}) \pmod{t^N}$$

lifts to an actual solution

$$(a_1, \ldots, a_n, b) \in k'[[t]]^{n+1}.$$

Then $f(a_1, \ldots, a_n) = 0$, and $b a_j = 1$, so the tuple $(a_1, \ldots, a_n)$ is nonzero. Hence f has a nontrivial zero in

$$k'((t))^n \subset k((t))^{\mathrm{nr}\,n}.$$

Thus $k((t))^{\mathrm{nr}}$ is C_1. If k is algebraically closed, then $k((t))^{\mathrm{nr}} = k((t))$, giving the final assertion. □

Remark 15.7 **(Greenberg approximation: structure of the proof)**

We indicate how the geometric facts above enter the proof of Greenberg's approximation theorem.

Step 1: Reduce to reduced irreducible pieces. Let

$$V = V(f_1, \ldots, f_m) \subset \mathbf{A}^n_{k((t))}.$$

One first replaces the ideal generated by the f_i by its radical and then separates the irreducible components. In positive characteristic, reducedness need not be preserved by arbitrary base change, as the preceding example shows; thus the argument keeps track of geometric reducedness when smoothness is used.

Step 2: Isolate the bad locus. Assume now that V is a variety of dimension d. For suitable subsets $I \subset \{1, \ldots, m\}$ with $|I| = n - d$, let V_I be the closed subset cut out by the equations in I. Besides V, the space V_I may have other d-dimensional components; call their union V_I^+. Also let $W \subset V$ be the singular locus. The separably-generated criterion and the smooth-locus corollaries give the needed dimension drop: the smooth locus is open and dense, hence W is a proper closed subset of V; similarly, $V \cap V_I^+$ is a proper closed subset of V. These closed subsets have dimension $< d$.

Step 3: Use induction on the bad locus. If an approximate solution modulo t^N also satisfies the equations defining W, then the induction hypothesis applied to W produces an actual solution. Similarly, if it lies approximately on some $V \cap V_I^+$, induction on that smaller-dimensional closed subset gives the desired actual solution.

Step 4: Work at a smooth complete-intersection point. After excluding the previous cases, the approximate point is near the smooth part of the complete intersection cut out by $n - d$ equations. At such a point a suitable Jacobian determinant has controlled

t-adic valuation. The refined Hensel–implicit-function theorem, Appendix Proposition A.5.6, then lifts the approximate point to an actual $k[[t]]$-valued point. This completes the induction on $\dim V$ [**3**, Proof of Theorem 6.2.12, pp. 145–146].

Corollary 15.8. — Let C be a smooth projective geometrically irreducible curve over a finite field $\mathbf{F}$, and let $K = \mathbf{F}(C)$. Then every central simple algebra over K is split by a finite cyclic extension of K; in particular, every Brauer class over K is represented by a cyclic algebra.

Sketch. — After base change to an algebraic closure $\overline{\mathbf{F}}$, the field $\overline{\mathbf{F}}(C)$ is the function field of a curve over an algebraically closed field. Hence it is C_1 by Tsen's theorem, and Proposition 13.3 gives

$$\mathrm{Br}(\overline{\mathbf{F}}(C)) = 0.$$

Thus every central simple algebra over $\mathbf{F}(C)$ is split after extending constants to some finite field $\mathbf{F}'/\mathbf{F}$. Finite extensions of finite fields are cyclic, so $\mathbf{F}'(C)/\mathbf{F}(C)$ is cyclic. The final statement follows from the cyclic-algebra description of relative Brauer classes. □

Lecture 17

This lecture begins the study of cohomological dimension. We define cd_p, prove useful equivalent criteria for $cd_p(G) \leq n$, and record how divisibility and primary decomposition enter vanishing arguments.

Theorem 16.1 **(Lang–Serre).** — *Let F be a non-archimedean local field. Then the maximal unramified extension F^{ur} is a C_1-field.*

Proof. — We use this as an external input in full generality. In equal characteristic, this is the theorem sketched above as Theorem 15.2. In mixed characteristic it is Lang's theorem on the C_1-property of maximal unramified extensions. For the form needed here, see Serre's discussion of fields of dimension at most one, Gille–Szamuely's central-simple algebra formulation, and Harari's Brauer-group application [**7**, Chap. II, §3.2]; [**3**, Chap. 6, §6.2]; [**4**, Example 6.20(c), pp. 92–93 and Chap. 8 introduction, p. 109]. □

Remark 16.2. — The tempting reduction-and-Hensel argument is not enough by itself. A nontrivial zero of the reduction $\bar{f}$ need not be a smooth zero, and Hensel lifting requires a smoothness condition. The inequality $\deg(f) < n$ is used in the deeper C_1 theorem to guarantee enough zeros; it is not merely a formal condition in the lifting step. For example, $x^2 - py^2$ has no nontrivial zero in $\mathbf{Q}_p^{\mathrm{ur}}$, since otherwise $\sqrt{p}$ would lie in the unramified extension.

Corollary 16.3. — Let F be a non-archimedean local field. For every algebraic extension K/F^{ur} and every prime $\ell \neq \mathrm{char}(F)$, one has

$$cd_\ell(G_K) \leq 1.$$

In particular, if $I_F = G_{F^{\mathrm{ur}}} = \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{ur}})$, then $cd_\ell(I_F) \leq 1$ for every $\ell \neq \mathrm{char}(F)$.

Proof. — The C_1 property is stable under algebraic extensions, so every algebraic extension of F^{ur} is again C_1. By Serre's theorem, a C_1-field has dimension at most 1; equivalently, for every prime ℓ different from the characteristic, the absolute Galois group has ℓ-cohomological dimension at most 1. □

16.1 Cohomological Dimension. —

Definition 16.4. — Let G be a profinite group and p a prime. The p-cohomological dimension of G, denoted $cd_p(G)$, is the smallest integer n such that

$$H^q(G, A) = 0$$

for every $q > n$ and every discrete p-primary torsion G-module A. If no such n exists, we set $cd_p(G) = \infty$.

Remark 16.5. — Equivalently, for every discrete torsion G-module A, the p-primary part of $H^q(G, A)$ is zero for all $q > n$. This follows from the primary decomposition of torsion groups and from the compatibility of continuous cohomology with filtered colimits.

Proposition 16.6. — *The following are equivalent:*

1. *$cd_p(G) \leq n$;*
2. *$H^q(G, A) = 0$ for all $q > n$ and all discrete p-primary torsion G-modules A;*
3. *$H^{n+1}(G, A) = 0$ for every simple discrete G-module A killed by p.*

Proof. — The equivalence of (1) and (2) is just the definition. Also, (2) implies (3). Assume (3). If A is a finite p-primary discrete G-module, then A admits a finite composition series whose successive quotients are simple modules killed by p. The long exact cohomology sequence gives $H^{n+1}(G, A) = 0$ by induction on the length of A.

For a general p-primary torsion module A, every element is contained in a finite G-submodule: the action is continuous and A is discrete, so each orbit is finite. Hence A is the filtered union of its finite p-primary G-submodules. Since continuous cohomology commutes with filtered direct limits (Theorem 4.5), we obtain $H^{n+1}(G, A) = 0$.

It remains to get vanishing in all degrees $q > n + 1$. Embed A into the standard induced module $M_G(A)$:

$$0 \to A \to M_G(A) \to M_G(A)/A \to 0.$$

Here $M_G(A)$ is again p-primary, because every continuous function $G \to A$ is locally constant and therefore has finite image. Since $H^q(G, M_G(A)) = 0$ for $q \geq 1$, the long exact sequence gives

$$H^{q+1}(G, A) \simeq H^q(G, M_G(A)/A).$$

Dimension shifting, starting from the already proved vanishing in degree $n + 1$, then gives

$$H^q(G, A) = 0 \quad \text{for all } q > n.$$

□

Definition 16.7. — An abelian group A is p-divisible if multiplication by p is surjective.

Proposition 16.8. — *If $cd_p(G) \leq n$ and A is a discrete p-divisible G-module, then*

$$H^q(G,A)[p^\infty] = 0 \quad \textit{for all } q > n.$$

Proof. — From the exact sequence

$$0 \to A[p] \to A \xrightarrow{\cdot p} A \to 0,$$

we get a long exact sequence in cohomology. Since $A[p]$ is p-torsion, we have $H^q(G, A[p]) = 0$ for $q > n$. Hence, multiplication by p is injective on $H^q(G,A)$ for $q > n$, which implies the p-primary part is zero. □

Corollary 16.9. — If $cd_p(G) \leq n$ for every prime p, and $A \in \mathscr{C}_G$ is divisible, then $H^q(G,A) = 0$ for $q > n$.

Proof. — The uniquely divisible part of A is a $\mathbf{Q}$-vector space, and its higher continuous cohomology vanishes by the usual averaging argument at finite level and passage to the limit. The torsion divisible part is a direct sum of p-divisible groups, so the preceding proposition kills each primary component of $H^q(G,A)$ for $q > n$. □

Definition 16.10. — The *strict p-cohomological dimension* of G, denoted $scd_p(G)$, is the smallest integer n such that

$$H^q(G,A)[p^\infty] = 0$$

for every discrete G-module A and every $q > n$. Thus strict cohomological dimension removes the torsion hypothesis on the coefficients.

Proposition 16.11. —

$$cd_p(G) \leq scd_p(G) \leq cd_p(G) + 1.$$

Proof. — The inequality $cd_p(G) \leq scd_p(G)$ is immediate. We prove $scd_p(G) \leq cd_p(G) + 1$: let A be a discrete G-module. Consider the decomposition of the multiplication-by-p map $A \xrightarrow{\cdot p} A$, which yields two exact sequences:

$$0 \to N \to A \to I \to 0, \qquad 0 \to I \to A \to Q \to 0,$$

where

$$N = A[p], \quad I = pA, \quad Q = A/pA.$$

Note that N and Q are p-primary torsion G-modules.

Let $q > cd_p(G) + 1$. Since N and Q are p-primary torsion, the definition of $cd_p(G)$ implies: $H^q(G,N) = 0$, $H^{q-1}(G,Q) = 0$.

From the long exact cohomology sequences associated with the above short exact sequences, we deduce:

$$H^q(G,A) \hookrightarrow H^q(G,I), \qquad H^q(G,I) \hookrightarrow H^q(G,A).$$

Hence multiplication by p:

$$H^q(G,A) \xrightarrow{\cdot p} H^q(G,A)$$

is injective. Therefore, the p-primary component of $H^q(G,A)$ is zero. $\square$

Example 16.12. — Take $G = \widehat{\mathbf{Z}}$. One has $cd_p(G) = 1$ for every p. On the other hand,

$$H^2(G,\mathbf{Z}) \cong H^1(G,\mathbf{Q}/\mathbf{Z}) = \mathbf{Q}/\mathbf{Z},$$

whence $scd_p(G) = 2$.

Proposition 16.13. — *Let G be a profinite group of cohomological dimension n. Then the strict cohomological dimension of G is n if and only if for every open subgroup U of G, we have $H^{n+1}(U,\mathbf{Z}) = 0$.*

Proof. — This is Serre's criterion as stated by Harari [**4**, Proposition 5.14, p. 83]. The necessity is immediate: if $scd(G) = n$, then every open subgroup U also has strict cohomological dimension at most n, so $H^{n+1}(U,\mathbf{Z}) = 0$.

Conversely, assume that $H^{n+1}(U,\mathbf{Z}) = 0$ for every open subgroup $U \subset G$. Since $cd(G) = n$, Proposition 16.11 gives $scd(G) \leq n+1$; hence the only remaining degree to kill is $n+1$. By Shapiro's lemma, for every open normal subgroup U and every $r \geq 0$,

$$H^{n+1}\big(G, C(G/U,\mathbf{Z}^r)\big) \cong H^{n+1}(U,\mathbf{Z}^r) = 0.$$

Here $C(G/U,\mathbf{Z}^r)$ is the coinduced module from U to G; because U is open and normal, it is just the finite-rank permutation module $\mathbf{Z}[G/U]^r$ with the natural G-action. Let M be a finitely generated discrete G-module. Then some open normal subgroup U acts trivially on M, and there is an exact sequence of G-modules

$$0 \to B \to \mathbf{Z}[G/U]^r \to M \to 0.$$

The long exact cohomology sequence gives $H^{n+1}(G,M) = 0$, because $H^{n+1}(G,\mathbf{Z}[G/U]^r) = 0$ by the preceding paragraph and $H^{n+2}(G,B) = 0$ by $scd(G) \leq n+1$. Finally, every discrete G-module is a filtered union of finitely generated discrete submodules, and continuous cohomology commutes with such filtered colimits. Hence $H^{n+1}(G,A) = 0$ for every discrete G-module A. Therefore $scd(G) \leq n$, and since $cd(G) = n \leq scd(G)$, we get $scd(G) = n$. $\square$

Lecture 18

This lecture continues cohomological dimension with restriction to subgroups and reduction to p-Sylow subgroups. Next we study the cohomological dimensions of subgroups and extensions:

Proposition 17.1. — *Let H be a closed subgroup of a profinite group G. Then:*

$$cd_p(H) \leq cd_p(G), \qquad scd_p(H) \leq scd_p(G).$$

Moreover, equality holds in the following cases:

1. *If $(G : H)$ is prime to p;*
2. *If H is open in G and $cd_p(G) < \infty$.*

Proof. — We prove the first two statements for cd_p; the proof for scd_p is analogous.

Step 1: $cd_p(H) \leq cd_p(G)$. Let A be a discrete p-primary torsion H-module. Then $\mathrm{Ind}_H^G A$ is a discrete p-primary torsion G-module, and by Shapiro's lemma 28.14,

$$H^q(G, \mathrm{Ind}_H^G A) \cong H^q(H, A).$$

If $q > cd_p(G)$, the left-hand side is zero. Hence $cd_p(H) \leq cd_p(G)$.

Step 2: equality when $(G : H)$ is prime to p. By Corollary 28.23, the restriction map $\mathrm{Res} : H^q(G, A) \to H^q(H, A)$ is injective on the p-primary component when $(G : H)$ is prime to p. If $H^q(H, A)(p) = 0$, then $H^q(G, A)(p) = 0$, hence $cd_p(G) \leq cd_p(H)$.

The equality for open H under the hypothesis $cd_p(G) < \infty$ is the standard invariance of finite p-cohomological dimension under passage to open subgroups. □

The supernatural order of a profinite group is denoted $|G| = |G : 1|$. If G is a pro-p group, then $|G| = p^\nu$, where $\nu = \infty$ exactly when G is infinite.

***Definition 17.2* (p-Sylow Group).** — A closed subgroup $P \subset G$ is called a p-Sylow subgroup if P is a pro-p group and the supernatural index $[G : P]$ is prime to p. Equivalently, P is as large as possible among closed pro-p subgroups of G [**4**, Definition 4.8, p. 68].

Proposition 17.3. — *Let G be profinite and p be a prime. Then*

1. *p-Sylow subgroups exist.*
2. *Any closed pro-p subgroup of G is contained in a p-Sylow subgroup.*
3. *Any two p-Sylow subgroups are conjugate.*

Proof. — We recall Harari's proof [**4**, Proposition 4.10, pp. 68–69]. Let $\mathscr{D}$ be the directed set of open normal subgroups of G. For each $U \in \mathscr{D}$, the finite group G/U has a nonempty finite set of classical p-Sylow subgroups. These finite sets are compatible under the transition maps $G/V \to G/U$ for $V \subset U$, and Lemma 4.9 of Harari says that the inverse limit of such nonempty finite sets is nonempty. Thus one can choose a compatible system $(P_U)_U$ of p-Sylow subgroups of the finite groups G/U. Then

$$P := \varprojlim_U P_U \subset G$$

is a closed pro-p subgroup, and $[G : P]$ is prime to p, so P is a p-Sylow subgroup of G.

For conjugacy, let P and Q be two p-Sylow subgroups of G. For each U, the finite set of elements $\overline{g} \in G/U$ satisfying $\overline{g} P_U \overline{g}^{-1} = Q_U$ is nonempty by the classical Sylow theorem. Taking the inverse limit gives an element $g \in G$ with $gPg^{-1} = Q$.

Finally, if $H \subset G$ is a closed pro-p subgroup, then for each U the image $H_U \subset G/U$ is contained in some classical p-Sylow subgroup of G/U. The inverse limit of the nonempty finite sets of such Sylow subgroups gives a compatible system whose inverse limit is a p-Sylow subgroup of G containing H. □

Corollary 17.4. — Let G_p be a p-Sylow subgroup of G. Then

$$cd_p(G) = cd_p(G_p) = cd(G_p), \qquad scd_p(G) = scd_p(G_p) = scd(G_p).$$

Proof. — Since $[G : G_p]$ is prime to p, Proposition 17.1 gives $cd_p(G) = cd_p(G_p)$ and $scd_p(G) = scd_p(G_p)$. For a pro-p group, cohomology with coefficients of order prime to p has no positive-degree obstruction, so the ordinary dimension is detected on p-primary torsion modules. Hence $cd_p(G_p) = cd(G_p)$ and $scd_p(G_p) = scd(G_p)$. □

Remark 17.5 **(Local-field application).** — Let F be a non-archimedean local field with residue characteristic p, let F^{tr} be the maximal tamely ramified extension, and put

$$P_F = \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{tr}}).$$

Then P_F is the wild inertia group and is pro-p. Therefore Corollary 17.4 applied to the profinite group P_F says

$$cd_p(P_F) = cd(P_F), \qquad scd_p(P_F) = scd(P_F).$$

Moreover, if A is a finite discrete P_F-module of order prime to p, then $H^q(P_F, A) = 0$ for $q > 0$: every finite quotient of P_F is a finite p-group, and multiplication by the order of such a quotient is invertible on A. This is the input used later in the Hochschild–Serre sequence for

$$1 \to P_F \to G_F \to G_F^{\mathrm{tr}} \to 1,$$

where it allows one to replace G_F by the tame quotient G_F^{tr} for prime-to-p coefficient modules [**7**, Chap. II, §5]; [**4**, Chap. 7, §7.4].

Example 17.6. — Harari's example is the following [**4**, Example 5.12, p. 83]. For $G = \widehat{\mathbf{Z}}$ and a prime p, the p-Sylow is $\mathbf{Z}_p$, hence Corollary 17.4 gives

$$cd_p(\widehat{\mathbf{Z}}) = cd_p(\mathbf{Z}_p) = 1.$$

On the other hand

$$H^2(\mathbf{Z}_p, \mathbf{Z}) \cong H^1(\mathbf{Z}_p, \mathbf{Q}/\mathbf{Z}) \cong \mathrm{Hom}_c(\mathbf{Z}_p, \mathbf{Q}/\mathbf{Z}) \cong \mathbf{Q}_p/\mathbf{Z}_p \neq 0,$$

where the first isomorphism comes from $0 \to \mathbf{Z} \to \mathbf{Q} \to \mathbf{Q}/\mathbf{Z} \to 0$ and the fact that $\mathbf{Q}$ is uniquely divisible. Thus $scd_p(\widehat{\mathbf{Z}}) = scd_p(\mathbf{Z}_p) = 2$. Later this is contrasted with the absolute Galois group of a p-adic field, for which $cd = scd = 2$.

17.1 p-torsion modules for pro-p-groups. —

Lemma 17.7. — *Let G be a pro-p group and $A \neq 0$ a simple discrete G-module such that $A = A[p^\infty]$. Then $A \simeq \mathbf{F}_p$ with trivial G-action.*

Proof. — Since A is p-primary torsion and simple, the subgroup $A[p]$ is a nonzero G-stable submodule; hence $A = A[p]$. Thus A is naturally a $\mathbf{F}_p$-vector space.

Let $v \in A$, $v \neq 0$. Since A is discrete, its stabilizer

$$G_v = \{g \in G \mid g \cdot v = v\}$$

is open in G. Hence the orbit $G \cdot v$ is finite.

Let $W = \langle G \cdot v \rangle_{\mathbf{F}_p}$. Then W is a finite-dimensional G-stable subspace of A, hence by simplicity $W = A$. Thus A is finite-dimensional over $\mathbf{F}_p$.

The action of G factors through a finite quotient, so we may assume G is a finite p-group.

Consider the action of G on $A \setminus \{0\}$. If $d = \dim_{\mathbf{F}_p}(A)$, then $|A \setminus \{0\}| = p^d - 1$, which is not divisible by p.

Each orbit has size dividing $|G|$, hence is a power of p. Therefore, any nontrivial orbit has size divisible by p. Since the total number of elements is not divisible by p, there must exist an orbit of size 1. Thus there exists $v \neq 0$ such that

$$g \cdot v = v \quad \text{for all } g \in G.$$

Then $\mathbf{F}_p v$ is a nonzero G-stable subspace of A, so by simplicity $A = \mathbf{F}_p v$. Hence $A \simeq \mathbf{F}_p$ and the action is trivial. □

Corollary 17.8. — Let G be a pro-p group. If $H^{n+1}(G, \mathbf{F}_p) = 0$, then $cd_p(G) \leq n$.

Proof. — Let M be a discrete p-primary torsion G-module. We must show $H^{n+1}(G, M) = 0$.

First assume M has finite length. Then M admits a finite filtration whose successive quotients are simple G-modules. By the previous lemma, these are all isomorphic to $\mathbf{F}_p$ with trivial action. The result follows by induction on the length of M using the long exact sequence in cohomology.

For general M, write $M = \varinjlim M_i$, where each M_i has finite length. Since the cohomology of discrete modules commutes with direct limits (lemma 4.5), we get

$$H^{n+1}(G, M) = \varinjlim H^{n+1}(G, M_i) = 0.$$

Thus $cd_p(G) \leq n$. □

Theorem 17.9. — *Let k be a field with $p \neq \mathrm{char}(k)$ and let $n \geq 0$. The following are equivalent:*

1. *$cd_p(G_k) \leq n$;*
2. *for every algebraic extension K/k,*

$$H^{n+1}(G_K, \mu_p) = 0;$$

3. *the same vanishing holds for every finite separable extension K/k with $p \nmid [K : k]$.*

Proof. — This is Serre's criterion for p-cohomological dimension [**7**, Chap. II, §3.3]. The implication (1) ⇒ (2) is immediate because μ_p is a discrete p-primary module, and (2) ⇒ (3) is clear.

Assume (3). Let $G = G_k$ and let H be a p-Sylow subgroup of G. Set $K = (k^{\mathrm{sep}})^H$, so that $G_K = H$.

Write $K = \bigcup_i K_i$, where

$$K_i = (k^{\mathrm{sep}})^{HU_i}$$

and U_i runs through the open normal subgroups of G. Then each K_i/k is finite separable and

$$[K_i : k] = [G : HU_i]$$

is prime to p, because H maps to a p-Sylow subgroup of the finite group G/U_i. By (3),

$$H^{n+1}(G_{K_i}, \mu_p) = 0.$$

Since $K = \bigcup_i K_i$, continuous cohomology with finite coefficients commutes with this filtered limit of fields, and hence

$$H^{n+1}(H, \mu_p) = \varinjlim_i H^{n+1}(G_{K_i}, \mu_p) = 0.$$

Since H is a pro-p group and μ_p has order p, by the lemma the action of H on μ_p is trivial, so $\mu_p \simeq \mathbf{F}_p$ as H-modules. Therefore

$$H^{n+1}(H, \mathbf{F}_p) = 0.$$

By Corollary 17.8, $cd_p(H) \le n$. Finally, since H is a p-Sylow subgroup of G, Corollary 17.4 gives

$$cd_p(G) = cd_p(H),$$

hence $cd_p(G) \le n$. □

Lecture 19

This lecture applies the preceding cohomological-dimension criteria to local fields. We use exact sequences of Galois groups, the maximal unramified extension, and Hochschild–Serre to obtain the dimension bounds needed for later local duality use.

18.1 Cohomological dimension of G_F for local fields. — First we record the extension inequality for cohomological dimension.

Proposition 18.1. — *Let*

$$1 \to N \to G \to Q \to 1$$

be an exact sequence of profinite groups. Then:

$$cd_p(G) \le cd_p(N) + cd_p(Q).$$

Proof. — Let A be any discrete p-primary G-module. We use the Hochschild–Serre spectral sequence:

$$E_2^{i,j} = H^i(Q, H^j(N, A)) \;\Rightarrow\; H^{i+j}(G, A).$$

By definition of $cd_p(N)$ and $cd_p(Q)$, we have:

$$H^j(N, A) = 0 \quad \text{for } j > cd_p(N),$$

and

$$H^i(Q, M) = 0 \quad \text{for all } p\text{-primary } Q\text{-modules } M \text{ and } i > cd_p(Q).$$

Thus $E_2^{i,j} = 0$ whenever $i > cd_p(Q)$ or $j > cd_p(N)$.

Fix $n > cd_p(N) + cd_p(Q)$. For any $i + j = n$, either $i > cd_p(Q)$ or $j > cd_p(N)$, hence $E_2^{i,j} = 0$. It follows that all terms on the diagonal $i + j = n$ vanish at the E_2-page, hence also at the abutment. Therefore $H^n(G, A) = 0$. Since this holds for all A, we obtain $cd_p(G) \leq cd_p(N) + cd_p(Q)$. $\square$

Let F be a non-archimedean local field and let $\ell \neq \mathrm{char}(F)$ be a prime. We now compute $cd_\ell(G_F)$ from the C_1-property of the maximal unramified extension and the extension inequality for cohomological dimension.

Proposition 18.2. — *Let F be a non-archimedean local field and let $\ell \neq \mathrm{char}(F)$. Then*

$$cd_\ell(G_F) = 2.$$

Proof. — This is the non-circular Serre argument [**7**, Chap. I, §3.3 and Chap. II, §3.2–3.3]. Let F^{ur} be the maximal unramified extension and put $I_F = G_{F^{\mathrm{ur}}} = \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{ur}})$. By Corollary 16.3,

$$cd_\ell(I_F) = cd_\ell(G_{F^{\mathrm{ur}}}) \leq 1.$$

The quotient is the Galois group of the residue field:

$$1 \longrightarrow I_F \longrightarrow G_F \longrightarrow \mathrm{Gal}(\bar{k}_F/k_F) \longrightarrow 1,$$

and $\mathrm{Gal}(\bar{k}_F/k_F) \cong \widehat{\mathbf{Z}}$, so $cd_\ell(\widehat{\mathbf{Z}}) = 1$. Applying the extension inequality, Proposition 18.1, gives

$$cd_\ell(G_F) \leq cd_\ell(I_F) + cd_\ell(\widehat{\mathbf{Z}}) \leq 2.$$

For the opposite inequality, use Kummer theory. Since $\ell \neq \mathrm{char}(F)$,

$$H^2(G_F, \mu_\ell) \cong \mathrm{Br}(F)[\ell].$$

The local invariant isomorphism $\mathrm{Br}(F) \cong \mathbf{Q}/\mathbf{Z}$ constructed from the valuation sequence in Lecture 8 gives $\mathrm{Br}(F)[\ell] \cong \mathbf{Z}/\ell\mathbf{Z} \neq 0$. Hence $H^2(G_F, \mu_\ell) \neq 0$, so $cd_\ell(G_F) \geq 2$. Therefore $cd_\ell(G_F) = 2$. $\square$

18.2 Euler–Poincaré characteristic formula. — Let $\mathscr{A}$ be the category of finite abelian groups. Define $K_0(\mathscr{A})$ as the abelian group generated by symbols $[A]$ for $A \in \mathscr{A}$, subject to the relations

$$[B] = [A] + [C] \quad \text{for every short exact sequence } 0 \to A \to B \to C \to 0.$$

Proposition 18.3. — *There is a canonical isomorphism*

$$K_0(\mathscr{A}) \cong \bigoplus_p \mathbf{Z}\,[\mathbf{Z}/p\mathbf{Z}],$$

where the sum runs over all primes.

Proof. — Every finite abelian group admits a composition series whose simple quotients are of the form $\mathbf{Z}/p\mathbf{Z}$. Hence the classes $[\mathbf{Z}/p\mathbf{Z}]$ generate $K_0(\mathscr{A})$. For a finite p-group A,

$$[A] = (\text{length of } A) \cdot [\mathbf{Z}/p\mathbf{Z}],$$

and this length is $\log_p |A|$. The primary decomposition of finite abelian groups therefore gives the displayed isomorphism. □

Let G be a profinite group and A a finite discrete G-module. Assume:

- $H^i(G, A)$ is finite for all i;
- $H^i(G, A) = 0$ for $i \gg 0$.

Definition 18.4. — The Euler–Poincaré characteristic is

$$\chi(G, A) := \sum_{i \geq 0} (-1)^i [H^i(G, A)] \in K_0(\mathscr{A}).$$

Proposition 18.5. — *Let F be a non-archimedean local field with residue characteristic p, and let A be a finite G_F-module whose order is prime to p. Then*

$$\chi(G_F, A) = 0, \qquad \textit{i.e.} \qquad [H^0(G_F, A)] - [H^1(G_F, A)] + [H^2(G_F, A)] = 0$$

in $K_0(\mathscr{A})$. Equivalently,

$$|H^1(G_F, A)| = |H^0(G_F, A)|\, |H^2(G_F, A)|.$$

Proof. — This is Harari's Proposition 10.12 in additive notation [**4**, Proposition 10.12, pp. 137–138]. We recall the argument because it is exactly the one used below. Let

$$I_F = \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{ur}})$$

be the inertia subgroup. Since $|A|$ is prime to p, wild inertia has no higher cohomology on A, and Harari's Lemma 10.13 gives

$$H^i(I_F, A) = 0 \qquad (i \geq 2),$$

with $H^0(I_F, A)$ and $H^1(I_F, A)$ finite. The quotient

$$G_F/I_F \simeq \mathrm{Gal}(\bar{k}_F/k_F) \simeq \widehat{\mathbf{Z}}$$

has cohomological dimension 1. The Hochschild–Serre spectral sequence gives

$$H^0(G_F, A) = H^0(\widehat{\mathbf{Z}}, H^0(I_F, A)),$$

$$H^2(G_F, A) = H^1(\widehat{\mathbf{Z}}, H^1(I_F, A)),$$

and an exact sequence

$$0 \to H^1(\widehat{\mathbf{Z}}, H^0(I_F, A)) \to H^1(G_F, A) \to H^0(\widehat{\mathbf{Z}}, H^1(I_F, A)) \to 0.$$

For every finite $\widehat{\mathbf{Z}}$-module M, Lemma 10.14 of Harari gives

$$|H^0(\widehat{\mathbf{Z}}, M)| = |H^1(\widehat{\mathbf{Z}}, M)|.$$

Applying this to $M = H^0(I_F, A)$ and $M = H^1(I_F, A)$ gives the displayed Euler characteristic formula on each primary component, hence the equality in $K_0(\mathscr{A})$. $\square$

Remark 18.6. — For a p-adic field $F/\mathbf{Q}_p$, Tate's full Euler–Poincaré formula is

$$\frac{|H^0(G_F, A)|\,|H^2(G_F, A)|}{|H^1(G_F, A)|} = |A[p^\infty]|^{-[F:\mathbf{Q}_p]}.$$

Thus the proposition above is the prime-to-p special case. The full formula is stated in Serre [**7**, Chap. II, §5.7, Theorem 5]; Harari states it in [**4**, Chap. 10, §10.3, p. 137] and then proves the prime-to-p case as Proposition 10.12.

Example 18.7. — Let $G = \widehat{\mathbf{Z}}$ and let A be a finite discrete G-module. For every prime ℓ, one has $\mathrm{cd}_\ell(G) = 1$, hence

$$H^i(G, A) = 0 \qquad (i \geq 2).$$

Let φ be a topological generator of G. Then

$$H^0(G, A) = A^G = \ker(\varphi - 1), \qquad H^1(G, A) = A/(\varphi - 1)A.$$

Since A is finite, we have:

$$|A| = |\ker(\varphi - 1)| \cdot |(\varphi - 1)A|, \quad |A/(\varphi - 1)A| = \frac{|A|}{|(\varphi - 1)A|}.$$

Combining these gives:

$$|H^0(G, A)| = |H^1(G, A)|, \quad \chi(G, A) = 0.$$

Proposition 18.8. — *Let*

$$1 \longrightarrow N \longrightarrow G \longrightarrow \Gamma := G/N \longrightarrow 1$$

be an exact sequence of groups, where N and Γ are cyclic/procyclic and $\mathrm{cd}_p(N) = \mathrm{cd}_p(\Gamma) = 1$. *Then for every finite discrete p-primary G-module A,*

$$[H^0(G, A)] - [H^1(G, A)] + [H^2(G, A)] = 0 \qquad \textit{in } K_0(\mathscr{A}).$$

Proof. — Let $\Gamma = G/N$. The Hochschild–Serre spectral sequence

$$E_2^{i,j} = H^i(\Gamma, H^j(N, A)) \;\Rightarrow\; H^{i+j}(G, A)$$

degenerates in low degrees because $\mathrm{cd}_p(N) = \mathrm{cd}_p(\Gamma) = 1$. Hence we get an exact sequence

$$0 \longrightarrow H^1(\Gamma, A^N) \longrightarrow H^1(G, A) \longrightarrow H^1(N, A)^\Gamma \longrightarrow 0$$

and an identification

$$H^2(G, A) \cong H^1(\Gamma, H^1(N, A)).$$

Also

$$H^0(G, A) = (A^N)^\Gamma = H^0(\Gamma, A^N).$$

Now let $C = \langle \sigma \rangle$ be cyclic/procyclic and M a finite p-primary C-module. Since $\mathrm{cd}_p(C) = 1$,

$$H^0(C, M) = M^C = \ker(\sigma - 1), \qquad H^1(C, M) = M/(\sigma - 1)M.$$

Because M is finite, $|M^C| = |M/(\sigma - 1)M|$, hence

$$[H^0(C, M)] = [H^1(C, M)] \quad \text{in } K_0(\mathscr{A}),$$

because both groups are p-primary. Apply this to $M = A^N$ and $M = H^1(N, A)$. Then

$$[H^1(\Gamma, A^N)] = [H^0(\Gamma, A^N)] = [H^0(G, A)]$$

$$[H^1(N, A)^\Gamma] = [H^1(\Gamma, H^1(N, A))] = [H^2(G, A)].$$

Taking classes in the short exact sequence for $H^1(G, A)$ gives

$$[H^1(G, A)] = [H^0(G, A)] + [H^2(G, A)].$$

□

18.3 The maximal tamely ramified extension of a local field. — Let F be a non-archimedean local field, with residue characteristic p, residue field k_F, and separable closure F^{sep}.

Definition 18.9. — The *maximal unramified extension* $F^{\mathrm{ur}} \subset F^{\mathrm{sep}}$ is the compositum of all finite unramified extensions of F.

The *maximal tamely ramified extension* $F^{\mathrm{tr}} \subset F^{\mathrm{sep}}$ is the compositum of all finite tamely ramified extensions of F.

Recall that a finite extension E/F is *tamely ramified* if its ramification index $e(E/F)$ is prime to p. Set

$$G_F := \mathrm{Gal}(F^{\mathrm{sep}}/F), \qquad G_F^{\mathrm{tr}} := \mathrm{Gal}(F^{\mathrm{tr}}/F), \qquad P_F := \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{tr}}).$$

Then P_F is the wild inertia subgroup of G_F.

Proposition 18.10 **(Structure of the tame Galois group)**

There is a canonical exact sequence

$$1 \longrightarrow I_F^{\mathrm{t}} \longrightarrow G_F^{\mathrm{tr}} \longrightarrow \mathrm{Gal}(F^{\mathrm{ur}}/F) \longrightarrow 1,$$

where

$$I_F^{\mathrm{t}} := \mathrm{Gal}(F^{\mathrm{tr}}/F^{\mathrm{ur}}) \cong \prod_{\ell \neq p} \mathbf{Z}_\ell(1), \quad \mathrm{Gal}(F^{\mathrm{ur}}/F) \cong \widehat{\mathbf{Z}}.$$

Hence

$$G_F^{\mathrm{tr}} \cong \left(\prod_{\ell \neq p} \mathbf{Z}_\ell(1) \right) \rtimes \widehat{\mathbf{Z}}.$$

Proof. — Every finite tamely ramified extension becomes, after an unramified extension, obtained by adjoining an e-th root of a uniformizer, with $(e,p)=1$. Therefore the tame inertia is the inverse limit of the cyclic groups of roots of unity of order prime to p, which gives

$$I_F^{\mathrm{t}} \cong \prod_{\ell \neq p} \mathbf{Z}_\ell(1).$$

The quotient by inertia is the Galois group of the maximal unramified extension, which is topologically generated by the arithmetic Frobenius, so it is $\widehat{\mathbf{Z}}$.

If Frob denotes arithmetic Frobenius and ζ_n is a primitive n-th root of unity with $(n,p)=1$, then

$$\mathrm{Frob}(\zeta_n) = \zeta_n^q,$$

where $q = \#k_F$. This gives the semidirect product structure. □

Remark 18.11. — The prime-to-p Euler characteristic formula in Proposition 18.5 can also be read from this tame description. If $|A|$ is prime to p, then wild inertia $P_F = \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{tr}})$ has no higher cohomology on A. Hochschild–Serre therefore reduces the cohomology of G_F with coefficients in A to the tame quotient

$$G_F^{\mathrm{tr}} \cong \left(\prod_{\ell \neq p} \mathbf{Z}_\ell(1)\right) \rtimes \widehat{\mathbf{Z}}.$$

Both factors have cohomological dimension 1, then apply Proposition 18.1.

Lecture 20

This lecture records the structure of unramified, tamely ramified, and wildly ramified extensions. The goal is to make explicit the inertia subgroup, wild inertia, and tame quotient that were used in the cohomological-dimension arguments. We will show that the wild inertia group is a free pro-p-group. Let F be a non-archimedean local field, with residue characteristic p, residue field k_F, and separable closure F^{sep}. Fix a uniformizer $\pi \in F$, and write $q = \#k_F$. Let

$$G_F := \mathrm{Gal}(F^{\mathrm{sep}}/F), \qquad I_F := \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{ur}}), \qquad P_F := \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{tr}}),$$

where $F^{\mathrm{ur}} \subset F^{\mathrm{sep}}$ is the maximal unramified extension and $F^{\mathrm{tr}} \subset F^{\mathrm{sep}}$ is the maximal tamely ramified extension. Finally put

$$I_F^{\mathrm{t}} := \mathrm{Gal}(F^{\mathrm{tr}}/F^{\mathrm{ur}}) = I_F/P_F.$$

We first describe F^{tr} and the tame quotient explicitly.

For every integer $n \geq 1$ with $(n,p)=1$, choose a primitive n-th root of unity $\xi_n \in F^{\mathrm{sep}}$, compatibly in the sense that whenever $d \mid n$ one has $\xi_n^{n/d} = \xi_d$.

Lemma 19.1. — *The maximal unramified extension F^{ur} contains all roots of unity of order prime to p.*

Proof. — The residue field of F^{ur} is an algebraic closure of k_F, hence contains all roots of unity of order prime to p. By Teichmüller lifting, these roots of unity lift uniquely to F^{ur}. □

Choose a compatible system $\{\pi_n\}_{(n,p)=1} \subset F^{\mathrm{sep}}$ such that

$$\pi_n^n = \pi, \qquad \pi_{mn}^m = \pi_n \quad (m, n \geq 1,\ (mn, p) = 1).$$

Define

$$L := \bigcup_{(n,p)=1} F^{\mathrm{ur}}(\pi_n).$$

Lemma 19.2. — *One has $L = F^{\mathrm{tr}}$.*

Proof. — For each $(n,p) = 1$, the extension $F^{\mathrm{ur}}(\pi_n)/F^{\mathrm{ur}}$ is totally ramified of degree n, hence tame. Therefore $L \subseteq F^{\mathrm{tr}}$.

Conversely, let E/F be a finite tamely ramified extension. After a finite unramified extension of F, the extension E/F becomes totally ramified of degree prime to p, hence is generated by adjoining an n-th root of a uniformizer. Since F^{ur} contains all prime-to-p roots of unity, every finite tame extension embeds into L. Thus $F^{\mathrm{tr}} = L$. □

Lemma 19.3. — *There is a canonical isomorphism*

$$\mathrm{Gal}(F^{\mathrm{tr}}/F^{\mathrm{ur}}) \cong \varprojlim_{(n,p)=1} \mu_n \cong \prod_{\ell \neq p} \mathbf{Z}_\ell(1).$$

Proof. — An automorphism of $F^{\mathrm{ur}}(\pi_n)$ over F^{ur} is uniquely determined by the image of π_n, and must be of the form

$$\pi_n \longmapsto \xi_n^a \pi_n \qquad (a \in \mathbf{Z}/n\mathbf{Z}).$$

Hence

$$\mathrm{Gal}(F^{\mathrm{ur}}(\pi_n)/F^{\mathrm{ur}}) \cong \mu_n.$$

Passing to the inverse limit over $(n,p) = 1$ gives the result. □

Let τ be the element of tame inertia characterized by

$$\tau(\pi_n) = \xi_n \pi_n \qquad \text{for all } (n,p) = 1.$$

Then τ is a topological generator of $\mathrm{Gal}(F^{\mathrm{tr}}/F^{\mathrm{ur}})$.

Let $\mathrm{Frob} \in \mathrm{Gal}(F^{\mathrm{tr}}/F)$ be the lift of arithmetic Frobenius which fixes the chosen compatible system $\{\pi_n\}$.

Lemma 19.4. — *For every $(n,p) = 1$,*

$$\mathrm{Frob}(\xi_n) = \xi_n^q.$$

Consequently,

$$\mathrm{Frob}\, \tau\, \mathrm{Frob}^{-1} = \tau^q.$$

Proof. — The arithmetic Frobenius acts on the residue field by $x \mapsto x^q$, so it acts on prime-to-p roots of unity by the same rule: $\mathrm{Frob}(\xi_n) = \xi_n^q$. Now compute on π_n:

$$\mathrm{Frob}\,\tau\,\mathrm{Frob}^{-1}(\pi_n) = \mathrm{Frob}\,\tau(\pi_n) = \mathrm{Frob}(\xi_n\pi_n) = \xi_n^q\pi_n = \tau^q(\pi_n).$$

Thus $\mathrm{Frob}\,\tau\,\mathrm{Frob}^{-1} = \tau^q$. □

Theorem 19.5. — *After the choice of π and the compatible system $\{\pi_n\}$, there is an isomorphism*

$$\mathrm{Gal}(F^{\mathrm{tr}}/F) \cong \left(\prod_{\ell \neq p} \mathbf{Z}_\ell(1)\right) \rtimes \widehat{\mathbf{Z}},$$

where $\widehat{\mathbf{Z}} \cong \mathrm{Gal}(F^{\mathrm{ur}}/F)$ is generated by arithmetic Frobenius and acts on tame inertia by $x \mapsto x^q$. Equivalently, $\mathrm{Gal}(F^{\mathrm{tr}}/F)$ *is topologically generated by τ and* Frob *with relation*

$$\mathrm{Frob}\,\tau\,\mathrm{Frob}^{-1} = \tau^q.$$

Proof. — By Lemma 19.2, F^{tr} is obtained from F^{ur} by adjoining the chosen prime-to-p roots π_n of the uniformizer. Hence the kernel of

$$\mathrm{Gal}(F^{\mathrm{tr}}/F) \longrightarrow \mathrm{Gal}(F^{\mathrm{ur}}/F)$$

is

$$I_F^{\mathrm{t}} = \mathrm{Gal}(F^{\mathrm{tr}}/F^{\mathrm{ur}}) \cong \prod_{\ell \neq p} \mathbf{Z}_\ell(1).$$

The chosen lift Frob of arithmetic Frobenius fixes every π_n, so it gives a section of the projection to $\mathrm{Gal}(F^{\mathrm{ur}}/F) \cong \widehat{\mathbf{Z}}$. The preceding lemma computes its conjugation action on tame inertia as $x \mapsto x^q$, which gives the asserted semidirect product and the displayed relation. □

We now recall the cohomological criterion for free pro-p groups. Let G be a pro-p group, and let $\mathbf{F}_p$ denote the trivial discrete G-module of order p.

Theorem 19.6. — *Let G be a pro-p group. The following are equivalent:*

1. *G is a free pro-p group;*
2. *$H^2(G, \mathbf{F}_p) = 0$;*
3. *$\mathrm{cd}_p(G) \leq 1$.*

In the local-field application, the group of interest is often the wild inertia group P_F, which is a pro-p group. The theorem above will be applied to P_F.

19.1 Wild inertia via ramification filtration. — Let F be a non-archimedean local field with separable closure F^{sep}. Let $F^{\mathrm{tr}} \subset F^{\mathrm{sep}}$ be the maximal tamely ramified extension, and set $P_F := \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{tr}})$. First we recall the ramification filtration. Let E/F be a finite Galois extension with Galois group $G = \mathrm{Gal}(E/F)$. Let $\mathscr{O}_E$ be the ring of integers of E, $\mathfrak{p}_E$ its maximal ideal, and define

$$U_E^0 := \mathscr{O}_E^\times, \qquad U_E^i := 1 + \mathfrak{p}_E^i \quad (i \geq 1).$$

For $i \geq -1$, define the lower ramification groups:

$$G_i := \{\sigma \in G \mid v_E(\sigma(x) - x) \geq i + 1 \text{ for all } x \in \mathscr{O}_E\}.$$

Lemma 19.7. — *For all $i \geq 0$, the map*

$$\theta_i : G_i \longrightarrow U_E^i / U_E^{i+1}, \qquad \sigma \mapsto \frac{\sigma(\pi)}{\pi} \bmod U_E^{i+1},$$

is a group homomorphism with kernel G_{i+1}.

Proof. — Fix a uniformizer π of E. If $\sigma \in G_i$, then

$$\frac{\sigma(\pi)}{\pi} \in U_E^i,$$

where $U_E^0 = \mathscr{O}_E^\times$ and $U_E^i = 1 + \mathfrak{p}_E^i$ for $i \geq 1$. Thus θ_i is well-defined.

$$\theta_i(\sigma\tau) = \frac{\sigma(\tau(\pi))}{\pi}.$$

Write $\tau(\pi) = u_\tau \pi$ with $u_\tau \in U_E^i$. Since $\sigma \in G_i$, one has $\sigma(u_\tau) \equiv u_\tau \pmod{U_E^{i+1}}$. Therefore

$$\frac{\sigma(\tau(\pi))}{\pi} = \sigma(u_\tau)\frac{\sigma(\pi)}{\pi} \equiv \theta_i(\sigma)\theta_i(\tau) \pmod{U_E^{i+1}},$$

so θ_i is a homomorphism. Its kernel consists of those σ with

$$\frac{\sigma(\pi)}{\pi} \equiv 1 \pmod{U_E^{i+1}},$$

or equivalently $v_E(\sigma(\pi) - \pi) \geq i + 2$. By the standard uniformizer criterion for the lower ramification filtration, this is equivalent to $\sigma \in G_{i+1}$. □

Corollary 19.8. — For all $i \geq 1$, the quotient G_i/G_{i+1} is a p-group.

Proof. — We have an injection

$$G_i/G_{i+1} \hookrightarrow U_E^i/U_E^{i+1}.$$

But

$$U_E^i/U_E^{i+1} \cong \mathfrak{p}_E^i/\mathfrak{p}_E^{i+1} \cong k_E$$

as additive groups, hence is a vector space over $\mathbf{F}_p$. Therefore it is a p-group. □

Proposition 19.9. — *The quotient G_0/G_1 has order prime to p.*

Proof. — The map θ_0 gives an injection

$$G_0/G_1 \hookrightarrow U_E^0/U_E^1 \cong k_E^\times.$$

Since $\# k_E^\times = q_E - 1$, this quotient has order prime to p. □

Proposition 19.10. — *The subgroup G_1 is the unique p-Sylow subgroup of the inertia group G_0. Equivalently, G_1 is the wild inertia subgroup of E/F.*

Proof. — From the previous results:

- G_0/G_1 has order prime to p,
- G_i/G_{i+1} is a p-group for all $i \geq 1$.

Since the ramification filtration is eventually trivial, G_1 is a p-group. The quotient G_0/G_1 has order prime to p, so G_1 is a p-Sylow subgroup of G_0. It is normal in G_0, hence is the unique p-Sylow subgroup of G_0. □

Corollary 19.11. — If E/F is finite Galois, then the fixed field E^{G_1} is the maximal tamely ramified subextension of E/F.

Let F^{sep} be the separable closure of F. Taking the inverse limit over all finite Galois extensions E/F, we obtain

$$P_F = \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{tr}}) = \varprojlim \mathrm{Gal}(E/E^{G_1}) = \varprojlim G_1.$$

Proposition 19.12. — *The group P_F is a pro-p group.*
Proof. — Each group G_1 in the inverse system above is a finite p-group. Therefore their inverse limit P_F is pro-p. □

Proposition 19.13. —

$$\mathrm{cd}_p(P_F) \leq 1.$$

Proof. — In Serre's notation, P_F is the Sylow p-subgroup U_p of the inertia group $U = \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{ur}})$. The structure theorem for wild inertia gives that U_p is a free pro-p group. Free pro-p groups have p-cohomological dimension at most 1, so $\mathrm{cd}_p(P_F) \leq 1$. □

Theorem 19.14. — *The wild inertia group*

$$P_F = \mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{tr}})$$

is a free pro-p group.
Proof. — This follows from $\mathrm{cd}_p(P_F) \leq 1$ and Theorem 19.6. □

19.2 Frattini subgroup of a finite p-group, proof of theorem 19.6. —

Definition 19.15. — Let G be a finite p-group. The *Frattini subgroup* of G is

$$\Phi(G) := \bigcap_{\substack{H \leq G \\ H \text{ maximal}}} H.$$

Lemma 19.16. — *Let G be a finite p-group and $H \leq G$. The following are equivalent:*

1. *H is a maximal subgroup of G,*
2. *$H \lhd G$ and $[G : H] = p$,*
3. *there exists a surjective homomorphism $\chi : G \to \mathbf{F}_p$ such that $H = \ker(\chi)$.*

Proposition 19.17. — *Let G be a finite p-group. Then*

$$\Phi(G) = G^p[G, G].$$

Proof. — First, note that any homomorphism $\chi : G \to \mathbf{F}_p$ kills:

– commutators $[G, G]$, since $\mathbf{F}_p$ is abelian,
– p-th powers G^p, since $\mathbf{F}_p$ has exponent p.

Thus

$$G^p[G,G] \subset \ker(\chi)$$

for every such χ, hence

$$G^p[G,G] \subset \Phi(G).$$

Conversely, suppose $x \notin G^p[G,G]$. Then its image in the quotient

$$G/G^p[G,G]$$

is nontrivial. But this quotient is an abelian group of exponent p, hence an $\mathbf{F}_p$-vector space. Therefore there exists a linear map

$$\overline{\chi} : G/G^p[G,G] \to \mathbf{F}_p$$

such that $\overline{\chi}(x) \neq 0$. Composing with the projection gives a homomorphism

$$\chi : G \to \mathbf{F}_p$$

such that $\chi(x) \neq 0$, so $x \notin \ker(\chi)$. Hence $x \notin \Phi(G)$, proving the reverse inclusion. □

Corollary 19.18. —

$$G/\Phi(G) \cong G/(G^p[G,G])$$

is an $\mathbf{F}_p$-vector space.

Proposition 19.19. — *Let $\varphi : G_1 \to G_2$ be a homomorphism of finite p-groups. The following are equivalent:*

1. *φ is surjective,*
2. *the induced map*
$$G_1 \to G_2/\Phi(G_2)$$
is surjective,
3. *the induced map*
$$G_1/\Phi(G_1) \to G_2/\Phi(G_2)$$
is surjective.

Lemma 19.20. — *Let $\varphi : G_1 \to G_2$ be a homomorphism of pro-p groups. Then*

$$\varphi \text{ is surjective} \iff H^1(G_2, \mathbf{F}_p) \longrightarrow H^1(G_1, \mathbf{F}_p) \text{ is injective.}$$

Proof. — Using

$$H^1(G, \mathbf{F}_p) \cong \mathrm{Hom}(G/\Phi(G), \mathbf{F}_p),$$

the map on cohomology is dual to

$$G_1/\Phi(G_1) \to G_2/\Phi(G_2).$$

Thus injectivity on H^1 is equivalent to surjectivity on Frattini quotients, which is equivalent to surjectivity of φ. □

Let

$$1 \longrightarrow R \longrightarrow F \longrightarrow G \longrightarrow 1$$

be a minimal presentation of G, where F is a free pro-p group and the induced map

$$H^1(G, \mathbf{F}_p) \longrightarrow H^1(F, \mathbf{F}_p)$$

is an isomorphism. Such a presentation is obtained by choosing a basis of the dual of $H^1(G, \mathbf{F}_p)$ and mapping the corresponding free pro-p group onto G. Applying cohomology with coefficients in $\mathbf{F}_p$, we obtain the 5-term exact sequence:

$$0 \to H^1(G, \mathbf{F}_p) \to H^1(F, \mathbf{F}_p) \to H^1(R, \mathbf{F}_p)^G \to H^2(G, \mathbf{F}_p) \to H^2(F, \mathbf{F}_p).$$

Since F is free pro-p, we have $H^2(F, \mathbf{F}_p) = 0$, hence:

$$0 \to H^1(G, \mathbf{F}_p) \to H^1(F, \mathbf{F}_p) \to H^1(R, \mathbf{F}_p)^G \to H^2(G, \mathbf{F}_p) \to 0.$$

Since $H^1(G, \mathbf{F}_p) \to H^1(F, \mathbf{F}_p)$ is an isomorphism, we have:

$$H^1(R, \mathbf{F}_p)^G \;\cong\; H^2(G, \mathbf{F}_p).$$

Lemma 19.21. — *If $R \neq 1$, then $H^1(R, \mathbf{F}_p)^G \neq 0$.*

Proof. — This is the relation-module form of the Frattini argument [**7**, Chap. I, §4.3, Proposition 26, pp. 33–34]. Since R is a nontrivial closed normal subgroup of the free pro-p group F, the quotient

$$R/(R^p[R, F])$$

is nonzero. Its $\mathbf{F}_p$-dual is precisely

$$H^1(R, \mathbf{F}_p)^G = \mathrm{Hom}_{\mathrm{cont}}(R/(R^p[R, F]), \mathbf{F}_p),$$

so the invariant group is nonzero. □

Theorem 19.22. — *Let G be a pro-p group. Then*

$$G \textit{ is free pro-}p \iff H^2(G, \mathbf{F}_p) = 0.$$

Proof. — If $G \cong F/R$ as above and $H^2(G, \mathbf{F}_p) = 0$, then $H^1(R, \mathbf{F}_p)^G = 0$. Thus $R = 1$, so $G \cong F$ is free.

Conversely, free pro-p groups have trivial H^2. □

Together with the definition of p-cohomological dimension, this proves Theorem 19.6: if $\mathrm{cd}_p(G) \leq 1$, then $H^2(G, \mathbf{F}_p) = 0$, and the theorem just proved gives the converse through freeness.

Lecture 21

This lecture introduces dualising modules for profinite groups and gives the first formulation of local duality. The local-field goal is to identify the dualising module of G_K with $\mathbf{Q}/\mathbf{Z}(1)$.

20.1 Dualising Modules and First Form of Local Duality. — Our goal is the following:

Theorem 20.1 **(Local Tate duality, ℓ-primary form)**

Let K be a local field, let $\ell \neq \mathrm{char}(K)$ be a prime, and write $G_K = \mathrm{Gal}(\overline{K}/K)$. Let A be a finite discrete G_K-module killed by a power of ℓ. Set

$$A^\vee := \mathrm{Hom}(A, \mu_{\ell^\infty}(\overline{K})),$$

where G_K acts on $A^\vee$ by

$$(g \cdot f)(a) = g\big(f(g^{-1}a)\big) \qquad (g \in G_K,\ f \in A^\vee,\ a \in A).$$

Then for each $i = 0, 1, 2$, the cup product followed by evaluation gives a canonical perfect pairing

$$H^i(G_K, A) \times H^{2-i}(G_K, A^\vee) \xrightarrow{\cup} H^2(G_K, A \otimes A^\vee) \xrightarrow{\mathrm{ev}} H^2(G_K, \mu_{\ell^\infty}(\overline{K})) \cong \mathbf{Q}_\ell/\mathbf{Z}_\ell.$$

This is the ℓ-primary piece of the full local duality theorem proved below. Harari states the theorem for every finite Γ_K-module when K is p-adic, with Cartier dual $M' = \mathrm{Hom}_{\mathbf{Z}}(M, \overline{K}^\times) = \mathrm{Hom}(M, \mu)$; for local fields of characteristic p, the same statement holds for modules of order prime to p [**4**, Theorem 10.9 and Remark 10.10, pp. 135–136].

Example 20.2. — Let $\mathbf{F}/\mathbf{F}_p$ be a finite extension and let

$$\chi : G_K \longrightarrow \mathbf{F}^\times$$

be a character. Write $\mathbf{F}_\chi$ for the one-dimensional $\mathbf{F}$-vector space on which G_K acts through χ. Then

$$(\mathbf{F}_\chi)^\vee = \mathrm{Hom}(\mathbf{F}_\chi, \mu_{p^\infty}(\overline{K})) \cong \mathbf{F}_{\chi^{-1}\omega},$$

where ω is the mod-p cyclotomic character. More explicitly, $\omega : G_K \to \mathbf{F}_p^\times \hookrightarrow \mathbf{F}^\times$ is defined by

$$g(\zeta) = \zeta^{\omega(g)} \qquad (\zeta \in \mu_p).$$

The displayed isomorphism uses the trace identification $\mathrm{Hom}_{\mathbf{F}_p}(\mathbf{F}, \mathbf{F}_p) \simeq \mathbf{F}$; under this identification the contragredient action contributes χ^{-1} and the action on μ_p contributes ω.

Hence local duality gives a perfect pairing

$$H^2(G_K, \mathbf{F}_\chi) \times H^0(G_K, \mathbf{F}_{\chi^{-1}\omega}) \longrightarrow \mathbf{Q}_p/\mathbf{Z}_p.$$

In particular,

$$\dim_{\mathbf{F}} H^0(G_K, \mathbf{F}_{\chi^{-1}\omega}) = \begin{cases} 1, & \text{if } \chi = \omega, \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently, $H^2(G_K, \mathbf{F}_\chi)$ is dual to a one-dimensional space exactly when $\chi = \omega$, and is zero otherwise.

We first recall the notion of a dualising module. Let G be a profinite group of cohomological dimension $n < \infty$. If B is a discrete torsion abelian group, put

$$B^* := \mathrm{Hom}(B, \mathbf{Q}/\mathbf{Z}).$$

Since $\mathbf{Q}/\mathbf{Z}$ is an injective abelian group, the functor $B \mapsto B^*$ is exact.

We consider the contravariant functor

$$A \longmapsto H^n(G, A)^*$$

from the category $\mathscr{C}_G^f$ of finite discrete G-modules to the category of abelian groups. Under mild hypotheses, this functor is representable by a discrete torsion G-module; more precisely, this follows from the following general result.

Theorem 20.3. — *Let $\mathscr{C}$ be a noetherian abelian category. Let $F : \mathscr{C} \longrightarrow \mathbf{Ab}$ be a contravariant left exact functor. Then F is Ind-representable: i.e., there exists a filtered inductive system $(I_i)_{i \in I}$ of objects of $\mathscr{C}$ such that F is naturally isomorphic to the functor*

$$A \longmapsto \varinjlim_i \mathrm{Hom}_{\mathscr{C}}(A, I_i).$$

This is the homological algebra input used in Harari's Theorem 10.1 [**4**, Theorem 10.1, pp. 131–132].

Corollary 20.4. — Let G be a profinite group of cohomological dimension $n = cd(G)$. Assume that for every $A \in \mathscr{C}_G^f$, the group $H^n(G, A)$ is finite. Then there exists a discrete torsion G-module I such that the functors

$$A \longmapsto \mathrm{Hom}_{\mathbf{Z}}\big(H^n(G, A), \mathbf{Q}/\mathbf{Z}\big) \qquad \text{and} \qquad A \longmapsto \mathrm{Hom}_G(A, I)$$

are naturally isomorphic. In particular, for every finite discrete G-module A there is a perfect pairing

$$\mathrm{Hom}_G(A, I) \times H^n(G, A) \longrightarrow \mathbf{Q}/\mathbf{Z}.$$

We call I the dualising module of G.

Proof. — Apply Theorem 20.3 to the noetherian abelian category $\mathscr{C}_G^f$. Since $cd(G) = n$, the functor $A \mapsto H^n(G, A)$ is right exact; hence, by exactness of $(-)^*$, the contravariant functor $A \mapsto H^n(G, A)^*$ is left exact. Then there exists an inductive system (I_j) of finite discrete G-modules such that

$$H^n(G, A)^* \simeq \varinjlim_j \mathrm{Hom}_G(A, I_j)$$

for all $A \in \mathscr{C}_G^f$. Setting $I = \varinjlim_j I_j$ gives

$$H^n(G, A)^* \simeq \mathrm{Hom}_G(A, I).$$

□

***Example 20.5* (The case $G = \widehat{\mathbf{Z}}$).** — Let $G = \widehat{\mathbf{Z}}$, and let $\gamma \in G$ be a topological generator. Let A be a finite discrete G-module. Then $H^1(G, A) \cong A/(\gamma - 1)A$. Hence

$$H^1(G, A)^* = \mathrm{Hom}_{\mathbf{Z}}\big(H^1(G, A), \mathbf{Q}/\mathbf{Z}\big) \cong \mathrm{Hom}_{\mathbf{Z}}\big(A/(\gamma - 1)A, \mathbf{Q}/\mathbf{Z}\big).$$

Since $\mathbf{Q}/\mathbf{Z}$ is given the trivial G-action, a homomorphism $A \to \mathbf{Q}/\mathbf{Z}$ factors through $A/(\gamma - 1)A$ if and only if it is G-equivariant. Therefore

$$\mathrm{Hom}_{\mathbf{Z}}\big(A/(\gamma - 1)A, \mathbf{Q}/\mathbf{Z}\big) \cong \mathrm{Hom}_G(A, \mathbf{Q}/\mathbf{Z}).$$

So the functor $A \mapsto H^1(G, A)^*$ is represented by the *discrete torsion G-module* $\mathbf{Q}/\mathbf{Z}$ *with trivial action*; equivalently, the dualising module of $\widehat{\mathbf{Z}}$ is $\mathbf{Q}/\mathbf{Z}$.

Proposition 20.6. — *Let G be a profinite group with $cd(G) = n$, and let I be a dualising module for G. If U is an open subgroup of G, then I viewed as a U-module is a dualising module for U. Moreover, the following diagram is commutative:*

$$\begin{tikzcd} H^n(G, A)^* \arrow[r, "\sim"] \arrow[d, "\mathrm{Cor}^*"'] & \mathrm{Hom}_G(A, I) \arrow[d] \\ H^n(U, A|_U)^* \arrow[r, "\sim"'] & \mathrm{Hom}_U(A|_U, I|_U) \end{tikzcd}$$

for every finite discrete G-module A, where the right vertical arrow is the canonical restriction map.

This is Harari's restriction compatibility for dualising modules [**4**, Proposition 10.4, pp. 133–134].

Proof. — Since U is open in G, we have $cd(U) = cd(G) = n$, by Proposition 17.1.

Let $B \in \mathscr{C}_U^f$. By Shapiro's lemma 28.14,

$$H^n(U, B) \cong H^n\big(G, \mathrm{Ind}_U^G B\big).$$

Using the defining property of the dualising module I for G, we get

$$H^n(U, B)^* \cong \mathrm{Hom}_G\big(\mathrm{Ind}_U^G B, I\big) \cong \mathrm{Hom}_U(B, I|_U),$$

so $I|_U$ is dualising for U.

For the compatibility statement, let $A \in \mathscr{C}_G^f$. The corestriction $\mathrm{Cor} : H^n(U, A|_U) \to H^n(G, A)$ is obtained by applying $H^n(G, -)$ to the canonical G-morphism

$$\pi : \mathrm{Ind}_U^G(A|_U) \longrightarrow A, \qquad f \longmapsto \sum_{g \in G/U} g \cdot f(g^{-1}).$$

Dualising and using the description above, the dual of Cor identifies with the canonical restriction map

$$\mathrm{Hom}_G(A, I) \longrightarrow \mathrm{Hom}_U(A|_U, I|_U),$$

by the remark after theorem 3.7. This gives the commutative diagram. $\square$

We now return to the local field.

20.2 The Dualising Module of a Local Field. —
Proposition 20.7. — *Let F be a p-adic field, and let $\Gamma_F = \mathrm{Gal}(\overline{F}/F)$. Then the dualising module of Γ_F is canonically isomorphic to $\mu := \mathbf{Q}/\mathbf{Z}(1)$, the discrete Γ_F-module of all roots of unity in $\overline{F}^{\times}$.*
Proof. — We follow Harari's proof [**4**, Proposition 10.5, p. 134]. Let I be the dualising module of Γ_F and put

$$I_n = \ker(I \xrightarrow{\cdot n} I).$$

For every open subgroup $U \subset \Gamma_F$, Proposition 20.6 says that $I|_U$ is the dualising module of U. Hence

$$\mathrm{Hom}_U(\mu_n, I_n) = \mathrm{Hom}_U(\mu_n, I) \cong H^2(U, \mu_n)^*.$$

By the Kummer exact sequence and the invariant map for the finite extension $F_U = \overline{F}^U$,

$$H^2(U, \mu_n) \cong \mathrm{Br}(F_U)[n] \cong \frac{1}{n}\mathbf{Z}/\mathbf{Z} \cong \mathbf{Z}/n\mathbf{Z}.$$

These isomorphisms are compatible as U varies, by Proposition 20.6 and the compatibility of the invariant map with restriction and corestriction. Hence

$$\mathrm{Hom}(\mu_n, I_n) \cong \mathbf{Z}/n\mathbf{Z},$$

and the action of Γ_F on this Hom group is trivial. Choose a generator $f_n \in \mathrm{Hom}_{\Gamma_F}(\mu_n, I_n)$. Since f_n has order n, it is injective. If it were not surjective, an element of I_n outside its image would define a homomorphism $\mu_n \to I_n$ not generated by f_n, contradicting $\mathrm{Hom}(\mu_n, I_n) \cong \mathbf{Z}/n\mathbf{Z}$. Thus f_n is an isomorphism. The compatibility for $n \mid m$ identifies these isomorphisms with the inclusions $\mu_n \subset \mu_m$ and $I_n \subset I_m$. Passing to the direct limit gives

$$I = \varinjlim_n I_n \cong \varinjlim_n \mu_n = \mu = \mathbf{Q}/\mathbf{Z}(1).$$

□

Theorem 20.8. — *Let K be a p-adic field. Then $G_K = \mathrm{Gal}(\overline{K}/K)$ is of strict cohomological dimension 2.*

This is Harari's Theorem 10.6 [**4**, p. 134].
Proof. — We already know that $cd(G_K) = 2$ for a p-adic field. By Proposition 16.13, it remains to show that for every open subgroup $U \subset G_K$,

$$H^3(U, \mathbf{Z}) = 0.$$

Let $U \subset G_K$ be an open subgroup. Then we have $H^3(U, \mathbf{Z}) \cong H^2(U, \mathbf{Q}/\mathbf{Z})$. By Proposition 20.7, the dualising module of U is $\mu = \mathbf{Q}/\mathbf{Z}(1)$, hence

$$H^2(U, \mathbf{Q}/\mathbf{Z}) \cong \mathrm{Hom}_U(\mathbf{Q}/\mathbf{Z}, \mu)^*.$$

Therefore it suffices to prove that $\mathrm{Hom}_U(\mathbf{Q}/\mathbf{Z}, \mu) = 0$: let L/K be the *finite* extension corresponding to U, so that $U = \mathrm{Gal}(\overline{K}/L)$. Let $f : \mathbf{Q}/\mathbf{Z} \to \mu$ be a U-equivariant homomorphism. Since $\mathbf{Q}/\mathbf{Z}$ is divisible, its image $\mathrm{Im}(f)$ is a divisible subgroup of μ. Moreover, f is U-equivariant, so its image is contained in the invariants: $\mathrm{Im}(f) \subset \mu^U = H^0(L, \mu)$. But L is again a p-adic field, hence it contains only finitely many roots of unity. Therefore $H^0(L, \mu)$ is a finite group.

A finite abelian group has no nonzero divisible subgroup, so we must have $\mathrm{Im}(f) = 0$, and hence $f = 0$. $\square$

Remark 20.9. — The same argument shows that if K is a local field of characteristic $p > 0$, then for every prime $\ell \neq p$ one has

$$scd_\ell(G_K) = 2.$$

Lecture 22

This lecture defines the cup product carefully enough for local duality. After recording the cochain-level construction and its signs, we state the finite-module form of local Tate duality. First we recall the cup product. Let G be a profinite group and A, B discrete G-modules.

Recall that homogeneous continuous cochains are

$$C^q(G, A) = \{\alpha : G^{q+1} \to A \mid \alpha \text{ is continuous and } \alpha(gg_0, \dots, gg_q) = g \cdot \alpha(g_0, \dots, g_q)\}.$$

The differential $d : C^q(G, A) \to C^{q+1}(G, A)$ is given by

$$(d\alpha)(g_0, \dots, g_{q+1}) = \sum_{i=0}^{q+1} (-1)^i \alpha(g_0, \dots, \widehat{g_i}, \dots, g_{q+1}).$$

***Definition 21.1* (Cup product).** — Let $\alpha \in C^p(G, A)$ and $\beta \in C^q(G, B)$. Define

$$\alpha \cup \beta \in C^{p+q}(G, A \otimes B) \quad \text{by}$$

$$(\alpha \cup \beta)(g_0, \dots, g_{p+q}) = \alpha(g_0, \dots, g_p) \otimes \beta(g_p, \dots, g_{p+q}).$$

***Proposition 21.2* (Leibniz rule).** — *For $\alpha \in C^p(G, A)$ and $\beta \in C^q(G, B)$, we have*

$$d(\alpha \cup \beta) = (d\alpha) \cup \beta + (-1)^p \alpha \cup (d\beta).$$

Proof. — Compute:

$$d(\alpha \cup \beta)(g_0, \dots, g_{p+q+1}) = \sum_{i=0}^{p+q+1} (-1)^i (\alpha \cup \beta)(g_0, \dots, \widehat{g_i}, \dots, g_{p+q+1}).$$

Split the sum into two parts.

(1) Terms with $0 \leq i \leq p$: these affect only the α part:

$$\sum_{i=0}^{p} (-1)^i \alpha(g_0, \dots, \widehat{g_i}, \dots, g_{p+1}) \otimes \beta(g_{p+1}, \dots, g_{p+q+1}) = ((d\alpha) \cup \beta)(g_0, \dots, g_{p+q+1}).$$

(2) Terms with $p+1 \leq i \leq p+q+1$: set $j = i-(p+1)$. Then these terms become:

$$(-1)^p \sum_{j=0}^{q+1} (-1)^j \alpha(g_0, \ldots, g_p) \otimes \beta(g_p, \ldots, \widehat{g_{p+j}}, \ldots, g_{p+q+1}),$$

which equals

$$(-1)^p (\alpha \cup d\beta)(g_0, \ldots, g_{p+q+1}).$$

Combining both parts gives the result. □

Corollary 21.3. — If α and β are cocycles, then $\alpha \cup \beta$ is a cocycle. If one is a cocycle and the other a coboundary, then $\alpha \cup \beta$ is a coboundary.

Corollary 21.4. — The cup product induces a pairing on cohomology:

$$H^p(G, A) \times H^q(G, B) \longrightarrow H^{p+q}(G, A \otimes B).$$

We record two special cases.

1. $p = 0$**.** Then $H^0(G, A) = A^G$. For $a \in A^G$, define $\alpha(g_0) = a$. Then

   $$(\alpha \cup \beta)(g_0, \ldots, g_q) = a \otimes \beta(g_0, \ldots, g_q),$$

   so we obtain a map

   $$H^q(G, B) \to H^q(G, A \otimes B), \quad \beta \mapsto a \cup \beta,$$

   following the definition, it is induced by the map

   $$f : B \to A \otimes B, \qquad b \mapsto a \otimes b.$$

   Thus $\alpha \cup \beta = f_*(\beta)$.
2. $p = q = 0$**.** We recover the natural pairing

   $$A^G \times B^G \to (A \otimes B)^G, \quad (a, b) \mapsto a \otimes b.$$

Remark 21.5. — The cup product is the natural extension of the tensor product to cohomology. More generally, if $A \times B \to C$ is a bilinear G-equivariant map, it factors through $A \otimes B \to C$; hence, it induces

$$H^p(G, A) \times H^q(G, B) \to H^{p+q}(G, A \otimes B) \to H^{p+q}(G, C).$$

Proposition 21.6. — *Let*

$$0 \to A' \to A \to A'' \to 0, \qquad 0 \to B' \to B \to B'' \to 0$$

be exact sequences of G-modules. Let C be a G-module and let $\varphi : A \times B \to C$ be a G-equivariant bilinear map such that $\varphi(A' \times B') = 0$. Then φ induces pairings

$$\varphi_1 : A' \times B'' \to C, \qquad \varphi_2 : A'' \times B' \to C.$$

For $\alpha \in H^p(G, A'')$ and $\beta \in H^{q-1}(G, B'')$, the connecting homomorphisms satisfy

$$(\delta\alpha) \cup \beta + (-1)^p \alpha \cup (\delta\beta) = 0$$

in $H^{p+q}(G, C)$, where the two cup products use the two induced pairings above.

Proof. — Choose cocycles $a'' \in Z^p(G, A'')$ and $b'' \in Z^{q-1}(G, B'')$ representing α and β, and lift them to cochains $a \in C^p(G, A)$ and $b \in C^{q-1}(G, B)$. Then da and db come from cochains $a' \in C^{p+1}(G, A')$ and $b' \in C^q(G, B')$, representing $\delta\alpha$ and $\delta\beta$. The Leibniz rule gives

$$a' \cup b'' + (-1)^p a'' \cup b' = d(a \cup b).$$

The right-hand side is a coboundary, so the asserted relation holds in cohomology. □

We now return to the dualising object. Let G be a profinite group with $cd(G) = n$. Recall that there exists a discrete torsion G-module I and a natural isomorphism

$$H^n(G, A)^* \;\cong\; \mathrm{Hom}_G(A, I), \qquad \forall A \in \mathscr{C}_G^f.$$

For $A \in \mathscr{C}_G^f$, set $A' := \mathrm{Hom}_{\mathbf{Z}}(A, I)$, endowed with the G-action $(g \cdot f)(a) = g \cdot f(g^{-1}a)$. Then

$$H^0(G, A') = \mathrm{Hom}_G(A, I).$$

Let $i : H^n(G, I) \longrightarrow \mathbf{Q}/\mathbf{Z}$ be the morphism corresponding to $\mathrm{Id}_I \in \mathrm{Hom}_G(I, I)$ via

$$H^n(G, I)^* \cong \mathrm{Hom}_G(I, I).$$

For any $A \in \mathscr{C}_G^f$, denote the pairing

$$\langle\ ,\ \rangle_A : \mathrm{Hom}_G(A, I) \times H^n(G, A) \longrightarrow \mathbf{Q}/\mathbf{Z}.$$

Proposition 21.7. — *The pairing $\langle\ ,\ \rangle_A$ coincides with the cup-product pairing*

$$H^0(G, A') \times H^n(G, A) \xrightarrow{\ \cup\ } H^n(G, A' \otimes A) \xrightarrow{\ \mathrm{ev}\ } H^n(G, I) \xrightarrow{\ i\ } \mathbf{Q}/\mathbf{Z}.$$

Proof. — The statement follows from the naturality of the duality isomorphism in A: let $\alpha \in H^n(G, A)$ and $\rho \in \mathrm{Hom}_G(A, I) = H^0(G, A')$. Consider the induced maps

$$\rho_* : H^n(G, A) \longrightarrow H^n(G, I), \qquad \rho^* : \mathrm{Hom}_G(I, M) \longrightarrow \mathrm{Hom}_G(A, M)$$

for any G-module M, where ρ^* is precomposition by ρ. The naturality of the duality pairing gives

$$\langle \rho, \alpha \rangle_A = i\big(\rho_*(\alpha)\big).$$

We conclude by observing that $\rho^*(\mathrm{Id}_I) = \rho$ and that, for any $\alpha \in H^n(G, A)$, the cup product $\rho \cup \alpha$ is exactly $\rho_*(\alpha)$, by the degree 0 case of the cup product. □

21.1 Local Tate Duality. — We now apply the abstract dualising-module formalism to the absolute Galois group of a local field. The main point is that the abstract dualising module is the concrete module of roots of unity, so the abstract pairing becomes the familiar cup product followed by the invariant map. Let F be a p-adic field, $G = \mathrm{Gal}(\overline{F}/F)$, and let $M \in \mathscr{C}_G^f$ be a finite discrete G-module. Define its Cartier dual by

$$M' := \mathrm{Hom}_{\mathbf{Z}}(M, \overline{F}^{\times}) = \mathrm{Hom}(M, \mu).$$

If M is killed by n, then $M' = \mathrm{Hom}(M, \mu_n)$. For local fields of characteristic $p > 0$, the same statement below holds for finite modules M whose order is prime to p [**4**, Remark 10.10, p. 136].

***Theorem 21.8* (Local Tate Duality).** — *For each $i = 0, 1, 2$, the cup-product*

$$H^i(G, M) \times H^{2-i}(G, M') \xrightarrow{\cup} H^2(G, M \otimes M') \longrightarrow H^2(G, \overline{F}^\times) \xrightarrow{\mathrm{inv}_F} \mathbf{Q}/\mathbf{Z}$$

defines a perfect pairing of finite groups [**4**, Theorem 10.9, pp. 135–136].

Example 21.9. — Let $M = \mathbf{Z}/n\mathbf{Z}$ with trivial G-action. Then

$$M' = \mathrm{Hom}(\mathbf{Z}/n\mathbf{Z}, \mu_n) \simeq \mu_n.$$

The duality gives, in particular, a perfect pairing

$$H^1(G, \mathbf{Z}/n\mathbf{Z}) \times H^1(G, \mu_n) \longrightarrow \mathbf{Q}/\mathbf{Z}.$$

Using Kummer theory, one has

$$H^1(G, \mu_n) \simeq F^\times / F^{\times n},$$

so this recovers the classical local duality between cyclic extensions and norm subgroups.

Proof of Theorem 21.8. — We follow Harari's proof [**4**, Theorem 10.9, pp. 135–136]. The finiteness of all groups $H^i(G, M)$ was proved earlier for finite modules over a p-adic field. Since G has cohomological dimension 2, Corollary 20.4 gives a dualising module I and Proposition 21.7 identifies the degree $(0, 2)$ duality pairing with cup product followed by the canonical map $H^2(G, I) \to \mathbf{Q}/\mathbf{Z}$.

By Proposition 20.7, $I = \mu = \mathbf{Q}/\mathbf{Z}(1)$. Therefore, for every finite G-module A, the pairing

$$H^0(G, A') \times H^2(G, A) \longrightarrow \mathbf{Q}/\mathbf{Z}$$

is perfect and is exactly the cup-product pairing. Taking $A = M$ proves the theorem for $i = 2$, and taking $A = M'$ and using $(M')' = M$ proves it for $i = 0$.

It remains to treat $i = 1$. It is enough to prove that the homomorphism

$$H^1(G, M) \longrightarrow H^1(G, M')^*$$

induced by cup product is injective; applying the same argument to M' then gives the reverse inequality of cardinalities, hence bijectivity. Choose an exact sequence of finite G-modules

$$0 \longrightarrow M \longrightarrow B \longrightarrow C \longrightarrow 0$$

such that the map $H^1(G, M) \to H^1(G, B)$ is zero. Harari constructs such a sequence by embedding M into the induced module $I_G(M)$, using $H^1(G, I_G(M)) = 0$, and then taking a finite submodule $B \subset I_G(M)$ large enough because $H^1(G, M)$ is finite [**4**, p. 136].

Cup-product compatibility with connecting morphisms gives the following exact commutative diagram, up to the usual sign:

$$\begin{tikzcd} H^0(G,B) \arrow[r] \arrow[d, "\sim"'] & H^0(G,C) \arrow[r, "\delta"] \arrow[d, "\sim"'] & H^1(G,M) \arrow[r] \arrow[d] & 0 \\ H^2(G,B')^* \arrow[r] & H^2(G,C')^* \arrow[r, "\delta^*"'] & H^1(G,M')^* & \end{tikzcd}$$

Here the first two vertical arrows are isomorphisms by the already proved degree $(0,2)$ case. Since $H^1(G,M) \to H^1(G,B)$ is zero, the connecting map $H^0(G,C) \to H^1(G,M)$ is surjective. A diagram chase therefore shows that the third vertical arrow is injective. Applying the same argument to M' gives the opposite cardinality inequality, so the injective map $H^1(G,M) \to H^1(G,M')^*$ is an isomorphism. Thus the $i = 1$ pairing is perfect. □

Lecture 23-24

These lectures prove the Euler–Poincare characteristic formula for finite G_F-modules, where $F/\mathbf{Q}_p$ is finite. The proof combines the structure of local-field Galois groups, restriction–inflation arguments, low-degree cohomology computations, and local Tate duality.

22.1 Euler–Poincare Characteristic Formula. — Let $F/\mathbf{Q}_p$ be a finite extension, denote $\Gamma = \mathrm{Gal}(\overline{F}/F)$. Recall that for $A \in \mathscr{C}_F^{\mathrm{fin}}$, the groups $H^i(\Gamma, A)$ are finite, and they vanish for $i > 2$.

Definition 22.1. — The Euler–Poincaré characteristic of A is the strictly positive rational number

$$\chi(A) = \frac{|H^0(\Gamma,A)| \cdot |H^2(\Gamma,A)|}{|H^1(\Gamma,A)|}.$$

Normalize the norm on F so that $\|x\|_F = \frac{1}{|\mathscr{O}_F/x\mathscr{O}_F|}$, so that

$$\|\pi\|_F = \frac{1}{|\mathscr{O}_F/\pi\mathscr{O}_F|} = \frac{1}{p^{[F:\mathbf{Q}_p]}}.$$

Theorem 22.2. — *(Local Euler–Poincaré characteristic)*

$$\chi_F(A) = \|a\|_F,$$

for any $A \in \mathscr{C}_F^{\mathrm{fin}}$, where $a = |A|$.

Remark 22.3. — If $p \nmid |A|$, then $\|A\|_F = 1$. Also, for local fields of characteristic p, the group $\mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{tr}})$ is pro-p, hence does not play a role cohomologically when $p \nmid |A|$ (If A is a finite G_F-module of order prime to p, then $H^i(P,A) = 0 \quad (i > 0)$.) Moreover,

$$H^i(G_F, A) \simeq H^i\big(\mathrm{Gal}(F^{\mathrm{tr}}/F), A^{\mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{tr}})}\big).$$

Let $\mathbf{F}$ be a finite extension of $\mathbf{F}_p$ and A a finite $\mathbf{F}_p[G_F]$-module. Then

$$|A| = p^{\dim_{\mathbb{F}_p} A}, \quad |H^i(G_F, A)| = p^{\dim_{\mathbb{F}_p} H^i(G_F,A)}.$$

The Euler–Poincaré characteristic is

$$\chi_F(A) = \frac{|H^0(G_F, A)|\,|H^2(G_F, A)|}{|H^1(G_F, A)|} = |A|^{-[F:\mathbf{Q}_p]}.$$

Taking logarithms gives

$$\dim_{\mathbb{F}_p} H^0(G_F, A) - \dim_{\mathbb{F}_p} H^1(G_F, A) + \dim_{\mathbb{F}_p} H^2(G_F, A) = -[F : \mathbf{Q}_p] \cdot \dim_{\mathbb{F}_p} A.$$

Proof. — Reduction to the absolute irreducible case: If

$$0 \to A \to B \to C \to 0$$

is exact in $\mathrm{Rep}_{\mathbb{F}_p}(G_F)$, then by the long exact sequence in cohomology and $\mathrm{cd}(G_F) = 2$,

$$\chi_F(B) = \chi_F(A)\,\chi_F(C).$$

Thus we may assume A is irreducible, i.e. it has no nonzero G_F-invariant submodule.

By the remark above, we may also assume $p \mid |A|$, hence A is an $\mathbb{F}_p$-vector space: If $A \in \mathscr{C}^{\mathrm{fin}}_{G_F}$ and V is a finite $\mathbb{F}_p$-vector space with trivial G_F-action, denote $A \otimes_{\mathbb{F}_p} V$ has the diagonal G_F-action, trivial on V. Then the map

$$A^{G_F} \otimes_{\mathbb{F}_p} V \longrightarrow \big(A \otimes_{\mathbb{F}_p} V\big)^{G_F}$$

is an isomorphism, since the action on V is trivial. Hence

$$H^i\big(G_F, A \otimes_{\mathbb{F}_p} V\big) \simeq H^i(G_F, A) \otimes_{\mathbb{F}_p} V.$$

In particular, $h_F(A \otimes V) = h_F(A)^{\dim_{\mathbb{F}_p} V}$, equivalently, in additive Euler characteristic notation,

$$\chi(A \otimes V) = (\dim_{\mathbb{F}_p} V)\chi(A).$$

If $F/\mathbb{F}_p$ is a finite extension, then the result for $A \otimes_{\mathbb{F}_p} F$ implies the result for A. $\square$

Lemma 22.4. — *Let K be a p-adic field, and let A be an irreducible finite discrete $\mathbb{F}_p[G_K]$-module. Then there exists a finite extension $k/\mathbb{F}_p$, finite unramified extensions $E_1, \ldots, E_r$ of K, and continuous characters*

$$\chi_i : G_{E_i} \longrightarrow k^\times$$

such that

$$A \otimes_{\mathbb{F}_p} k \simeq \bigoplus_{i=1}^{r} \mathrm{Ind}_{G_{E_i}}^{G_K} \chi_i.$$

Proof. — Let K^{tr} be the maximal tamely ramified extension of K, and put $P_K = \mathrm{Gal}(K^{\mathrm{sep}}/K^{\mathrm{tr}})$. Thus P_K is the wild inertia subgroup of G_K. It is a pro-p group and is normal in G_K. Since $A \neq 0$, we have $A^{P_K} \neq 0$.

Indeed, the action of P_K on the finite module A factors through a finite p-group, and every nonzero representation of a finite p-group over $\mathbb{F}_p$ has nonzero invariants. Since $P_K \triangleleft G_K$, the subspace A^{P_K} is G_K-stable. By irreducibility of A, it follows that

$$A^{P_K} = A.$$

Hence the action of G_K on A factors through the tame quotient

$$G_K/P_K \simeq \mathrm{Gal}(K^{\mathrm{tr}}/K).$$

Recall the structure of the tame quotient. If q is the cardinality of the residue field of K, then

$$G_K/P_K \simeq \langle \varphi, \tau \mid \varphi\tau\varphi^{-1} = \tau^q \rangle,$$

where τ is a topological generator of tame inertia and φ is a lift of arithmetic Frobenius. In particular, the image of τ in $\mathrm{Aut}_{\mathbb{F}_p}(A)$ has finite order prime to p.

Choose a finite extension $k/\mathbb{F}_p$ large enough so that:

1. $A_k := A \otimes_{\mathbb{F}_p} k$ is semisimple as a $k[G_K]$-module;
2. the action of tame inertia on every irreducible constituent of A_k is split, i.e. diagonalizable over k;
3. the relevant Frobenius eigenvalues occurring below are contained in k.

It is enough to prove the assertion for each irreducible $k[G_K]$-constituent B of A_k.

Let B be such a constituent. Since P_K acts trivially on A, it also acts trivially on B. Thus B is a representation of the tame quotient. The image of tame inertia on B is finite of order prime to p, hence semisimple. Therefore

$$B = \bigoplus_{\lambda} B_\lambda,$$

where

$$B_\lambda = \{v \in B : \tau v = \lambda v\}.$$

The relation

$$\varphi\tau\varphi^{-1} = \tau^q$$

shows that Frobenius permutes these eigenspaces. More precisely, if $v \in B_\lambda$, then

$$\tau(\varphi v) = \lambda^q \varphi v, \quad \Longrightarrow \varphi(B_\lambda) = B_{\lambda^q}.$$

Choose an eigenvalue λ occurring in B, and let $s \geq 1$ be the smallest integer such that $\lambda^{q^s} = \lambda$. Then the stabilizer of the character of tame inertia defined by λ is the subgroup generated by inertia and φ^s. Let $H \subseteq G_K$ be its inverse image in G_K. Then H is an open subgroup of G_K, contains the full inertia subgroup, and therefore

$$H = G_E$$

for a finite unramified extension E/K, with $[E : K] = s$.

The eigenspace B_λ is stable under H. Since H/I_K is generated topologically by the image of φ^s, and since the image of H in $\mathrm{Aut}_k(B_\lambda)$ is finite, after enlarging k if necessary we may choose a one-dimensional H-stable subspace

$$W \subset B_\lambda.$$

The action of H on W is given by a continuous character

$$\chi : H = G_E \longrightarrow k^\times.$$

The inclusion $W \hookrightarrow B$ induces, by Frobenius reciprocity, a nonzero G_K-equivariant morphism

$$\mathrm{Ind}_{G_E}^{G_K} \chi \longrightarrow B.$$

Since B is irreducible, this morphism is surjective.

We now compare dimensions. The distinct G_K-translates of W lie in the distinct tame-inertia eigenspaces

$$B_\lambda,\ B_{\lambda^q},\ \ldots,\ B_{\lambda^{q^{s-1}}},$$

and hence are linearly independent. Therefore the image of $\mathrm{Ind}_{G_E}^{G_K} \chi$ in B has dimension at least s. On the other hand,

$$\dim_k \mathrm{Ind}_{G_E}^{G_K} \chi = [G_K : G_E] = [E : K] = s.$$

Thus the above surjection is an isomorphism:

$$B \simeq \mathrm{Ind}_{G_E}^{G_K} \chi.$$

Applying this argument to every irreducible constituent B of $A \otimes_{\mathbb{F}_p} k$, we obtain

$$A \otimes_{\mathbb{F}_p} k \simeq \bigoplus_i \mathrm{Ind}_{G_{E_i}}^{G_K} \chi_i,$$

where each E_i/K is finite unramified and each $\chi_i : G_{E_i} \to k^\times$ is a continuous character. $\square$

Continue on Theorem 22.2. — It remains to prove the Euler–Poincaré formula for modules of the form

$$A = \mathrm{Ind}_{G_E}^{G_K} \chi,$$

where E/K is finite unramified and $\chi : G_E \to k^\times$ is a one-dimensional k-valued character. By Shapiro's lemma,

$$H^i\big(G_K, \mathrm{Ind}_{G_E}^{G_K} \chi\big) \simeq H^i(G_E, \chi)$$

for all i. Therefore

$$\chi_K\big(\mathrm{Ind}_{G_E}^{G_K} \chi\big) = \chi_E(\chi).$$

Moreover,

$$\Big|\mathrm{Ind}_{G_E}^{G_K} \chi\Big| = |\chi|^{[E:K]},$$

where $|\chi|$ denotes the order of the underlying additive group of the one-dimensional k-vector space.

Thus, assuming the formula for one-dimensional characters over E, we get

$$\chi_E(\chi) = |\chi|^{-[E:\mathbb{Q}_p]} = \left(|\chi|^{[E:K]}\right)^{-[K:\mathbb{Q}_p]} = \left|\operatorname{Ind}_{G_E}^{G_K} \chi\right|^{-[K:\mathbb{Q}_p]}.$$

Hence the Euler–Poincaré formula for $\operatorname{Ind}_{G_E}^{G_K} \chi$ follows from the one-dimensional case over E. □

Let F be a finite extension of $\mathbb{Q}_p$, and let $k/\mathbb{F}_p$ be a finite extension. For a finite $k[G_F]$-module V, write

$$\chi_F(V) := \prod_{i=0}^{2} |H^i(G_F, V)|^{(-1)^i} = \frac{|H^0(G_F, V)|\,|H^2(G_F, V)|}{|H^1(G_F, V)|}.$$

We prove that, for every continuous character $\psi : G_F \longrightarrow k^\times$, one has

$$\chi_F(k(\psi)) = |k(\psi)|^{-[F:\mathbb{Q}_p]} = |k|^{-[F:\mathbb{Q}_p]},$$

where $k(\psi)$ denotes the one-dimensional k-vector space on which G_F acts through ψ.

***Lemma 22.5* (Restriction through a prime-to-p extension)**

Let E/F be a finite Galois extension such that $p \nmid [E : F]$, and put

$$\Delta = \mathrm{Gal}(E/F).$$

Then, for every finite p-primary discrete G_F-module M, restriction induces a natural isomorphism

$$H^i(G_F, M) \xrightarrow{\sim} H^i(G_E, M)^\Delta$$

for all $i \geq 0$.

Proof. — The Hochschild–Serre spectral sequence gives

$$H^r\big(\Delta, H^s(G_E, M)\big) \Longrightarrow H^{r+s}(G_F, M).$$

Since M is p-primary, each $H^s(G_E, M)$ is also p-primary. Because $|\Delta|$ is prime to p, we have

$$H^r(\Delta, N) = 0 \qquad (r > 0)$$

for every p-primary Δ-module N. Hence the spectral sequence degenerates at the row $r = 0$, giving

$$H^i(G_F, M) \simeq H^i(G_E, M)^\Delta.$$

□

Continue on Theorem 22.2, one dimensional case. — Now let

$$\psi : G_F \longrightarrow k^\times$$

be a continuous character, and let E/F be the finite Galois extension cut out by ψ, so that

$$G_E = \ker(\psi), \qquad \Delta = \mathrm{Gal}(E/F) \simeq \mathrm{im}(\psi) \subset k^\times.$$

Since $k^\times$ has order prime to p, the group Δ has order prime to p. Therefore the restriction lemma applies.

Since G_E acts trivially on $k(\psi)$, we get

$$H^i(G_E, k(\psi)) \simeq H^i(G_E, \mathbb{F}_p) \otimes_{\mathbb{F}_p} k(\psi).$$

Thus

$$H^i(G_F, k(\psi)) \simeq \big(H^i(G_E, \mathbb{F}_p) \otimes_{\mathbb{F}_p} k(\psi)\big)^{\Delta}.$$

Equivalently, after tensoring $H^i(G_E, \mathbb{F}_p)$ with k,

$$H^i(G_F, k(\psi)) \simeq \big(H^i(G_E, k)\big)_{\psi^{-1}},$$

where the right-hand side denotes the ψ^{-1}-isotypic component for the natural action of Δ on $H^i(G_E, k)$. Hence

$$\log_{|k|} \chi_F(k(\psi)) = \sum_{i=0}^{2} (-1)^i \dim_k\big(H^i(G_E, k)_{\psi^{-1}}\big).$$

It remains to compute this alternating multiplicity. We shall prove the following identity in the Grothendieck group $K_0(k[\Delta])$:

$$[H^0(G_E, k)] - [H^1(G_E, k)] + [H^2(G_E, k)] = -[F : \mathbb{Q}_p]\,[k[\Delta]].$$

Once this identity is known, taking the ψ^{-1}-isotypic component gives

$$\sum_{i=0}^{2} (-1)^i \dim_k\big(H^i(G_E, k)_{\psi^{-1}}\big) = -[F : \mathbb{Q}_p],$$

because the regular representation $k[\Delta]$ contains each one-dimensional character of Δ with multiplicity 1. Therefore

$$\chi_F(k(\psi)) = |k|^{-[F:\mathbb{Q}_p]}.$$

It remains to prove the Grothendieck-group identity. By extension of scalars from $\mathbb{F}_p$ to k, it is enough to prove

$$[H^0(G_E, \mathbb{F}_p)] - [H^1(G_E, \mathbb{F}_p)] + [H^2(G_E, \mathbb{F}_p)] = -[F : \mathbb{Q}_p]\,[\mathbb{F}_p[\Delta]]$$

in $K_0(\mathbb{F}_p[\Delta])$.

Local Tate duality gives Δ-equivariant perfect pairings

$$H^i(G_E, \mathbb{F}_p) \times H^{2-i}(G_E, \mu_p) \longrightarrow H^2(G_E, \mu_p) \simeq \mathbb{F}_p.$$

Thus the desired identity is equivalent, up to the duality involution on $K_0(\mathbb{F}_p[\Delta])$, to

$$[H^2(G_E, \mu_p)] - [H^1(G_E, \mu_p)] + [H^0(G_E, \mu_p)] = -[F : \mathbb{Q}_p]\,[\mathbb{F}_p[\Delta]].$$

By Kummer theory,

$$H^1(G_E, \mu_p) \simeq E^\times / E^{\times p}.$$

Moreover,

$$H^0(G_E, \mu_p) = \mu_p(E), \qquad H^2(G_E, \mu_p) \simeq \mathbb{Z}/p\mathbb{Z},$$

where the second group is trivial as a Δ-module. The valuation exact sequence

$$0 \longrightarrow \mathscr{O}_E^\times \longrightarrow E^\times \xrightarrow{v_E} \mathbb{Z} \longrightarrow 0$$

gives

$$E^\times/E^{\times p} \simeq \mathscr{O}_E^\times/\mathscr{O}_E^{\times p} \oplus \mathbb{Z}/p\mathbb{Z}.$$

The summand $\mathbb{Z}/p\mathbb{Z}$, generated by the class of a uniformizer, cancels with $H^2(G_E, \mu_p)$. Hence it is enough to prove

$$[\mathscr{O}_E^\times/\mathscr{O}_E^{\times p}] - [\mu_p(E)] = [F : \mathbb{Q}_p]\,[\mathbb{F}_p[\Delta]]$$

in $K_0(\mathbb{F}_p[\Delta])$.

For a finitely generated $\mathbb{Z}_p[\Delta]$-module M, the exact sequence

$$0 \longrightarrow M[p] \longrightarrow M \xrightarrow{p} M \longrightarrow M/pM \longrightarrow 0$$

shows that

$$[M/pM] - [M[p]]$$

depends only on the rational representation $M \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$.

Apply this to $M = \mathscr{O}_E^\times$, we have

$$\mathscr{O}_E^\times[p] = \mu_p(E), \quad M/pM = \mathscr{O}_E^\times/\mathscr{O}_E^{\times p}.$$

After passing to a sufficiently small principal unit subgroup $U_E^n = 1 + \mathfrak{p}_E^n$, the p-adic logarithm gives a Δ-equivariant isomorphism

$$U_E^n \simeq \mathfrak{p}_E^n.$$

Since $\mathfrak{p}_E^n$ is a full $\mathbb{Z}_p$-lattice in E, we have

$$U_E^n \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \simeq E$$

as $\mathbb{Q}_p[\Delta]$-modules.

Since $[E : F]$ is prime to p, the extension E/F is tamely ramified. By the normal basis theorem,

$$E \simeq F[\Delta]$$

as $F[\Delta]$-modules. Hence, as $\mathbb{Q}_p[\Delta]$-modules,

$$E \simeq \mathbb{Q}_p[\Delta]^{[F:\mathbb{Q}_p]}.$$

It follows that

$$[\mathscr{O}_E^\times/\mathscr{O}_E^{\times p}] - [\mu_p(E)] = [F : \mathbb{Q}_p]\,[\mathbb{F}_p[\Delta]].$$

This proves the required Grothendieck-group identity, and therefore

$$\chi_F(k(\psi)) = |k|^{-[F:\mathbb{Q}_p]}.$$

The one-dimensional case is proved. □

Lecture 25

This lecture begins the Tate-cohomological formulation of local class field theory. For a finite Galois extension L/K, we reinterpret $K^\times/N_{L/K}L^\times$ and $\mathrm{Gal}(L/K)^{\mathrm{ab}}$ as Tate cohomology groups and introduce the fundamental class.

23.1 Tate Cohomology and Local Reciprocity. — Let L/K be a finite Galois extension of local fields, and put

$$G = \mathrm{Gal}(L/K), \qquad n = [L : K].$$

The cohomological construction of local reciprocity starts from the two elementary identifications

$$\widehat{H}^{-2}(G, \mathbf{Z}) = H_1(G, \mathbf{Z}) \cong G^{\mathrm{ab}},$$

and

$$\widehat{H}^0(G, L^\times) = (L^\times)^G / N_G(L^\times) = K^\times / N_{L/K}(L^\times).$$

The first is the standard computation $H_1(G, \mathbf{Z}) \cong G^{\mathrm{ab}}$, and the second is the definition of $\widehat{H}^0$, since $(L^\times)^G = K^\times$ and $N_G = N_{L/K}$.

There is also a distinguished class

$$u_{L/K} \in H^2(G, L^\times) = \widehat{H}^2(G, L^\times),$$

called the *fundamental class.* Under the identification

$$H^2(G, L^\times) \cong \mathrm{Br}(L/K) \subset \mathrm{Br}(K)$$

and the local invariant map $\mathrm{inv}_K : \mathrm{Br}(K) \to \mathbf{Q}/\mathbf{Z}$, it is characterized by

$$\mathrm{inv}_K(u_{L/K}) = \frac{1}{n}.$$

Equivalently, $\mathrm{Br}(L/K)$ is the cyclic subgroup of $\mathrm{Br}(K) \simeq \mathbf{Q}/\mathbf{Z}$ killed by n, and $u_{L/K}$ is its generator with invariant $1/n$.

The cup product in Tate cohomology gives

$$\widehat{H}^{-2}(G, \mathbf{Z}) \times H^2(G, L^\times) \longrightarrow \widehat{H}^0(G, L^\times), \qquad (\alpha, u) \longmapsto \alpha \cup u.$$

***Theorem 23.1* (Local Reciprocity).** — *The map*

$$\Theta_{L/K} : \widehat{H}^{-2}(G, \mathbf{Z}) \longrightarrow \widehat{H}^0(G, L^\times), \qquad \alpha \longmapsto \alpha \cup u_{L/K},$$

is an isomorphism. In concrete terms, it gives an isomorphism

$$G^{\mathrm{ab}} \xrightarrow{\sim} K^\times / N_{L/K}(L^\times).$$

Following Serre's convention, this cup-product isomorphism is the inverse of the local reciprocity map. We denote the inverse by

$$\theta_{L/K} : K^\times / N_{L/K} L^\times \xrightarrow{\sim} G^{\mathrm{ab}}.$$

For $a \in K^\times$, its image is written $(a, L/K)$ and is called the local reciprocity map, or norm-residue symbol [**2**, Serre, Local class field theory, §2.2, pp. 16–17]. Thus the fundamental class and cup product first identify the abelianized Galois group with the norm quotient; the Artin map is the inverse identification.

23.2 Group Homology and Tate Cohomology. — Let G be a group and let A be a left G-module. Let

$$P_\bullet \longrightarrow \mathbf{Z} \longrightarrow 0$$

be a projective resolution of the trivial right $\mathbf{Z}[G]$-module $\mathbf{Z}$. Define

$$H_q(G,A) := H_q(P_\bullet \otimes_{\mathbf{Z}[G]} A), \qquad q \geq 0.$$

This is independent of the chosen projective resolution, and

$$H_q(G,A) \cong \mathrm{Tor}_q^{\mathbf{Z}[G]}(\mathbf{Z}, A).$$

Lemma 23.2. — *For every G-module A,*

$$H_0(G,A) \cong A_G := A/I_G A,$$

where $I_G = \ker(\mathbf{Z}[G] \to \mathbf{Z})$ is the augmentation ideal.
Proof. — Since $H_0(G,A) = \mathbf{Z} \otimes_{\mathbf{Z}[G]} A$, the defining relation in the tensor product is $ga \sim a$. Hence the quotient is exactly $A/I_G A$. □

Lemma 23.3. — *Let $\mathrm{Ind}^G(X) = \mathbf{Z}[G] \otimes_{\mathbf{Z}} X$, where X is an abelian group with trivial action before induction. Then*

$$H_q(G, \mathrm{Ind}^G(X)) = 0 \quad (q \geq 1), \qquad H_0(G, \mathrm{Ind}^G(X)) \cong X.$$

Proof. — By Shapiro's lemma for homology, applied to the trivial subgroup, one has

$$H_q(G, \mathrm{Ind}^G(X)) \cong H_q(\{1\}, X).$$

□

Remark 23.4 **(Dimension shifting for homology).** — Let

$$\pi : \mathrm{Ind}^G(A) = \mathbf{Z}[G] \otimes_{\mathbf{Z}} A \longrightarrow A, \qquad g \otimes a \longmapsto ga,$$

and set $B = \ker(\pi)$. Since $\mathrm{Ind}^G(A)$ has no higher homology, the long exact homology sequence gives

$$H_q(G,A) \cong H_{q-1}(G,B) \qquad (q \geq 2).$$

Assume now that G is finite. The norm element

$$N_G = \sum_{g \in G} g \in \mathbf{Z}[G]$$

acts on A by

$$N_G(a) = \sum_{g \in G} ga.$$

It induces a map

$$\overline{N}_G : A_G = A/I_G A \longrightarrow A^G.$$

Definition 23.5. — For $i \in \mathbf{Z}$, the Tate cohomology groups are

$$\widehat{H}^i(G,A) = \begin{cases} H^i(G,A), & i \geq 1, \\ A^G/N_G A, & i = 0, \\ \ker(\overline{N}_G), & i = -1, \\ H_{-i-1}(G,A), & i \leq -2. \end{cases}$$

Equivalently,

$$\widehat{H}^{-1}(G,A) = \ker(\overline{N}_G) = \frac{\ker(N_G : A \to A)}{I_G A}.$$

This is Sharifi's Definition 1.6.5 [**8**, pp. 18–19].

Theorem 23.6. — *For every short exact sequence of G-modules*

$$0 \to A \to B \to C \to 0,$$

there is a natural long exact sequence

$$\cdots \to \widehat{H}^i(G,A) \to \widehat{H}^i(G,B) \to \widehat{H}^i(G,C) \xrightarrow{\delta} \widehat{H}^{i+1}(G,A) \to \cdots.$$

This is Tate's long exact sequence [**8**, Theorem 1.6.6, p. 19].

Proposition 23.7. — *If A is an induced G-module, then*

$$\widehat{H}^i(G,A) = 0 \qquad \textit{for all } i \in \mathbf{Z}.$$

Proof. — By Shapiro's lemma, the ordinary cohomology groups $H^i(G,A)$ vanish for $i \geq 1$, and the homology groups $H_q(G,A)$ vanish for $q \geq 1$. Thus only $i = 0$ and $i = -1$ remain.

It is enough to take $A = \mathrm{Ind}^G(X) = \mathbf{Z}[G] \otimes_{\mathbf{Z}} X$. The invariants are

$$A^G = N_G \mathbf{Z}[G] \otimes_{\mathbf{Z}} X = N_G A,$$

so $\widehat{H}^0(G,A) = 0$. Moreover

$$A_G \cong X,$$

and under this identification the norm map $\overline{N}_G : A_G \to A^G$ sends x to $N_G \otimes x$, which is injective. Hence $\widehat{H}^{-1}(G,A) = \ker(\overline{N}_G) = 0$. This proves the proposition. □

Lecture 26

This lecture extends cup products to Tate cohomology. We explain complete resolutions, the compatibility with ordinary cup products, and the restriction–corestriction formalism needed for Tate–Nakayama.

24.1 Complete Resolutions. — Let G be a finite group. We use the following precise construction of a complete resolution.

Theorem 24.1. — *Let*

$$P_\bullet \xrightarrow{\alpha} \mathbf{Z}$$

be a projective resolution of $\mathbf{Z}$ *by* G*-modules which are free of finite rank over* $\mathbf{Z}$*. Put*

$$P_i^* = \mathrm{Hom}_{\mathbf{Z}}(P_i, \mathbf{Z}), \qquad (g\varphi)(x) = \varphi(g^{-1}x),$$

and let

$$\widehat{\alpha} : \mathbf{Z} \longrightarrow P_0^*$$

be the $\mathbf{Z}$*-dual of* $\alpha : P_0 \to \mathbf{Z}$*. Define* $Q_\bullet$ *by*

$$\cdots \longrightarrow P_1 \longrightarrow P_0 \xrightarrow{\widehat{\alpha}\circ\alpha} P_0^* \longrightarrow P_1^* \longrightarrow \cdots,$$

where P_0 *is placed in degree* 0*; equivalently,*

$$Q_i = P_i \quad (i \geq 0), \qquad Q_i = P^*_{-1-i} \quad (i < 0).$$

Then $Q_\bullet$ *is exact, and for every* G*-module* A*,*

$$\widehat{H}^i(G, A) = H^i\big(\mathrm{Hom}_{\mathbf{Z}[G]}(Q_\bullet, A)\big) \qquad (i \in \mathbf{Z}).$$

Proof. — Since each P_i is projective over $\mathbf{Z}[G]$, it is free over $\mathbf{Z}$. Hence $\mathbf{Z}$-dualising preserves exactness of

$$\cdots \to P_1 \to P_0 \xrightarrow{\alpha} \mathbf{Z} \to 0.$$

Thus the dual sequence

$$0 \to \mathbf{Z} \xrightarrow{\widehat{\alpha}} P_0^* \to P_1^* \to \cdots$$

is exact. At the middle map, let $\beta = \widehat{\alpha} \circ \alpha$. Then

$$\mathrm{im}(P_1 \to P_0) = \ker(\alpha) = \ker(\beta),$$

because $\widehat{\alpha}$ is injective, and

$$\mathrm{im}(\beta) = \mathrm{im}(\widehat{\alpha}) = \ker(P_0^* \to P_1^*),$$

because α is surjective. Hence $Q_\bullet$ is exact.

For $i \geq 1$, the complex $\mathrm{Hom}_{\mathbf{Z}[G]}(Q_\bullet, A)$ is just the usual cochain complex computing $H^i(G, A)$, hence it computes $\widehat{H}^i(G, A)$. For $i \leq -2$, the negative part of $Q_\bullet$, after applying $\mathrm{Hom}_{\mathbf{Z}[G]}(-, A)$, identifies with the homology complex

$$P_\bullet \otimes_{\mathbf{Z}[G]} A$$

through the standard finite-rank duality isomorphism

$$P_i \otimes_{\mathbf{Z}[G]} A \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Z}[G]}(P_i^*, A).$$

Thus the cohomology in degree $i \leq -2$ is

$$H_{-i-1}(G, A) = \widehat{H}^i(G, A).$$

It remains to check the two middle degrees. The middle part of $\mathrm{Hom}_{\mathbf{Z}[G]}(Q_\bullet, A)$ is identified with

$$P_1 \otimes_{\mathbf{Z}[G]} A \longrightarrow P_0 \otimes_{\mathbf{Z}[G]} A \longrightarrow \mathrm{Hom}_{\mathbf{Z}[G]}(P_0^*, A) \longrightarrow \mathrm{Hom}_{\mathbf{Z}[G]}(P_1^*, A).$$

Under the augmentations, the cokernel of the first map is A_G, and the kernel of the last map is A^G. The middle map is exactly the norm map

$$\overline{N}_G : A_G \longrightarrow A^G.$$

Therefore the cohomology in degrees -1 and 0 is respectively

$$\ker(\overline{N}_G) = \widehat{H}^{-1}(G, A), \qquad \mathrm{coker}(\overline{N}_G) = A^G/N_G A = \widehat{H}^0(G, A).$$

This proves the theorem. □

A complex $Q_\bullet$ obtained in this way is called a *complete resolution* of $\mathbf{Z}$. Concretely, it is formed by putting a finite-rank projective resolution

$$\cdots \to P_1 \to P_0 \to \mathbf{Z} \to 0$$

on the left and its $\mathbf{Z}$-dual

$$0 \to \mathbf{Z} \to P_0^* \to P_1^* \to \cdots$$

on the right, joined by the middle map

$$P_0 \xrightarrow{\widehat{\alpha}\circ\alpha} P_0^*.$$

Remark 24.2. — Abstractly, a complete resolution of $\mathbf{Z}$ is a doubly infinite exact complex of projective $\mathbf{Z}[G]$-modules

$$\cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow P_{-1} \longrightarrow P_{-2} \longrightarrow \cdots$$

which agrees in nonnegative degrees with a projective resolution of $\mathbf{Z}$ and whose negative part is the dual resolution. For a G-module A, Tate cohomology is computed as

$$\widehat{H}^n(G, A) = H^n\big(\mathrm{Hom}_{\mathbf{Z}[G]}(P_\bullet, A)\big), \qquad n \in \mathbf{Z}.$$

The reason complete resolutions are useful is that they let us use the same cochain language in every degree. A class in $\widehat{H}^i(G, A)$ is represented by a homogeneous G-equivariant map $P_i \to A$, modulo the usual coboundary relation. Thus the negative-degree classes can be multiplied with positive-degree classes without leaving the cochain formalism.

24.2 Cup Products in Tate Cohomology. — The ordinary cup product at the cochain level was already defined in Lecture 22: see Definition 21.1, Proposition 21.2, and Corollary 21.3 and the following corollary. In this lecture we only explain how the same formalism extends from ordinary cohomology to Tate cohomology.

Remark 24.3. — If $a \in H^0(G, A) = A^G$ and $\beta \in H^q(G, B)$, then

$$a \cup \beta = f_*(\beta),$$

where $f : B \to A \otimes_{\mathbf{Z}} B$ is the G-homomorphism

$$x \longmapsto a \otimes x.$$

In particular, for $q = 0$, the cup product is simply

$$A^G \times B^G \longrightarrow (A \otimes_{\mathbf{Z}} B)^G, \qquad (a, b) \longmapsto a \otimes b.$$

More generally, if $\varphi : A \times B \to C$ is a G-equivariant bilinear pairing, then φ factors through $A \otimes_{\mathbf{Z}} B$, and we write

$$H^i(G, A) \times H^j(G, B) \longrightarrow H^{i+j}(G, C)$$

again as a cup product. This is Harari's Remark 2.22 [**4**, p. 44].

Proposition 24.4. — *Cup products are compatible with connecting homomorphisms as follows.*

1. *Let*

$$0 \to A' \to A \to A'' \to 0, \qquad 0 \to C' \to C \to C'' \to 0$$

be exact sequences of G-modules, and suppose a pairing $A \times B \to C$ induces pairings $A' \times B \to C'$ and $A'' \times B \to C''$. Then

$$(\delta\alpha'') \cup \beta = \delta(\alpha'' \cup \beta).$$

2. *Let*

$$0 \to B' \to B \to B'' \to 0, \qquad 0 \to C' \to C \to C'' \to 0$$

be exact sequences of G-modules, and suppose a pairing $A \times B \to C$ induces pairings $A \times B' \to C'$ and $A \times B'' \to C''$. Then

$$\alpha \cup (\delta\beta'') = (-1)^i \delta(\alpha \cup \beta''), \qquad \alpha \in H^i(G, A).$$

Proof. — We prove the second formula; the first is the same without the sign. Let $a \in Z^i(G, A)$ represent α, and let $b'' \in Z^j(G, B'')$ represent β''. Choose a cochain $b \in C^j(G, B)$ lifting b''. Then db takes values in B' and represents $\delta\beta''$. Since $da = 0$, Lemma 21.2 gives

$$d(a \cup b) = (-1)^i a \cup db.$$

The cochain $a \cup b$ lifts $a \cup b''$, so $d(a \cup b)$ represents $\delta(\alpha \cup \beta'')$. Hence

$$\alpha \cup (\delta\beta'') = (-1)^i \delta(\alpha \cup \beta'').$$

□

We now pass from ordinary cohomology to Tate cohomology, following Harari [**4**, §2.6]. Let $Q_\bullet$ be the complete resolution of Theorem 24.1. A Tate i-cochain with values in A is an element of

$$\mathrm{Hom}_{\mathbf{Z}[G]}(Q_i, A).$$

To define cup products in all degrees, one chooses maps

$$\kappa_{i,j} : Q_{i+j} \longrightarrow Q_i \otimes_{\mathbf{Z}} Q_j$$

which extend the homogeneous diagonal maps in nonnegative degrees and satisfy the chain identity corresponding to Proposition 21.2. For

$$\varphi \in \mathrm{Hom}_{\mathbf{Z}[G]}(Q_i, A), \qquad \psi \in \mathrm{Hom}_{\mathbf{Z}[G]}(Q_j, B),$$

define

$$\varphi \cup \psi : Q_{i+j} \longrightarrow A \otimes_{\mathbf{Z}} B$$

as the composite

$$Q_{i+j} \xrightarrow{\kappa_{i,j}} Q_i \otimes_{\mathbf{Z}} Q_j \xrightarrow{\varphi\otimes\psi} A \otimes_{\mathbf{Z}} B.$$

Harari formulates the same construction using maps

$$\varphi_{p,q} : Q_{p+q} \longrightarrow Q_p \otimes_{\mathbf{Z}} Q_q$$

satisfying

$$\varphi_{p,q}\partial = (\partial \otimes 1)\varphi_{p+1,q} + (-1)^p(1 \otimes \partial)\varphi_{p,q+1}.$$

This identity is the complete-resolution analogue of Proposition 21.2. The cases $p, q \geq 0$ are the ordinary homogeneous formula; the mixed-sign cases are written explicitly in Harari's proof of Lemma 2.26 [**4**, pp. 46–47]. Sharifi states the same point by saying that, for the complex $Q_\bullet$ of Theorem 1.6.10, the proof of the ordinary cup product goes through after choosing the maps $\kappa_{i,j}$ [**8**, Theorem 1.9.12, p. 40].

Theorem 24.5. — *Let G be a finite group. There is a unique family of cup products*

$$\widehat{H}^i(G, A) \otimes_{\mathbf{Z}} \widehat{H}^j(G, B) \longrightarrow \widehat{H}^{i+j}(G, A \otimes_{\mathbf{Z}} B)$$

which is natural in A and B, and satisfies the following properties. First, in degree 0 the diagram

$$\begin{array}{ccc} H^0(G, A) \otimes_{\mathbf{Z}} H^0(G, B) & \xrightarrow{\ \cup\ } & H^0(G, A \otimes_{\mathbf{Z}} B) \\ \downarrow & & \downarrow \\ \widehat{H}^0(G, A) \otimes_{\mathbf{Z}} \widehat{H}^0(G, B) & \xrightarrow{\ \cup\ } & \widehat{H}^0(G, A \otimes_{\mathbf{Z}} B) \end{array}$$

commutes. Here the vertical maps are the natural quotient maps $A^G \to A^G/N_G A$, $B^G \to B^G/N_G B$, and $(A \otimes_{\mathbf{Z}} B)^G \to (A \otimes_{\mathbf{Z}} B)^G/N_G(A \otimes_{\mathbf{Z}} B)$. Second, if $0 \to A_1 \to A \to A_2 \to 0$ is exact and remains exact after tensoring with B, then

$$\delta(\alpha_2 \cup \beta) = (\delta\alpha_2) \cup \beta.$$

If $0 \to B_1 \to B \to B_2 \to 0$ *is exact and remains exact after tensoring with* A*, then*

$$\delta(\alpha \cup \beta_2) = (-1)^i \alpha \cup (\delta\beta_2), \qquad \alpha \in \widehat{H}^i(G, A).$$

Proof. — The existence is the complete-resolution construction just described. The identity for $\varphi_{p,q}$ gives

$$d(f \cup g) = df \cup g + (-1)^i f \cup dg$$

for Tate cochains of degrees i and j. Hence the product of two cocycles is a cocycle, and changing either cocycle by a coboundary changes the product by a coboundary.

Naturality follows because a G-homomorphism $A \to A'$ sends a cochain $f : Q_i \to A$ to the cochain $Q_i \to A'$, and similarly in the second variable. The formula defining $f \cup g$ is functorial in both coefficient modules.

In positive degrees, the complete resolution agrees with the usual projective resolution, and the maps $\kappa_{i,j}$ restrict to the ordinary homogeneous diagonal. Thus the product agrees with the ordinary cup product for $i, j \geq 1$. In degree 0, the formula is the ordinary product on invariants followed by the quotient defining $\widehat{H}^0$, which is precisely the displayed commutative diagram.

For the connecting maps, take an exact sequence

$$0 \to A_1 \to A \to A_2 \to 0$$

which remains exact after tensoring with B. Let α_2 be represented by a cocycle $f_2 : Q_i \to A_2$, and lift it to a cochain $f : Q_i \to A$. Then df takes values in A_1, and the class of df represents $\delta\alpha_2$. If $g : Q_j \to B$ represents β, then

$$d(f \cup g) = df \cup g + (-1)^i f \cup dg = df \cup g,$$

because g is a cocycle. This is exactly $\delta(\alpha_2 \cup \beta) = (\delta\alpha_2) \cup \beta$.

The second formula is the same calculation with the connecting map in the second variable. If $g_2 : Q_j \to B_2$ represents β_2 and $g : Q_j \to B$ is a lift, then dg represents $\delta\beta_2$. Since f is a cocycle,

$$d(f \cup g) = (-1)^i f \cup dg,$$

which gives

$$\delta(\alpha \cup \beta_2) = (-1)^i \alpha \cup (\delta\beta_2).$$

Finally, uniqueness is obtained by dimension shifting. Embed any G-module A into an induced module $I(A)$, for example the coinduced module of maps $G \to A$. The quotient A' in

$$0 \to A \to I(A) \to A' \to 0$$

satisfies $\widehat{H}^q(G, I(A)) = 0$ for all q by Proposition 23.7. Hence the connecting maps give canonical isomorphisms

$$\widehat{H}^{q-1}(G, A') \xrightarrow{\sim} \widehat{H}^q(G, A).$$

Repeatedly applying this reduction moves any degree to degree 0, where the product is specified by the displayed diagram. The two compatibility formulas therefore force the product in all degrees. □

***Remark 24.6* (Why the negative degrees matter).** — The product used in local reciprocity is genuinely a Tate product:

$$\widehat{H}^{-2}(G,\mathbf{Z}) \times H^2(G,L^\times) \longrightarrow \widehat{H}^0(G,L^\times).$$

The first factor is homological:

$$\widehat{H}^{-2}(G,\mathbf{Z}) = H_1(G,\mathbf{Z}) \cong G^{\mathrm{ab}},$$

while the second factor is ordinary cohomology. The complete resolution is what allows these two kinds of classes to be multiplied in one theory.

Remark 24.7. — The same notation is used when the tensor product is followed by a G-equivariant map $A \otimes_{\mathbf{Z}} B \to C$. Thus a bilinear G-equivariant pairing $A \times B \to C$ gives

$$\widehat{H}^i(G,A) \times \widehat{H}^j(G,B) \longrightarrow \widehat{H}^{i+j}(G,C).$$

This is the form used for the product with the fundamental class in Lecture 25.

***Proposition 24.8* (Graded commutativity).** — *Let $s_{A,B} : A \otimes_{\mathbf{Z}} B \to B \otimes_{\mathbf{Z}} A$ be the symmetry map $a \otimes b \mapsto b \otimes a$. For*

$$\alpha \in \widehat{H}^i(G,A), \qquad \beta \in \widehat{H}^j(G,B),$$

one has

$$s_{A,B,*}(\alpha \cup \beta) = (-1)^{ij}\, \beta \cup \alpha.$$

Proof. — Define another pairing with values in $A \otimes_{\mathbf{Z}} B$ by

$$\alpha \star \beta := (-1)^{ij} s_{B,A,*}(\beta \cup \alpha), \qquad \alpha \in \widehat{H}^i(G,A), \quad \beta \in \widehat{H}^j(G,B).$$

We show that $\star$ satisfies the characterizing properties of the Tate cup product in Theorem 24.5.

First, $\star$ is natural in A and B. In nonnegative degrees it agrees with the ordinary cup product, because ordinary cohomology satisfies

$$s_{A,B,*}(\alpha \cup \beta) = (-1)^{ij} \beta \cup \alpha$$

[**8**, Corollary 1.9.7, pp. 37–38].

It remains to check the connecting morphisms. Suppose

$$0 \to A_1 \to A \to A_2 \to 0$$

is exact and remains exact after tensoring with B. For $\alpha_2 \in \widehat{H}^i(G,A_2)$ and $\beta \in \widehat{H}^j(G,B)$, the connecting morphism is on the second factor of $\beta \cup \alpha_2$. Hence Theorem

24.5 gives

$$\begin{aligned}\delta(\alpha_2 \star \beta) &= (-1)^{ij} s_{B,A_1,*}\big(\delta(\beta \cup \alpha_2)\big)\\ &= (-1)^{ij}(-1)^j s_{B,A_1,*}\big(\beta \cup \delta\alpha_2\big)\\ &= (-1)^{(i+1)j} s_{B,A_1,*}\big(\beta \cup \delta\alpha_2\big)\\ &= (\delta\alpha_2) \star \beta.\end{aligned}$$

Now suppose

$$0 \to B_1 \to B \to B_2 \to 0$$

is exact and remains exact after tensoring with A. For $\beta_2 \in \widehat{H}^j(G, B_2)$, the connecting morphism is on the first factor of $\beta_2 \cup \alpha$. Thus

$$\begin{aligned}\delta(\alpha \star \beta_2) &= (-1)^{ij} s_{B_1,A,*}\big(\delta(\beta_2 \cup \alpha)\big)\\ &= (-1)^{ij} s_{B_1,A,*}\big((\delta\beta_2) \cup \alpha\big)\\ &= (-1)^i(-1)^{i(j+1)} s_{B_1,A,*}\big((\delta\beta_2) \cup \alpha\big)\\ &= (-1)^i \alpha \star (\delta\beta_2).\end{aligned}$$

Therefore $\star$ has the same defining properties as the Tate cup product. By the uniqueness in Theorem 24.5, $\alpha \star \beta = \alpha \cup \beta$. Applying $s_{A,B,*}$ to this equality gives

$$s_{A,B,*}(\alpha \cup \beta) = (-1)^{ij} \beta \cup \alpha.$$

□

***Proposition 24.9* (Restriction and corestriction).** — *Let $H \subset G$ be a subgroup. Tate cup products are compatible with restriction:*

$$\mathrm{Res}(\alpha \cup \beta) = \mathrm{Res}(\alpha) \cup \mathrm{Res}(\beta).$$

If H has finite index in G, then they are also compatible with corestriction in the projection-formula sense:

$$\mathrm{Cor}(\alpha) \cup \beta = \mathrm{Cor}\big(\alpha \cup \mathrm{Res}(\beta)\big),$$

for $\alpha \in \widehat{H}^i(H, A)$ and $\beta \in \widehat{H}^j(G, B)$.

Lecture 27-28

These lectures study cohomologically trivial modules and Tate cohomology criteria for finite groups. The central theme is reduction to p-Sylow subgroups and the use of dimension shifting. Finally we reach the Tate-Nakayama theorem, which is a main input in the local class field theory.

Let p be a prime number. Recall that a finite p-group is a finite group whose order is a power of p. Such a group is nilpotent; in particular, the center and the abelianization of a non-trivial p-group are non-trivial. A p-Sylow subgroup H of a finite group G is a p-subgroup whose index is prime to p.

We will use the special cohomological properties of finite p-groups to reduce parts of the cohomology of a finite group to the cohomology of its p-Sylow subgroups.

25.1 Cohomologically Trivial Modules. —

Lemma 25.1. — *Let G be a finite p-group, and let $A \neq 0$ be a torsion p-primary G-module. Then $A^G \neq 0$.*

Lemma 25.2. — *Let G be a finite group, let $H \subseteq G$ be a subgroup, and let A be a G-module. If $p \nmid [G : H]$, then for every $q > 0$, the restriction map*

$$\mathrm{Res} : H^q(G, A)\{p\} \longrightarrow H^q(H, A)$$

is injective. The same statement holds for all $q \in \mathbf{Z}$ if ordinary cohomology is replaced by Tate cohomology:

$$\mathrm{Res} : \widehat{H}^q(G, A)\{p\} \longrightarrow \widehat{H}^q(H, A).$$

Proof. — For group cohomology, one has (Lemma 28.22)

$$\mathrm{Cor} \circ \mathrm{Res} = [G : H] \cdot \mathrm{id}\,.$$

The same formula holds for Tate cohomology. Since $[G : H]$ is prime to p, multiplication by $[G : H]$ is an automorphism on the p-primary component. Thus restriction is injective on the p-primary component. □

Lemma 25.3. — *Let G be a finite group, and let A, B be G-modules.*

1. *If B is an induced G-module, then* $\mathrm{Hom}_{\mathbf{Z}}(A, B)$ *is also an induced G-module, where G acts by*
$$(g \cdot f)(a) = g \cdot f(g^{-1}a).$$
2. *If A is a projective G-module, then $A \otimes_{\mathbf{Z}} B$ is relatively injective*[6]*.*

We shall use the following shifting notation. For any G-module A, let

$$0 \longrightarrow A \longrightarrow I_G(A) \longrightarrow A_1 \longrightarrow 0$$

be the natural embedding into the induced module $I_G(A)$. Define inductively

$$A_q = (A_{q-1})_1 \qquad (q > 0).$$

Similarly, using the natural surjection

$$I_G(A) \longrightarrow A,$$

let A_{-1} be its kernel and define

$$A_q = (A_{q+1})_{-1} \qquad (q < 0).$$

Then dimension shifting gives

$$\widehat{H}^q(G, A) \simeq \widehat{H}^{q-r}(G, A_r) \quad \forall q, r \in \mathbf{Z}.$$

[6] We call a G-module relatively injective if it is a direct factor of the induced G-module $I_G(A)$ (where A is an abelian group).

Definition 25.4. — Let G be a finite group. A G-module A is called *cohomologically trivial* if, for every subgroup $H \subseteq G$ and every integer $n > 0$, one has

$$H^n(H, A) = 0.$$

Every induced G-module is cohomologically trivial, because its restriction to any subgroup is again induced.

Lemma 25.5. — *Let G be a finite group, and for each prime p, let G_p be a p-Sylow subgroup of G. A G-module A is cohomologically trivial if and only if A is cohomologically trivial as a G_p-module for every prime p.*

Proof. — One direction is immediate.

Conversely, let $H \subseteq G$ be any subgroup. For a prime p, choose a p-Sylow subgroup $H_p \subseteq H$. It is contained in some p-Sylow subgroup of G, which we may identify with G_p up to conjugacy. By the previous restriction lemma, for $n > 0$,

$$H^n(H, A)\{p\} \longrightarrow H^n(H_p, A)$$

is injective since $[H : H_p]$ is prime to p. If A is cohomologically trivial over every G_p, then $H^n(H_p, A) = 0$. Hence every p-primary component of $H^n(H, A)$ is zero, and therefore

$$H^n(H, A) = 0.$$

Thus A is cohomologically trivial. □

Theorem 25.6. — *Let G be a finite p-group, and let A be a p-torsion G-module. Assume that there exists an integer $q \in \mathbf{Z}$ such that $\widehat{H}^q(G, A) = 0$. Then A is an induced G-module. Thus A is cohomologically trivial. In particular, A is a free $\mathbf{F}_p[G]$ module.*

Proof. — We first prove the special case $H^1(G, A) = 0$. Let $\Lambda = \mathbb{F}_p[G]$. Choose an $\mathbb{F}_p$-basis I of A^G, and set $V = \bigoplus_{i \in I} \Lambda$. Then V is an induced G-module, and $V^G \simeq A^G$ (Since $\Lambda_G = \mathbf{F}_p$). Denote the isomorphism by j_G. We want to extend j_G to a G-morphism

$$j : A \longrightarrow V.$$

Consider the exact sequence $0 \longrightarrow A^G \longrightarrow A \longrightarrow A/A^G \longrightarrow 0$. Applying $\mathrm{Hom}_{\mathbb{F}_p}(-, V)$, we get an exact sequence of G-modules

$$0 \longrightarrow \mathrm{Hom}_{\mathbb{F}_p}(A/A^G, V) \longrightarrow \mathrm{Hom}_{\mathbb{F}_p}(A, V) \longrightarrow \mathrm{Hom}_{\mathbb{F}_p}(A^G, V) \longrightarrow 0.$$

The first term is induced by the previous lemma, because V is induced. Therefore its first cohomology vanishes. Taking G-invariants gives a surjection

$$\mathrm{Hom}_G(A, V) \twoheadrightarrow \mathrm{Hom}_G(A^G, V) = \mathrm{Hom}_G(A^G, V^G)$$

Thus j_G extends to a G-morphism $j : A \longrightarrow V$.

The kernel of j has no nonzero G-invariants, since j is an isomorphism on A^G. Since $\ker j$ is a p-torsion module for the p-group G, $\ker j \neq 0 \implies (\ker j)^G \neq 0$, a contradiction. Hence j is injective.

Let $C = \operatorname{coker}(j)$. From the exact sequence

$$0 \longrightarrow A \longrightarrow V \longrightarrow C \longrightarrow 0$$

we get

$$0 \longrightarrow A^G \longrightarrow V^G \longrightarrow C^G \longrightarrow H^1(G, A).$$

Since $A^G \to V^G$ is an isomorphism and $H^1(G, A) = 0$, it follows that $C^G = 0 \implies C = 0$. Thus j is an isomorphism, and $A \simeq V$ is induced.

Now assume only that $\widehat{H}^q(G, A) = 0$ for some $q \in \mathbf{Z}$. By dimension shifting,

$$\widehat{H}^q(G, A) \simeq H^1(G, A_{q-1}).$$

Thus $H^1(G, A_{q-1}) = 0$. By the special case just proved, A_{q-1} is induced. Therefore its Tate cohomology vanishes in every degree. Shifting back gives

$$H^1(G, A) \simeq \widehat{H}^{2-q}(G, A_{q-1}) = 0.$$

Applying the special case once more, we conclude that A itself is induced. Since induced modules are cohomologically trivial, the theorem follows. □

Theorem 25.7. — *Let G be a finite group, and for each prime $p \mid |G|$, let G_p be a p-Sylow subgroup of G. Let A be a G-module.*

(a) Assume that for every prime $p \mid |G|$, there exists an integer $q_p \in \mathbf{Z}$ such that

$$\widehat{H}^{q_p}(G_p, A) = 0, \qquad \widehat{H}^{q_p+1}(G_p, A) = 0.$$

Then A is cohomologically trivial.

Moreover, if A is free as a $\mathbf{Z}$-module, then A is projective as a $\mathbf{Z}[G]$-module (hence relatively injective).

(b) Conversely, if A is cohomologically trivial, then

$$\widehat{H}^q(H, A) = 0$$

for every subgroup $H \subseteq G$ and every $q \in \mathbf{Z}$.

In addition, there exists an exact sequence of G-modules

$$0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0,$$

where F is free over $\mathbf{Z}[G]$ and R is projective over $\mathbf{Z}[G]$.

Proof. — Choose a surjection from a free $\mathbf{Z}[G]$-module onto A:

$$0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0.$$

Thus F is free over $\mathbf{Z}[G]$, hence relatively injective, and therefore $\widehat{H}^i(H, F) = 0$ for every subgroup $H \subseteq G$ and every $i \in \mathbf{Z}$.

Fix a prime $p \mid |G|$. By assumption, for the p-Sylow G_p, there is an integer $q = q_p$ such that

$$\widehat{H}^q(G_p, A) = 0, \qquad \widehat{H}^{q+1}(G_p, A) = 0.$$

The long exact cohomology sequence attached to

$$0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0$$

then gives

$$\widehat{H}^{q+1}(G_p, R) = 0, \qquad \widehat{H}^{q+2}(G_p, R) = 0.$$

Since R is a subgroup of the free abelian group F, the group R is torsion-free. Hence multiplication by p gives an exact sequence

$$0 \longrightarrow R \xrightarrow{p} R \longrightarrow R/pR \longrightarrow 0.$$

Using the corresponding long exact sequence of Tate cohomology, the previous vanishing implies

$$\widehat{H}^{q+1}(G_p, R/pR) = 0.$$

The module R/pR is killed by p. Therefore we can apply Theorem 25.6 on R/pR, thus it is an induced G_p-module. □

We first prove the projectivity statement under the extra assumption that A is free over $\mathbf{Z}$. Define $M := \mathrm{Hom}_{\mathbf{Z}}(A, R)$ with the usual G-action $(g \cdot f)(a) = g\, f(g^{-1}a)$.

Lemma 25.8. — *With the notation above, one has*

$$H^1(G, M) = 0.$$

Proof. — Because A is free over $\mathbf{Z}$, reduction modulo p gives

$$M/pM \simeq \mathrm{Hom}_{\mathbf{Z}}(A, R/pR).$$

As R/pR is an induced G_p-module, the module $\mathrm{Hom}_{\mathbf{Z}}(A, R/pR)$ is induced as a G_p-module. Hence M/pM is induced as a G_p-module.

Now consider

$$0 \longrightarrow M \xrightarrow{p} M \longrightarrow M/pM \longrightarrow 0.$$

Since M/pM is induced over G_p, its Tate cohomology vanishes. The associated long exact sequence shows that multiplication by p is injective on $H^1(G_p, M)$. Equivalently, $H^1(G_p, M)[p] = 0$.

By Lemma 25.2, the restriction $H^1(G, M)\{p\} \hookrightarrow H^1(G_p, M)$ is injective, Therefore $H^1(G, M)\{p\} = 0$. Since this holds for every prime p, we conclude that $H^1(G, M) = 0$. □

Continue proof on Theorem 25.7. — First keep assuming that A is free over $\mathbf{Z}$. Since A is free, applying $\mathrm{Hom}_{\mathbf{Z}}(A, -)$ to $0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0$ gives an exact sequence of G-modules

$$0 \longrightarrow \mathrm{Hom}_{\mathbf{Z}}(A, R) \longrightarrow \mathrm{Hom}_{\mathbf{Z}}(A, F) \longrightarrow \mathrm{Hom}_{\mathbf{Z}}(A, A) \longrightarrow 0.$$

Taking G-invariants and using $H^1(G, M) = 0$, we obtain a surjection

$$\mathrm{Hom}_G(A, F) \twoheadrightarrow \mathrm{Hom}_G(A, A).$$

Thus the identity map of A lifts to a G-morphism $s : A \longrightarrow F$ such that the composite

$$A \xrightarrow{s} F \longrightarrow A$$

is the identity. Hence the surjection $F \to A$ splits, and $F \simeq A \oplus R$ as G-modules. Therefore A is a direct summand of a free $\mathbf{Z}[G]$-module, so A is projective over $\mathbf{Z}[G]$.

Now we prove part (a) without assuming that A is free over $\mathbf{Z}$. The module R is torsion-free, since it is a subgroup of the free abelian group F. Moreover, the vanishing already obtained gives, for every prime $p \mid |G|$, two consecutive vanishing Tate groups for R over G_p. Hence the projective case just proved applies to R. Therefore R is projective over $\mathbf{Z}[G]$ (in particular, relative injective). Hence, R cohomologically trivial.

The module F is free, hence also projective and cohomologically trivial. From the exact sequence $0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0$, the long exact cohomology sequence shows that A is cohomologically trivial. This proves part (a).

We now prove part (b). Suppose A is cohomologically trivial. Then A satisfies the hypotheses of part (a), for example by taking $q = 1$ for each Sylow subgroup. Therefore, for any surjection $F \longrightarrow A$ with F free over $\mathbf{Z}[G]$, its kernel R is projective over $\mathbf{Z}[G]$. This gives the desired exact sequence.

Finally, let $H \subseteq G$ be any subgroup. Since R is projective over $\mathbf{Z}[G]$, it is also projective over $\mathbf{Z}[H]$. Thus R is relatively injective as an H-module, and so $\widehat{H}^q(H, R) = 0$ for all $q \in \mathbf{Z}$. The same is true for F. The long exact sequence of Tate cohomology therefore gives

$$\widehat{H}^q(H, A) = 0, \quad \forall q \in \mathbf{Z}.$$

□

Remark 25.9. — The previous theorem shows that, for a finite group G, a G-module A is cohomologically trivial if and only if

$$\widehat{H}^q(H, A) = 0$$

for every subgroup $H \subseteq G$ and every $q \in \mathbf{Z}$.

25.2 Tate-Nakayama Theorem. — Throughout this subsection, G is a finite group. For every prime number p, we fix a p-Sylow subgroup $G_p \subseteq G$.

Lemma 25.10. — *Let A be a cohomologically trivial G-module, and let B be a torsion-free G-module. Then $A \otimes_{\mathbf{Z}} B$ is cohomologically trivial.*

Proof. — By the criterion for cohomologically trivial modules (previous theorem), there exists an exact sequence $0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0$ where F is free over $\mathbf{Z}[G]$ and R is a direct factor of a free $\mathbf{Z}[G]$-module.

Tensoring with B, and using that B is torsion-free, gives an exact sequence

$$0 \longrightarrow R \otimes B \longrightarrow F \otimes B \longrightarrow A \otimes B \longrightarrow 0.$$

The modules $F \otimes B$ and $R \otimes B$ are relatively injective, hence cohomologically trivial. Therefore the long exact cohomology sequence implies that $A \otimes B$ is also cohomologically trivial. $\square$

Proposition 25.11. — *Let*

$$f : A' \longrightarrow A$$

be a morphism of G-modules. Assume that for every prime p, there exists an integer n_p such that the induced map

$$f_*^i : \widehat{H}^i(G_p, A') \longrightarrow \widehat{H}^i(G_p, A)$$

is surjective for $i = n_p$, bijective for $i = n_p + 1$, and injective for $i = n_p + 2$.

Let B be a torsion-free G-module. Then, for every subgroup $H \subseteq G$ and every integer i, the morphism

$$\widehat{H}^i(H, A' \otimes B) \longrightarrow \widehat{H}^i(H, A \otimes B)$$

induced by $f \otimes \mathrm{id}_B$ is an isomorphism.

In particular, taking $B = \mathbf{Z}$, the map

$$\widehat{H}^i(H, A') \longrightarrow \widehat{H}^i(H, A)$$

is an isomorphism for all $H \subseteq G$ and all $i \in \mathbf{Z}$.

Proof. — We first treat the case where f is injective. Let $A'' = \mathrm{coker}(f)$, so that we have an exact sequence

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0.$$

The long exact sequence of Tate cohomology, together with the assumptions on f_*^i, gives

$$\widehat{H}^{n_p}(G_p, A'') = 0, \qquad \widehat{H}^{n_p+1}(G_p, A'') = 0$$

for every prime p.

By the cohomological triviality criterion, A'' is cohomologically trivial. Therefore, by the previous lemma, $A'' \otimes B$ is also cohomologically trivial. Since B is torsion-free, tensoring the above short exact sequence with B remains exact:

$$0 \longrightarrow A' \otimes B \longrightarrow A \otimes B \longrightarrow A'' \otimes B \longrightarrow 0.$$

The long exact Tate cohomology sequence now shows that

$$\widehat{H}^i(H, A' \otimes B) \xrightarrow{\sim} \widehat{H}^i(H, A \otimes B)$$

for all subgroups $H \subseteq G$ and all $i \in \mathbf{Z}$.

We now reduce the general case to the injective case. Embed A' into the induced module $I_G(A')$. Let

$$A^* := A \oplus I_G(A')$$

and define

$$\theta : A' \longrightarrow A^*, \qquad \theta(x) = (f(x), j(x)),$$

where $j : A' \hookrightarrow I_G(A')$ is the chosen embedding. Then θ is injective.

Since induced modules are cohomologically trivial, the Tate cohomology of A^* identifies with that of A. The same remains true after tensoring with B. Thus the injective case applied to θ proves the result for f. $\square$

Proposition 25.12. — *Let A, B, C be G-modules, and let $\varphi : A \times B \longrightarrow C$ be a G-equivariant bilinear map. Let $a \in \widehat{H}^q(G, A)$ for some $q \in \mathbf{Z}$.*

For a subgroup $H \subseteq G$, denote by $a_H = \mathrm{Res}^G_H(a)$ the restriction of a. For any G-module D, define

$$f(n, H, D) : \widehat{H}^n(H, B \otimes D) \longrightarrow \widehat{H}^{n+q}(H, C \otimes D)$$

to be cup-product with a_H, using the bilinear map induced by φ.

Assume that for every prime p, there exists an integer n_p such that $f(n, G_p, \mathbf{Z})$ is surjective for $n = n_p$, bijective for $n = n_p + 1$, and injective for $n = n_p + 2$.

Then, for every subgroup $H \subseteq G$, every integer n, and every torsion-free G-module D, the map

$$f(n, H, D) : \widehat{H}^n(H, B \otimes D) \longrightarrow \widehat{H}^{n+q}(H, C \otimes D)$$

is an isomorphism.

Proof. — We first prove the case $q = 0$. Then

$$a \in \widehat{H}^0(G, A) = A^G / N_G A$$

is represented by a G-invariant element of A, which we still denote by a. Define a G-morphism

$$f : B \longrightarrow C, \qquad b \longmapsto \varphi(a, b).$$

The maps $f(n, H, D)$ are precisely the maps on Tate cohomology induced by

$$f \otimes \mathrm{id}_D : B \otimes D \longrightarrow C \otimes D.$$

Thus the previous proposition applies and proves the assertion when $q = 0$.

For general q, we use dimension shifting. Let

$$A \hookrightarrow I_G(A), \qquad C \hookrightarrow I_G(C)$$

be the standard embeddings into induced modules, and put

$$A_1 = I_G(A)/A, \qquad C_1 = I_G(C)/C.$$

There is a G-equivariant bilinear map

$$I_G(A) \times B \longrightarrow I_G(C), \quad [(\alpha, b) \longmapsto \big(g \longmapsto \varphi(\alpha(g), gb)\big)].$$

It induces a bilinear map

$$\varphi_1 : A_1 \times B \longrightarrow C_1.$$

Since $I_G(A)$ is induced, its Tate cohomology vanishes. Hence the connecting homomorphism gives a lift $a = \delta(a_1)$ for some $a_1 \in \widehat{H}^{q-1}(G, A_1)$.

Cup-products are compatible with connecting homomorphisms. Therefore the cup-product map defined by a is, up to sign, the composite of the cup-product map defined by a_1 with the connecting isomorphism

$$\widehat{H}^{n+q-1}(H, C_1 \otimes D) \xrightarrow{\sim} \widehat{H}^{n+q}(H, C \otimes D).$$

The latter is an isomorphism because $I_G(C) \otimes D$ is cohomologically trivial. The dimension shifting argument reduces the assertion to the case $q = 0$. □

***Theorem 25.13* (Tate–Nakayama).** — *Let A be a G-module, and let*

$$a \in H^2(G, A).$$

Assume that for every prime p, the following two conditions hold:

(a) $H^1(G_p, A) = 0$*;*
(b) $H^2(G_p, A)$ *has cardinality* $m_p := |G_p|$*and is generated by the restriction* $a_p = \mathrm{Res}^G_{G_p}(a)$.

Then, for every torsion-free G-module D, every subgroup $H \subseteq G$, and every integer n, cup-product with $a_H = \mathrm{Res}^G_H(a)$ induces an isomorphism

$$\widehat{H}^n(H, D) \xrightarrow{\sim} \widehat{H}^{n+2}(H, A \otimes D).$$

In particular, taking $D = \mathbf{Z}$, cup-product with a_H gives an isomorphism

$$\widehat{H}^n(H, \mathbf{Z}) \xrightarrow{\sim} \widehat{H}^{n+2}(H, A).$$

Proof. — We apply the previous proposition with

$$B = \mathbf{Z}, \qquad C = A, \qquad q = 2,$$

and with $\varphi : A \times \mathbf{Z} \to A$ the evident bilinear map.

For every prime p, choose $n_p = -1$. We must check the three hypotheses of the previous proposition for the Sylow subgroup G_p.

For $n = -1$, the cup-product map is

$$\widehat{H}^{-1}(G_p, \mathbf{Z}) \longrightarrow H^1(G_p, A).$$

This map is surjective because the target is zero by assumption (a).

For $n = 0$, the cup-product map is

$$\widehat{H}^0(G_p, \mathbf{Z}) \longrightarrow H^2(G_p, A).$$

Since

$$\widehat{H}^0(G_p, \mathbf{Z}) \simeq \mathbf{Z}/m_p\mathbf{Z},$$

this map sends the canonical generator to a_p. By assumption (b), it is an isomorphism.

For $n = 1$, the source is

$$\widehat{H}^1(G_p, \mathbf{Z}) = H^1(G_p, \mathbf{Z}) = 0,$$

so the corresponding cup-product map is injective.

The hypotheses of the previous proposition are therefore satisfied. Hence cup-product with a_H gives isomorphisms

$$\widehat{H}^n(H, D) \xrightarrow{\sim} \widehat{H}^{n+2}(H, A \otimes D)$$

for every subgroup $H \subseteq G$, every integer n, and every torsion-free G-module D. Taking $D = \mathbf{Z}$ gives the stated special case. $\square$

Lecture 29

This lecture constructs the local reciprocity map from the finite-level Tate–Nakayama isomorphisms. We characterize it by characters, prove compatibility in finite towers, pass to the inverse limit, and compute its value on unramified extensions.

Throughout the local class field theory lectures, K denotes a non-archimedean local field, L/K denotes a finite extension, and $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$. For a finite unramified extension L/K, we write $F \in \mathrm{Gal}(L/K)$ for the arithmetic Frobenius, i.e. the automorphism inducing $x \mapsto x^{\#k_K}$ on the residue field. With this convention, the reciprocity map used below satisfies

$$(\pi_K, L/K) = F$$

for every uniformizer π_K of K.

26.1 Characterization by Characters. — Let L/K be a finite Galois extension of local fields, and put

$$G = \mathrm{Gal}(L/K).$$

The fundamental class $u_{L/K} \in H^2(G, L^\times)$ gives, by Tate–Nakayama, an isomorphism

$$G^{\mathrm{ab}} \cong \widehat{H}^{-2}(G, \mathbf{Z}) \xrightarrow{\cup u_{L/K}} \widehat{H}^0(G, L^\times) = K^\times / N_{L/K} L^\times.$$

We denote its inverse by

$$\theta_{L/K} : K^\times / N_{L/K} L^\times \longrightarrow G^{\mathrm{ab}}, \qquad \bar{a} \longmapsto (a, L/K) =: s_a.$$

Thus $N_{L/K} L^\times$ is the kernel of the induced map $K^\times \to G^{\mathrm{ab}}$.

Lemma 26.1. — *Let A be a finite abelian group and let $a, a' \in A$. If*

$$\chi(a) = \chi(a') \qquad \text{for every } \chi \in \mathrm{Hom}(A, \mathbf{Q}/\mathbf{Z}),$$

then $a = a'$.

Proof. — The natural map $A \to (A^\vee)^\vee$, with $A^\vee = \mathrm{Hom}(A, \mathbf{Q}/\mathbf{Z})$, is an isomorphism for finite abelian groups. $\square$

Before studying the compatibility formula, we need a lemma on cup-product calculation. If A is a G-module and $a \in A$ has norm zero (resp. is in A^G), we denote by $\bar{a}_0$ its class in $\widehat{H}^{-1}(G, A)$ (resp. by $\bar{a}^0$ its class in $\widehat{H}^0(G, A)$).

Proposition 26.2. — *Let G be a finite group and B a G-module. Let $f : G \longrightarrow B$ be a 1-cocycle, and write again $\bar{f} \in H^1(G,B)$ for its cohomology class. For $s \in G$, let $\bar{s} \in \widehat{H}^{-2}(G,\mathbf{Z})$ be the class corresponding to the image of s in G^{ab}. Then*

$$\bar{s} \cup \bar{f} = \overline{f(s)}_0 \in \widehat{H}^{-1}(G,B).$$

Proof. — The element $f(s)$ lies in $\widehat{H}^{-1}(G,B)$ because

$$N_G f(s) = \sum_{t\in G} t f(s) = \sum_{t\in G} f(ts) - \sum_{t\in G} f(t) = 0.$$

We compare both images under the isomorphism

$$\widehat{H}^{-1}(G,B) \xrightarrow{\sim} \widehat{H}^0(G, I_G \otimes_{\mathbf{Z}} B).$$

Let this isomorphism be denoted by d. If

$$x = \sum_{t\in G} t \otimes t \cdot f(s),$$

then $d(f(s)) = \bar{x}^0$. On the other hand, the defining property of the cup product and the relation $d(\bar{s}) = s - 1 \in I_G/I_G^2$ give

$$d(\bar{s} \cup f) = \bar{y}^0, \qquad y = -\sum_{t\in G} t(s-1) \otimes f(t).$$

Using the cocycle identity $f(ts) = f(t) + t \cdot f(s)$, one obtains

$$y = \sum_{t\in G} ts \otimes t \cdot f(s).$$

Hence

$$x - y = \sum_{t\in G} t(1-s) \otimes t \cdot f(s) = N_G\big((1-s) \otimes f(s)\big),$$

which is zero in $\widehat{H}^0(G, I_G \otimes_{\mathbf{Z}} B)$. Therefore $d(\bar{s} \cup \bar{f}) = d(\overline{f(s)}_0)$, and the isomorphism d gives $\bar{s} \cup \bar{f} = \overline{f(s)}_0 \in \widehat{H}^{-1}(G,B)$. □

For $\chi \in \mathrm{Hom}(G, \mathbf{Q}/\mathbf{Z}) = H^1(G, \mathbf{Q}/\mathbf{Z})$, let

$$d_\chi \in H^2(G, \mathbf{Z})$$

be the image of χ under the connecting homomorphism associated to

$$0 \longrightarrow \mathbf{Z} \longrightarrow \mathbf{Q} \longrightarrow \mathbf{Q}/\mathbf{Z} \longrightarrow 0.$$

***Proposition 26.3* (Character formula).** — *Assume L/K is finite abelian and put $G = \mathrm{Gal}(L/K)$. For $a \in K^\times$ and $\chi \in \mathrm{Hom}(G, \mathbf{Q}/\mathbf{Z})$, one has*

$$\chi\big(\theta_{L/K}(\bar{a})\big) = \mathrm{inv}_K(\bar{a} \cup d_\chi).$$

Proof. — Let $u = u_{L/K}$, $s = \theta_{L/K}(\bar{a})$, and $n = [L : K]$. By definition of $\theta_{L/K}$,

$$s \cup u = \bar{a} \in \widehat{H}^0(G, L^\times).$$

Equivalently $u \cup s = \bar{a}$, because the relevant degrees are -2 and 2. Using associativity and graded commutativity of cup products, with only even degrees involved after $s \cup d_\chi$, we get

$$\bar{a} \cup d_\chi = u \cup (s \cup d_\chi).$$

By Proposition 26.2,

$$s \cup \chi = \chi(s) \in \widehat{H}^{-1}(G, \mathbf{Q}/\mathbf{Z}).$$

If $\chi(s) = m/n$, then the connecting homomorphism

$$\widehat{H}^{-1}(G, \mathbf{Q}/\mathbf{Z}) \longrightarrow \widehat{H}^0(G, \mathbf{Z}) = \mathbf{Z}/n\mathbf{Z}$$

sends m/n to the class of m. Hence

$$s \cup d_\chi = m \in \widehat{H}^0(G, \mathbf{Z}) \implies \bar{a} \cup d_\chi = m\, u.$$

Applying the invariant map gives

$$\mathrm{inv}_K(\bar{a} \cup d_\chi) = m\ \mathrm{inv}_K(u) = \frac{m}{n} = \chi(s).$$

$\square$

***Proposition 26.4* (Compatibility in finite towers).** — *Let L'/K and L/K be finite Galois extensions of local fields with*

$$K \subset L' \subset L.$$

Let $G = \mathrm{Gal}(L/K)$ and $H = \mathrm{Gal}(L/L')$. Then the diagram

$$\begin{tikzcd} K^\times \arrow[r, "\theta_{L/K}"] \arrow[d, equal] & G^{\mathrm{ab}} \arrow[d, two heads] \\ K^\times \arrow[r, "\theta_{L'/K}"] & (G/H)^{\mathrm{ab}} \end{tikzcd}$$

commutes.

Proof. — It is enough to treat the abelian quotients. Let $\chi : (G/H)^{\mathrm{ab}} \to \mathbf{Q}/\mathbf{Z}$ be a character, and let $\widetilde{\chi}$ be its inflation to G^{ab}. For $a \in K^\times$, put $s = \theta_{L/K}(a)$, and let s' be its image in $(G/H)^{\mathrm{ab}}$. Then

$$\chi(s') = \widetilde{\chi}(s).$$

By the character formula and functoriality of cup products under inflation,

$$\widetilde{\chi}(s) = \mathrm{inv}_K(\bar{a} \cup d_{\widetilde{\chi}}) = \mathrm{inv}_K(\bar{a} \cup d_\chi) = \chi(\theta_{L'/K}(a)).$$

Characters separate points of finite abelian groups by Lemma 26.1, so the diagram commutes. $\square$

Definition 26.5. — The continuous homomorphism

$$\theta_K : K^\times \longrightarrow \mathrm{Gal}(K^{\mathrm{ab}}/K)$$

obtained from the finite-level reciprocity map $\theta_{L/K}$ for every finite abelian extension L/K, by passing to the limit over the finite abelian extensions L/K, is called the reciprocity map.

Its image is dense, and it induces an isomorphism:

$$\varprojlim_{L/K} K^\times/N_{L/K}L^\times \xrightarrow{\sim} \mathrm{Gal}(K^{\mathrm{ab}}/K),$$

where the limit is taken over finite abelian extensions L/K.

Our goal is to prove that for a p-adic field, the reciprocity map induces an isomorphism of the profinite completion of $K^\times$ with $\mathrm{Gal}(K^{\mathrm{ab}}/K)$. However, this requires the existence theorem, whose first proof in the case of a p-adic field will be given at the end. For now, we just deduce from the above results that $\varprojlim_{L/K} K^\times/N_{L/K}L^\times$, with L/K finite abelian, is a quotient of the profinite completion $\widehat{K^\times}$ of $K^\times$.

26.2 Unramified and Ramified Reciprocity. — We now compute the reciprocity map on the two most visible parts of $K^\times$: the valuation and the unit group.

***Proposition 26.6* (Unramified computation).** — *Let L/K be an unramified extension of degree f, and let $F \in \mathrm{Gal}(L/K)$ be the arithmetic Frobenius. Then for every $a \in K^\times$,*

$$(a, L/K) = F^{v_K(a)}.$$

Proof. — Put $G = \mathrm{Gal}(L/K)$, and let $\chi \in \mathrm{Hom}(G, \mathbf{Q}/\mathbf{Z})$. By Proposition 26.3,

$$\chi((a, L/K)) = \mathrm{inv}_K(\bar{a} \cup d_\chi).$$

For an unramified extension, the valuation sequence

$$0 \to \mathscr{O}_L^\times \to L^\times \xrightarrow{v_L} \mathbf{Z} \to 0$$

identifies $H^2(G, L^\times)$ with $H^2(G, \mathbf{Z})$, by Proposition 7.8. Under this identification, the invariant map is evaluation at F after applying

$$H^2(G, \mathbf{Z}) \xrightarrow{\delta^{-1}} H^1(G, \mathbf{Q}/\mathbf{Z}).$$

Since $v_L(a) = v_K(a)$, compatibility of cup products with v_L gives

$$\mathrm{inv}_K(\bar{a} \cup d_\chi) = v_K(a)\chi(F).$$

Thus

$$\chi((a, L/K)) = \chi(F^{v_K(a)})$$

for all χ, and Lemma 26.1 proves the formula. □

Corollary 26.7. — The projection of $\theta_K : K^\times \to \mathrm{Gal}(K^{\mathrm{ab}}/K)$ to the maximal unramified quotient is the valuation map:

$$\begin{tikzcd} K^\times \arrow[r, "\theta_K"] \arrow[d, "v_K"'] & \mathrm{Gal}(K^{\mathrm{ur}}/K) \arrow[d, "\sim"] \\ \mathbf{Z} \arrow[r] & \widehat{\mathbf{Z}}. \end{tikzcd}$$

The right vertical map sends the arithmetic Frobenius to 1, and the bottom map is the natural inclusion.

Corollary 26.8. — Let L/K be a finite abelian extension, let K' be the maximal unramified subextension of L/K, and put $I = \mathrm{Gal}(L/K')$. Then

$$\theta_{L/K}(\mathscr{O}_K^\times) = I.$$

Passing to the limit, $\theta_K(\mathscr{O}_K^\times)$ is the abelian inertia subgroup

$$\mathrm{Gal}(K^{\mathrm{ab}}/K^{\mathrm{ur}}) \subset \mathrm{Gal}(K^{\mathrm{ab}}/K).$$

Proof. — If $x \in \mathscr{O}_K^\times$, then $v_K(x) = 0$. By the unramified computation, the image of x in $\mathrm{Gal}(K'/K)$ is trivial, so $\theta_{L/K}(x) \in I$.

Conversely, let $t \in I$. Choose $a \in K^\times$ with $t = \theta_{L/K}(a)$. Let $m = [K' : K]$. The image of t in $\mathrm{Gal}(K'/K)$ is trivial, so the unramified formula gives $m \mid v_K(a)$. For $b \in L^\times$, one has

$$v_K(N_{L/K} b) = m\, v_L(b).$$

Choose $b \in L^\times$ with $v_L(b) = v_K(a)/m$. Then

$$u = a\, N_{L/K}(b)^{-1} \in \mathscr{O}_K^\times,$$

and $\theta_{L/K}(u) = \theta_{L/K}(a) = t$, since norms lie in the kernel of $\theta_{L/K}$. □

Lecture 30

The final lecture proves the existence theorem for p-adic fields. We study norm subgroups, introduce the Hilbert symbol used in the proof, and conclude that finite abelian extensions of K correspond exactly to finite-index subgroups of $K^\times$.

27.1 Norm Subgroups. —

Definition 27.1. — A subgroup M of $K^\times$ is called a norm subgroup if there exists a finite abelian extension L/K with $M = N_{L/K} L^\times$.

Example 27.2. — Let $m \geq 1$ be an integer, and let M_m be the set of elements $a \in K^\times$ with $v_K(a) \equiv 0 \pmod{m}$; it follows from a direct computation of norms that M_m is the norm group of the unramified extension of K of degree m.

Norm subgroups are closely related to the reciprocity map $\theta_K : K^\times \longrightarrow G_K^{\mathrm{ab}} = \mathrm{Gal}(K^{\mathrm{ab}}/K)$. By construction, θ_K is obtained by inverse limit from the isomorphisms $K^\times/N_{L/K}L^\times \to \mathrm{Gal}(L/K)$, where L runs through all finite abelian extensions of K. If we put $\widetilde{K} = \varprojlim_L K^\times/N_{L/K}L^\times$, we see that θ_K can be factored into

$$K^\times \xrightarrow{\ i\ } \widetilde{K} \xrightarrow{\ \widetilde{\theta}\ } G_K^{\mathrm{ab}},$$

where i is the natural map, and $\widetilde{\theta}$ is an isomorphism. Note that $\widetilde{K}$ is just the completion of $K^\times$ with respect to the topology defined by the norm subgroups.

This shows that norm subgroups of $K^\times$ and open subgroups of G_K^{ab} correspond to each other in a one-one way: if U is an open subgroup of G_K^{ab}, with fixed field L, we attach to U the norm subgroup $\theta_K^{-1}(U) = N_{L/K}L^\times$; if M is a norm subgroup of $K^\times$, we attach to it the adherence of $\theta_K(M)$; the corresponding field L_M is then the set of elements in K^{ab} which are invariant by the $\theta_K(a)$, for $a \in M$. We thus get a "Galois correspondence" between norm subgroups and finite abelian extensions; we state it as a proposition:

Proposition 27.3. — *(a) The map $L \mapsto N_{L/K}L^\times$ is a bijection of the set of finite abelian extensions of K onto the set of norm subgroups of $K^\times$.*

(b) This bijection reverses the inclusion.

(c)

$$N_{LL'/K}(LL')^\times = N_{L/K}L^\times \cap N_{L'/K}L'^\times$$

and

$$N_{L\cap L'/K}(L\cap L')^\times = N_{L/K}L^\times \cdot N_{L'/K}L'^\times.$$

(d) Any subgroup of $K^\times$ which contains a norm subgroup is a norm subgroup.

Lemma 27.4. — *Let E/K be a finite separable extension, and let L/K be the maximal abelian subextension of E/K inside E. Then*

$$N_{E/K}E^\times = N_{L/K}L^\times.$$

Proof. — The inclusion

$$N_{E/K}E^\times \subset N_{L/K}L^\times$$

is immediate from the transitivity of norms. We prove the reverse inclusion. Let F/K be a finite Galois closure of E/K, and put

$$G = \mathrm{Gal}(F/K), \qquad H = \mathrm{Gal}(F/E).$$

Let G' be the commutator subgroup of G. Since L/K is the maximal abelian subextension of E/K, Galois theory gives $\mathrm{Gal}(F/L) = HG'$. Take $a \in N_{L/K}L^\times$. By compatibility of finite reciprocity maps in towers, the element $\theta_{F/K}(a) \in G^{\mathrm{ab}} = G/G'$ lies in the image of

$$H^{\mathrm{ab}} = H/H' \longrightarrow G/G'.$$

The reciprocity map $\theta_{F/E} : E^\times \longrightarrow H^{\mathrm{ab}}$ is surjective, so there is $b \in E^\times$ such that

$$\theta_{F/K}(N_{E/K}b) = \theta_{F/K}(a).$$

Thus $a\, N_{E/K}(b)^{-1} \in \ker(\theta_{F/K}) = N_{F/K}F^\times$. Choose $c \in F^\times$ with

$$a = N_{E/K}(b)\, N_{F/K}(c).$$

But $N_{F/K} = N_{E/K} \circ N_{F/E}$, so the right-hand side lies in $N_{E/K}E^\times$. Therefore $N_{L/K}L^\times \subset N_{E/K}E^\times$. □

***Lemma 27.5* (Norm subgroup criterion).** — *Let K be a local field.*

1. *If a subgroup $U \subset K^\times$ contains $N_{L/K}L^\times$ for some finite abelian extension L/K, then there is a finite abelian subextension F/K of L/K such that*

$$U = N_{F/K}F^\times.$$

2. *If L and M are finite abelian extensions of K inside a fixed separable closure and*

$$N_{L/K}L^\times = N_{M/K}M^\times,$$

then $L = M$.

3. *For every finite abelian extension F/K, the group $N_{F/K}F^\times$ is open of finite index in $K^\times$.*

Proof. — For (1), apply finite-level reciprocity $\theta_{L/K} : K^\times \to \mathrm{Gal}(L/K)$. Since $N_{L/K}L^\times = \ker(\theta_{L/K})$, the image of U is a subgroup of $\mathrm{Gal}(L/K)$, hence equals $\mathrm{Gal}(L/F)$ for a unique intermediate abelian extension F/K. The compatibility of reciprocity for $K \subset F \subset L$ shows that the preimage of $\mathrm{Gal}(L/F)$ is exactly $N_{F/K}F^\times$. Hence $U = N_{F/K}F^\times$.

For (2), take the maximal abelian extension K^{ab} in the fixed separable closure. The preimages of $\mathrm{Gal}(K^{\mathrm{ab}}/L)$ and $\mathrm{Gal}(K^{\mathrm{ab}}/M)$ under the dense map $\theta_K : K^\times \to \mathrm{Gal}(K^{\mathrm{ab}}/K)$ are equal. Since the image is dense and these subgroups are closed, the two closed subgroups are equal; hence $L = M$.

For (3), finite-level reciprocity gives $K^\times/N_{F/K}F^\times \simeq \mathrm{Gal}(F/K)$, so the norm subgroup has finite index. It remains to see that it is open. The norm map $N_{F/K} : F^\times \to K^\times$ is continuous, because after choosing a K-basis of F, it is given by the determinant of multiplication by x, hence by a polynomial in the coordinates of x. It is also proper. Indeed, if e and f are the ramification index and residue degree of F/K, then

$$v_K(N_{F/K}x) = f\, v_F(x).$$

Thus the preimage of $\mathscr{O}_K^\times$ is contained in $\mathscr{O}_F^\times$, which is compact. Since $K^\times$ is the disjoint union of the translates $\varpi_K^m \mathscr{O}_K^\times$, the same argument shows that the preimage of any compact subset of $K^\times$ is compact. Hence the image $N_{F/K}F^\times$ is closed; being closed of finite index in the topological group $K^\times$, it is open. □

Lemma 27.6. — *If $U \subset K^\times$ has finite index n, then*

$$K^{\times n} \subset U.$$

Proof. — The quotient $K^\times/U$ has order n. Hence every element of the quotient is killed by n. □

27.2 Hilbert Symbol. — The crucial step in the proof of the existence theorem for p-adic fields uses Kummer extensions. We now recall the local symbol. Let k be a field, let $n \geq 1$ be prime to $\mathrm{char}(k)$, and assume that k contains a fixed primitive n-th root of unity. For $a \in k^\times$, Kummer theory gives a character

$$\chi_a : G_k \longrightarrow \mathbf{Z}/n\mathbf{Z} \subset \mathbf{Q}/\mathbf{Z},$$

corresponding to the class of $a \in k^\times/k^{\times n} \simeq H^1(k, \mu_n)$. For $a, b \in k^\times$, define

$$(a, b)_n := b \cup d_{\chi_a} \in H^2(k, \overline{k}^\times) = \mathrm{Br}(k).$$

This symbol is bilinear and takes values in $\mathrm{Br}(k)[n]$.

Proposition 27.7 **(Properties of the Hilbert symbol)**

Let k contain a primitive n-th root of unity, with n prime to $\mathrm{char}(k)$*. For $a, b \in k^\times$:*

1. *If $b \in N_{k(a^{1/n})/k} k(a^{1/n})^\times$, then*

$$(a, b)_n = 0.$$

2. *One has*

$$(a, -a)_n = 0, \qquad (a, b)_n = -(b, a)_n,$$

and, if $a \neq 1$, also

$$(a, 1 - a)_n = 0.$$

3. *If k is p-adic and $b \in k^\times$ satisfies*

$$(a, b)_n = 0 \qquad \textit{for every } a \in k^\times,$$

then $b \in k^{\times n}$.

Proof. — For (1), the character χ_a factors through $G = \mathrm{Gal}(k(a^{1/n})/k)$, and the cup product can be computed in

$$\widehat{H}^0(G, k(a^{1/n})^\times) = k^\times/N_{k(a^{1/n})/k} k(a^{1/n})^\times.$$

Thus a norm has zero symbol.

For (2), write $\alpha = a^{1/n}$. If $x^n - a \neq 0$, then $x^n - a$ is a norm from $k(\alpha)$. More precisely, let d be the largest divisor of n such that a has a d-th root in k, and put $m = n/d$. Then $k(\alpha)/k$ is cyclic of degree m. If ζ is the chosen primitive n-th root of unity, then, for fixed i, the conjugates of $x - \zeta^i\alpha$ are the factors $x - \zeta^j\alpha$ with $j \equiv i \pmod{d}$. Grouping the full factorization

$$x^n - a = \prod_{j=0}^{n-1} (x - \zeta^j \alpha)$$

into d Galois orbits gives

$$x^n - a = \prod_{i=0}^{d-1} N_{k(\alpha)/k}(x - \zeta^i \alpha).$$

Hence $(a, x^n - a)_n = 0$ by (1). Taking $x = 0$ and $x = 1$ gives $(a, -a)_n = 0$ and $(a, 1-a)_n = 0$. Bilinearity gives

$$(a,b)_n + (b,a)_n = (a, -ab)_n + (b, -ab)_n = (ab, -ab)_n = 0.$$

For (3), skew-symmetry gives $(b,a)_n = 0$ for every a. Equivalently, $\chi_b \cup a = 0$ for every $a \in k^\times$. By the character formula for the reciprocity map, $\chi_b(\theta_k(a)) = 0$ for every a. The image of θ_k is dense, so $\chi_b = 0$. By Kummer theory, the class of b in $k^\times / k^{\times n}$ is zero. □

Theorem 27.8 **(Existence theorem for p-adic fields)**

Let K be a finite extension of $\mathbf{Q}_p$, and fix a separable closure $\overline{K}$. If $U \subset K^\times$ is a finite-index subgroup, then there is a unique finite abelian extension L/K inside $\overline{K}$ such that

$$U = N_{L/K} L^\times.$$

Proof. — Uniqueness is Lemma 27.5(2).

We prove existence. Let $n = [K^\times : U]$. By Lemma 27.6, $K^{\times n} \subset U$. By Lemma 27.5(1), it is enough to show that $K^{\times n}$ contains a norm subgroup from some finite abelian extension of K. In fact we will show, after a harmless reduction, that $K^{\times n}$ itself is such a norm subgroup.

First reduce to the case $\mu_n \subset K$. Let $F = K(\mu_n)$. It suffices to prove the special case for fields containing μ_n: applying that special case to F, we get a finite abelian extension E/F such that

$$F^{\times n} = N_{E/F} E^\times \implies N_{E/K} E^\times = N_{F/K}(F^{\times n}) \subset K^{\times n}.$$

By Lemma 27.4, this norm group is the norm group of the maximal abelian subextension of E/K. Hence $K^{\times n}$ contains a norm subgroup, and Lemma 27.5(1) applies. Thus we may assume $\mu_n \subset K$.

Assume now that K contains μ_n. Let $\Gamma = \mathrm{Gal}(K^{\mathrm{ab}}/K)$. The quotient $\Gamma/n\Gamma$ is finite: its continuous character group is

$$\mathrm{Hom}_{\mathrm{cont}}(\Gamma/n\Gamma, \mathbf{Q}/\mathbf{Z}) = H^1(K, \mathbf{Z}/n\mathbf{Z}),$$

which is finite for a p-adic field. Hence there is a finite abelian extension L/K such that $\mathrm{Gal}(L/K) \simeq \Gamma/n\Gamma$. Let $\theta_n : K^\times \longrightarrow \Gamma/n\Gamma$ be the map induced by θ_K. Since the target is killed by n, $K^{\times n} \subset \ker(\theta_n)$. Conversely, let $b \in \ker(\theta_n)$. For every $a \in K^\times$, the Kummer character χ_a is n-torsion, so it factors through $\Gamma/n\Gamma$. Hence $\chi_a(\theta_K(b)) = 0$. By the character formula, this says

$$\mathrm{inv}_K((a,b)_n) = 0 \implies (a,b)_n = 0 \qquad \text{for all } a \in K^\times.$$

Proposition 27.7(3) gives $b \in K^{\times n}$. Thus

$$\ker(\theta_n) = K^{\times n}.$$

At the finite level, $\ker(\theta_n) = N_{L/K}L^\times$, so

$$K^{\times n} = N_{L/K}L^\times.$$

Since U contains $K^{\times n}$, Lemma 27.5(1) finally gives a finite abelian extension M/K with $U = N_{M/K}M^\times$. □

Corollary 27.9. — For a p-adic field K, the reciprocity map induces an isomorphism

$$\widehat{K^\times} \xrightarrow{\sim} \mathrm{Gal}(K^{\mathrm{ab}}/K),$$

where $\widehat{K^\times}$ denotes the profinite completion of $K^\times$. In particular, this group is isomorphic to the direct product $\widehat{\mathbf{Z}} \times \mathscr{O}_K^\times$. Furthermore, θ_K is injective.[7]

Proof. — By definition 26.5,

$$\mathrm{Gal}(K^{\mathrm{ab}}/K) \cong \varprojlim_{L/K} K^\times/N_{L/K}L^\times.$$

The existence theorem says that the norm groups $N_{L/K}L^\times$ are exactly the finite-index subgroups of $K^\times$. Therefore, the inverse limit is the profinite completion of $K^\times$. Since $K^\times \simeq \mathbf{Z} \times \mathscr{O}_K^\times$ and $\mathscr{O}_K^\times$ is already profinite, the natural map $K^\times \to \widehat{K^\times}$ is injective. Hence θ_K is injective. □

Corollary 27.10. — The reciprocity map induces an isomorphism

$$\mathscr{O}_K^\times \xrightarrow{\sim} \mathrm{Gal}(K^{\mathrm{ab}}/K^{\mathrm{ur}}).$$

This completes the local class field theory picture for p-adic fields: finite abelian extensions of K correspond to finite-index subgroups of $K^\times$, the Galois group is $K^\times/N_{L/K}L^\times$, the valuation controls the unramified quotient, and the unit group controls inertia.

Appendix

In this appendix we collect the homological algebra and change-of-groups tools used repeatedly in the main text: universal derived functors, acyclic resolutions, restriction, inflation, Shapiro's lemma, and Hochschild–Serre. The goal is to keep the lecture proofs self-contained while allowing the main narrative to refer back to these formal results when needed [**7**, Chap. I, §2]; [**2**, Chap. II, §4].

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[7] In other words, the universal norm group $\bigcap_L N_{L/K}L^\times$, taken over the finite abelian extensions L/K, is reduced to $\{1\}$.

28.1 Homological Algebra. —
Definition 28.1. — A (covariant) homological (resp. cohomological) δ-functor between $\mathscr{A}$ and $\mathscr{B}$ is a collection of additive functors $T_n : \mathscr{A} \to \mathscr{B}$ (resp. $T^n : \mathscr{A} \to \mathscr{B}$) for $n \geq 0$, together with morphisms

$$\delta_n : T_n(C) \to T_{n-1}(A)$$

(resp. $\delta^n : T^n(C) \to T^{n+1}(A)$) defined for each short exact sequence $0 \to A \to B \to C \to 0$ in $\mathscr{A}$. Here we make the convention that $T^n = T_n = 0$ for $n < 0$. These two conditions are imposed:

1. For each short exact sequence as above, there is a long exact sequence
$$\cdots \to T_{n+1}(C) \xrightarrow{\delta} T_n(A) \to T_n(B) \to T_n(C) \xrightarrow{\delta} T_{n-1}(A) \to \cdots$$
(resp.
$$\cdots \to T^{n-1}(C) \xrightarrow{\delta} T^n(A) \to T^n(B) \to T^n(C) \xrightarrow{\delta} T^{n+1}(A) \to \cdots).$$
In particular, T_0 is right exact, and T^0 is left exact.
2. For each morphism of short exact sequences from $0 \to A' \to B' \to C' \to 0$ to $0 \to A \to B \to C \to 0$, the δ's give a commutative diagram
$$\begin{array}{ccc} T_n(C') & \xrightarrow{\delta} & T_{n-1}(A') \\ \downarrow & & \downarrow \\ T_n(C) & \xrightarrow{\delta} & T_{n-1}(A) \end{array} \quad \text{resp.} \quad \begin{array}{ccc} T^n(C') & \xrightarrow{\delta} & T^{n+1}(A') \\ \downarrow & & \downarrow \\ T^n(C) & \xrightarrow{\delta} & T^{n+1}(A) \end{array}$$

Definition 28.2. — A *morphism* $S \to T$ *of* δ*-functors* is a system of natural transformations $S_n \to T_n$ (resp. $S^n \to T^n$) that commute with δ.

A homological δ-functor T is *universal* if, given any other δ-functor S and a natural transformation $f_0 : S_0 \to T_0$, there exists a unique morphism $\{f_n : S_n \to T_n\}$ of δ-functors that extends f_0.

A cohomological δ-functor T is *universal* if, given S and $f^0 : T^0 \to S^0$, there exists a unique morphism $T \to S$ of δ-functors extending f^0.
Definition 28.3. — Let $F : \mathscr{A} \to \mathscr{B}$ be a left exact functor between two abelian categories. If $\mathscr{A}$ has enough injectives, we can construct the right derived functors R^iF $(i \geq 0)$ of F as follows. If A is an object of $\mathscr{A}$, choose an injective resolution $A \to I^\bullet$ and define

$$R^iF(A) = H^i(F(I^\bullet)).$$

It is independent of the choice of the injective resolution of A. Note that since $0 \to F(A) \to F(I^0) \to F(I^1)$ is exact, we always have $R^0F(A) \cong F(A)$.

Since F also defines a right exact functor $F^{\mathrm{op}} : \mathscr{A}^{\mathrm{op}} \to \mathscr{B}^{\mathrm{op}}$, and $\mathscr{A}^{\mathrm{op}}$ has enough projectives, we can construct the left derived functors L_iF^{op} as well. Since $I^\bullet$ becomes a projective resolution of A in $\mathscr{A}^{\mathrm{op}}$, we see that

$$R^iF(A) = (L_iF^{\mathrm{op}})^{\mathrm{op}}(A).$$

Therefore all the results about right exact functors apply to left exact functors. In particular, the objects $R^iF(A)$ are independent of the choice of injective resolutions, $R^\bullet F$ is a universal cohomological δ-functor, and $R^iF(I) = 0$ for $i \neq 0$ whenever I is injective. Calling an object Q F-acyclic if $R^iF(Q) = 0$ $(i \neq 0)$, we see that the right derived functors of F can also be computed from F-acyclic resolutions.

Definition 28.4. — (Ext functors) For each R-module A, the functor $F(B) = \mathrm{Hom}_R(A, B)$ is left exact. Its right derived functors are called the Ext groups:

$$\mathrm{Ext}^i_R(A, B) = R^i\mathrm{Hom}_R(A, -)(B).$$

In particular, $\mathrm{Ext}^0(A, B)$ is $\mathrm{Hom}(A, B)$, and injectives are characterized by Ext via the following exercise.

Exercise 28.5. — Show that the following are equivalent.

1. B is an *injective* R-module.
2. $\mathrm{Hom}_R(-, B)$ is an exact functor.
3. $\mathrm{Ext}^i_R(A, B)$ vanishes for all $i \neq 0$ and all A (B is $\mathrm{Hom}_R(-, B)$-acyclic for all A).
4. $\mathrm{Ext}^1_R(A, B)$ vanishes for all A.

Exercise 28.6. — Show that the following are equivalent.

1. A is a *projective* R-module.
2. $\mathrm{Hom}_R(A, -)$ is an exact functor.
3. $\mathrm{Ext}^i_R(A, B)$ vanishes for all $i \neq 0$ and all B (A is $\mathrm{Hom}_R(-, B)$-acyclic for all B).
4. $\mathrm{Ext}^1_R(A, B)$ vanishes for all B.

The notion of derived functor has obvious variations for contravariant functors. For example, let F be a contravariant left exact functor from $\mathscr{A}$ to $\mathscr{B}$. This is the same as a covariant left exact functor from $\mathscr{A}^{\mathrm{op}}$ to $\mathscr{B}$, so if $\mathscr{A}$ has enough projectives (i.e., $\mathscr{A}^{\mathrm{op}}$ has enough injectives), we can define the right derived functors $R^\bullet F(A)$ to be the cohomology of $F(P_\bullet)$, $P_\bullet \to A$ being a projective resolution in $\mathscr{A}$. This too is a universal δ-functor with $R^0F(A) = F(A)$, and $R^iF(P) = 0$ for $i \neq 0$ whenever P is projective.

Proposition 28.7. — *For each R-module B, the functor $G(A) = \mathrm{Hom}_R(A, B)$ is contravariant and left exact. Its right derived functors are denoted $R^\bullet G(A)$. However, we will see in 2.7.6 that these are just the functors $\mathrm{Ext}^\bullet(A, B)$. That is,*

$$R^\bullet\mathrm{Hom}(-, B)(A) \cong R^\bullet\mathrm{Hom}(A, -)(B) = \mathrm{Ext}^\bullet(A, B).$$

Remark 28.8. — In our note, we use this result to calculate the group cohomology. This is because $\mathscr{C}_G$ has enough injectives but not for projectives: we consider the projective resolution for the trivial G module $\mathbf{Z}$, and the right derived functor for the covariant functor $\mathrm{Hom}_G(\mathbf{Z}, -)$; on the other hand, we consider the injective resolution for the discrete G module A, and the left derived functor for the contravariant functor $\mathrm{Hom}_G(-, A)$. Notice that $\mathrm{Hom}_G(\mathbf{Z}, A)$ with trivial G-module $\mathbf{Z}$ is A^G. By the previous discussion on homological algebra, we have

$$R^\bullet\mathrm{Hom}_G(-,A)(\mathbf{Z}) \cong R^\bullet\mathrm{Hom}_G(\mathbf{Z},-)(A) = \mathrm{Ext}^\bullet(\mathbf{Z},A).$$

Definition 28.9. — A G-module M is called *acyclic* if $H^i(G,M)=0$ for all $i>0$. Any injective G-module is acyclic.

Proposition 28.10. — *Let $I^\bullet$ be an acyclic resolution of M, i.e. there is an exact sequence $0\to M\to I^0\to I^1\to\cdots$ with I^i all acyclic. Then $H^i((I^\bullet)^G)\cong H^i(G,M)$ for all i.*

Proof. — Let $M_0:=M$, and we inductively construct $M_i := I^{i-1}/M_{i-1}\hookrightarrow I^i$. Thus we have an exact sequence $0\to M_i\to I^i\to M_{i+1}\to 0$, that induces $0\to H^0(G,M_i)\to H^0(G,I^i)\to H^0(G,M_{i+1})\to H^1(G,M_i)\to 0$ and $H^j(G,M_{i+1})\xrightarrow{\sim} H^{j+1}(G,M_i)$ for $j\geq 1$. By definition, $H^0(G,M_{i+1})=\mathrm{Ker}[d^{i+1}:I^{i+1}\to I^{i+2}]$ and hence $H^1(G,M_i)\cong H^{i+1}((I^\bullet)^G)$. We then deduce $H^i(G,M)\cong H^i(G,M_0)\cong H^{i-1}(G,M_1)\cong\cdots\cong H^1(G,M_{i-1})\cong H^i((I^\bullet)^G)$. □

Remark 28.11. — Let $I^\bullet$ be an acyclic resolution of M, and $J^\bullet$ be an injective resolution of M. By Lemma 2.1.7, the identity map on M induces a morphism of complexes of G-modules: $I^\bullet\to J^\bullet$. By an induction argument as in the above proof, one can show that the morphism $I^\bullet\to J^\bullet$ induces an isomorphism $H^i((I^\bullet)^G)\cong H^i((J^\bullet)^G)$ for $i\geq 0$.

28.2 Change of Groups: Restriction, Corestriction, Inflation. — Let A be a G-module and let G' be a group endowed with a morphism $f: G'\to G$. We can then endow A with a structure of a G'-module by setting

$$g'\cdot a := f(g')\cdot a,\quad g'\in G',\quad a\in A.$$

Denote by f^*A (or simply A when no confusion arises) this G'-module.

As A^G is a subgroup of $(f^*A)^{G'}$, we obtain a morphism of functors from $H^0(G,\cdot)$ to $H^0(G',f^*\cdot)$. The universal property of derived functors (Appendix, Theorem A.46, b) shows that for any integer $i\geq 0$, there is a unique family of morphisms of functors

$$f_i^*: H^i(G,\cdot)\longrightarrow H^i(G',\cdot)$$

compatible with the coboundaries (i.e., the maps δ^i) of the long exact cohomology sequences (in an evident way). We have thus obtained a morphism of cohomological functors (or morphism of δ-functors).

Let now A' be a G'-module and $u: A\to A'$ a morphism of abelian groups. If furthermore u is f-compatible, i.e., satisfies

$$u(f(g')\cdot a) = g'\cdot u(a)\quad g'\in G',\quad a\in A,$$

then u is a G'-homomorphism from f^*A to A' and induces a homomorphism $u_*: H^i(G',f^*A)\to H^i(G',A')$. Composing this morphism with f_i^*, we obtain a homomorphism

$$H^i(G,A)\longrightarrow H^i(G',A')$$

associated to the group homomorphism $f : G' \to G$ and to the f-compatible homomorphism $u : A \to A'$. The newly defined homomorphism has an obvious expression via the explicit definition of H^i using the cochains.

Furthermore, if we have a f-compatible morphism of short exact sequences from the sequence $0 \to A \to B \to C \to 0$ to $0 \to A' \to B' \to C' \to 0$, the corresponding morphisms of the H^i remain compatible with the maps δ^i of long exact sequences associated to these short exact sequences (we thus have a morphism of cohomological functors).

Definition 28.12. — (a) Let A be a G-module and H a subgroup of G. Taking for f the canonical injection of H into G, we obtain for $i \geq 0$ a homomorphism $\mathrm{Res} : H^i(G, A) \to H^i(H, A)$ that we call the *restriction homomorphism*.

(b) Let A be a G-module. For a normal subgroup H of G, the quotient G/H acts on A^H and the inclusion $A^H \to A$ is compatible with the canonical surjection $G \to G/H$. We deduce for all $i \geq 0$ a homomorphism $\mathrm{Inf} : H^i(G/H, A^H) \to H^i(G, A)$ that we call the *inflation homomorphism*.

Remark 28.13. — The restriction and inflation homomorphisms have a very simple expression in terms of the cocycles: in the case of the restriction one simply restricts the cocycle $G^i \to A$ to the subgroup H^i; in the case of inflation, one composes the canonical surjection $G^i \to (G/H)^i$ with any cocycle $(G/H)^i \to A^H \subset A$.

Let now H be a subgroup of a finite group G and let A be an H-module. Consider the G-module $I_G^H(A)$. Associating to any $u \in I_G^H(A)$ its value at 1, we obtain a morphism of abelian groups $I_G^H(A) \to A$ compatible with the injection $H \to G$. We deduce from the above, homomorphisms (compatible with long exact sequences)

$$H^i(G, I_G^H(A)) \longrightarrow H^i(H, A).$$

Theorem 28.14 **(Shapiro).** — *These are isomorphisms.*

Remark 28.15. — Let H be a subgroup of G, and suppose that B is a $\mathbf{Z}[H]$-module. We set

$$\mathrm{Ind}_H^G(B) = \mathbf{Z}[G] \otimes_{\mathbf{Z}[H]} B, \quad \text{and } \mathrm{CoInd}_H^G(B) = \mathrm{Hom}_{\mathbf{Z}[H]}(\mathbf{Z}[G], B),$$

with natural G actions. These modules are called induced and co-induced modules from H to G, respectively.

In fact, if H is of finite index in G, the notions of induced and coinduced from H to G coincide: we have a canonical isomorphism of G-modules

$$\chi : \mathrm{CoInd}_H^G(B) \xrightarrow{\sim} \mathrm{Ind}_H^G(B), \qquad \chi(\varphi) = \sum_{\bar{g} \in H \backslash G} g^{-1} \otimes \varphi(g),$$

where for each $\bar{g} \in H \setminus G$, the element $g \in G$ is an arbitrary choice of representative of $\bar{g}$.

We now present another construction that will be used later. Let G be a group and A a G-module. Let $t \in G$. Set $G' = G$, $A' = A$, and denote by $f : g \mapsto t^{-1}gt$ the inner automorphism associated with t^{-1}. The homomorphism of abelian groups $u : a \mapsto t \cdot a$

from A to A is then f-compatible, and hence induces for all $i \geq 0$ a homomorphism $\sigma_i : H^i(G, A) \mapsto H^i(G, A)$.

Proposition 28.16. — *The map σ_i is the identity.*

Remark 28.17. — Let A be a G-module. If H is a normal subgroup of G, we can make G act on $H^i(H, A)$ *via the conjugation action* of G on H. then Proposition 28.16 implies that H acts trivially on $H^i(H, A)$: G/H acts on $H^i(H, A)$.

We deduce that the groups $H^i(H, A)$ are obtained using the cohomology of the complex

$$0 \longrightarrow I_0^H \longrightarrow I_1^H \longrightarrow I_2^H \longrightarrow \cdots ,$$

they are hence naturally G/H-modules via the action of G/H on the I_j^H for $j \geq 0$. The corresponding action of G/H on the $H^i(H, A)$ coincides with that defined above (via the action by conjugation of G on H). Indeed, for $i = 0$, the two actions are given by $(t, a) \mapsto t \cdot a$ for $t \in G/H$ and $x \in A^H$; we deduce the result for any i by shifting.

Theorem 28.18 **(Restriction–Inflation).** — *Let H be a normal subgroup of G and let A be a G-module. Then the sequence*

$$0 \longrightarrow H^1(G/H, A^H) \xrightarrow{\text{Inf}} H^1(G, A) \xrightarrow{\text{Res}} H^1(H, A)$$

is exact.

Proof. — It is straightforward that the sequence is a complex. Let us first show the injectivity of Inf. Let $f : G/H \to A^H$ be a 1-cocycle cohomologous to 0 in $H^1(G, A)$. We can also view f as a map from G to A constant on each class mod H. Then there exists an $a \in A$ such that $f(s) = s \cdot a - a$ for all s in G, and we obtain $sa - a = sta - a$ for all t in H, hence $t \cdot a = a$ by setting $s = 1$. Lastly $a \in A^H$ and f is indeed zero in $H^1(G/H, A^H)$.

Let us now show that an element of Ker Res is in Im Inf. Let $f : G \to A$ be a 1-cocycle. If $\mathrm{Res}(f) = 0$, then there exists an $a \in A$ such that $f(t) = t \cdot a - a$ for all t in H. By if necessary replacing f by a cohomologous cocycle $t \mapsto f(t) - (t \cdot a - a)$, we may assume $f(t) = 0$ for all t in H. As $f(st) = f(s) + s \cdot f(t)$, we obtain that f factors as a map $\overline{f} : G/H \to A$. For s in H, the classes st and t in G/H are the same (recall that H is *normal* in G). The formula then gives $f(t) = f(st) = sf(t)$, which shows that $\overline{f}$ has coefficients in A^H. It is a cocycle that induces an element of $H^1(G/H, A^H)$ whose image by Inf is f. □

Remark 28.19. — This result can also be deduced from the Hochschild–Serre spectral sequence defined below.

As we have seen, the group G/H acts on the cohomology groups $H^i(H, A)$. We can then view the restriction–inflation sequence as the exact sequence of low degree terms of the spectral sequence given by the following theorem:

Theorem 28.20 **(Hochschild–Serre).** — *Let G be a group. Let H be a normal subgroup of G and A a G-module. Then we have a spectral sequence*

$$E_2^{pq} = H^p(G/H, H^q(H, A)) \implies H^{p+q}(G, A).$$

In particular, the theorem implies (by standard homological result) that for each $n > 0$, we have a filtration of $H^n(G, A)$ by a decreasing sequence $F^0 = H^n(G, A) \supseteq \cdots \supseteq F^{n+1} = 0$, where F^p/F^{p+1} (for $p = 0, \ldots, n$) is isomorphic to a subquotient $E_\infty^{p,n-p}$ of $H^p(G/H, H^{n-p}(H, A))$. The first terms of the exact sequence of low degree terms of this spectral sequence are written as

$$0 \to H^1(G/H, A^H) \xrightarrow{\text{Inf}} H^1(G, A) \xrightarrow{\text{Res}} H^1(H, A)^{G/H}$$
$$\to H^2(G/H, A^H) \xrightarrow{\text{Inf}} H^2(G, A)$$

Corollary 28.21. — Let G be a group. Let H be a normal subgroup of G and let A be a G-module. Let $n \geq 2$ be an integer, we make an additional assumption that

$$H^i(H, A) = 0 \quad \text{for } 1 \leq i \leq n-1.$$

Then the sequence

$$0 \longrightarrow H^n(G/H, A^H) \xrightarrow{\text{Inf}} H^n(G, A) \xrightarrow{\text{Res}} H^n(H, A)^{G/H}$$

is exact.

Proof. — The assumption tells us that the terms E_2^{pq} of the initial page of the spectral sequence are zero for $0 < q < n$. It follows that the only terms E_∞^{pq}, with $p + q = n$, that may be nonzero are the $E_\infty^{n,0}$ and $E_\infty^{0,n}$. The assumption immediately implies that $E_\infty^{n,0} = E_2^{n,0}$ and furthermore, $E_\infty^{0,n}$ is always a subgroup of $E_2^{0,n}$.

We conclude by writing the filtration

$$0 \subset F^n \subset H^n(G, A),$$

where $F^n = E_\infty^{n,0}$ and $H^n(G, A)/F^n = E_\infty^{0,n}$. □

Corestriction map: let H be a subgroup of a finite group G. Let A be a G-module. We will define homomorphisms $H^i(H, A) \to H^i(G, A)$ "in the opposite direction" of the restriction. We start with the case $i = 0$. The corestriction is then defined by the **norm**:

$$N_{G/H} : a \mapsto \sum_{s \in G/H} s \cdot a$$

from A^H to A^G, where G/H is the (finite, by assumption) set of left classes modulo H. It is immediate that $s \cdot a$ depends only on the class of s in G/H (since $a \in A^H$) and that $N_{G/H}(a) \in A^G$. In this way we obtain *corestriction* morphisms

$$\text{Cores} : H^i(H, A) \to H^i(G, A)$$

compatible with the usual morphisms of short exact sequences.

Theorem 28.22. — *Let $m = [G : H]$ be the index of H in G. Then the composite* $\text{Cores} \circ \text{Res}$ *is the multiplication by m in $H^i(G, A)$.*

Proof. — The statement is clear for $i = 0$. The general case is obtained by shifting (embedding A into an induced G-module I). □

Corollary 28.23. — Let $m = [G : H]$ be the index of H in G. Then the kernel of Res is killed by m.

In particular, if m prime to p, Res is injective on the p-primary part of $H^q(G, A)$.

Corollary 28.24. — Let G be a *finite* group of cardinality m. Let A be a G-module. Then for $i > 0$, the groups $H^i(G, A)$ are m-torsion.

In particular, if A is n-torsion with n prime to m, we obtain $H^i(G, A) = 0$ for $i > 0$.

Proof. — Apply Theorem 28.22 in the case $H = \{1\}$. □

Corollary 28.25. — Let G be a *finite* group. Then for any G-module of finite type A, $H^i(G, A)$ is finite for all $i > 0$.

Proof. — The cochain description shows that the groups $H^i(G, A)$ are of finite type. As by Corollary 28.24, these groups are torsion for $i > 0$, they are finite. □

Corollary 28.26. — Let G be a finite group. Let A be a uniquely divisible abelian group endowed with an action of G (e.g. $A = \mathbf{Q}$ with the trivial action by G). Then $H^i(G, A) = 0$ for all $i > 0$.

Proof. — Indeed, Corollary 28.24 says that for $i > 0$, the group $H^i(G, A)$ is torsion. On the other hand, the multiplication by n in A is, by assumption, an isomorphism for all $n > 0$ and hence multiplication by n in $H^i(G, A)$ is also an isomorphism. □

Example 28.27. — Consider $A = \mathbf{Q}/\mathbf{Z}$ with the trivial action of G. By the above corollary and the long exact cohomology sequence associated to the exact sequence

$$0 \to \mathbf{Z} \to \mathbf{Q} \to \mathbf{Q}/\mathbf{Z} \to 0,$$

we have $H^i(G, \mathbf{Q}/\mathbf{Z}) \simeq H^{i+1}(G, \mathbf{Z})$ for all $i > 0$. In particular, $H^2(G, \mathbf{Z}) \simeq H^1(G, \mathbf{Q}/\mathbf{Z})$ is the *character group* of G. We also have $H^1(G, \mathbf{Z}) = 0$ given that $\mathbf{Z}$ has no non-trivial finite subgroup.

28.3 Miscellaneous Proofs. — We have included some complicated proofs–though they are not essential to our main story–in this subsection of the appendix.

Proof of Proposition 4.4. — For each $q \geq 0$, the compatible transition maps give a canonical map

$$u_q : \varinjlim_i C^q(G_i, A_i) \longrightarrow C^q(G, A).$$

It is enough to show that u_q is an isomorphism, because the maps u_q are compatible with the differentials of the cochain complexes.

For injectivity, let $f_i : G_i^q \to A_i$ be a locally constant cochain whose image in $C^q(G, A)$ is zero. Since $G \to G_i$ is surjective, every value of f_i occurs as the value of the induced cochain on G^q. The image of f_i is finite, since G_i^q is compact and A_i is discrete. After replacing i by a larger index j, these finitely many values become zero in A_j. Thus f_i is zero in the inductive limit of cochains.

For surjectivity, let $f : G^q \to A$ be continuous. Its image is finite. Hence, after choosing a sufficiently large index i, all values of f lift to A_i. Since f is locally constant and G^q is compact, there is an open normal subgroup $N \subset G$ such that f is constant on the cosets of N^q. By the projective limit topology, after increasing the index to some $j \geq i$, there is an open normal subgroup $N_j \subset G_j$ whose inverse image in G is contained in N. In particular, the kernel of $G \to G_j$ is contained in N. Since $G \to G_j$ is surjective, f factors through G_j^q, and the chosen lifts of its finitely many values define a locally constant cochain $f_j : G_j^q \to A_j$ whose image under u_q is f. Thus u_q is surjective.

Consequently the cochain complexes commute with this filtered direct limit, and filtered direct limits are exact in abelian groups. Passing to cohomology gives

$$H^q(G, A) \cong \varinjlim_i H^q(G_i, A_i).$$

□

Proof of Theorem 20.3. — For each object $A \in \mathscr{C}$ and each element $x \in F(A)$, consider the pair (A, x). We say that (A, x) is *minimal* if for every epimorphism $A \twoheadrightarrow B$ which is not an isomorphism, the element x does not lie in the image of $F(B) \subseteq F(A)$.

If (A', x') and (A, x) are two such pairs, we say that (A', x') dominates (A, x) if there exists a morphism $u : A \to A'$ such that $x = F(u)(x')$. Since $\mathscr{C}$ is noetherian, every pair (A, x) is dominated by a minimal pair: indeed, one chooses a subobject $A_0 \subseteq A$ maximal with the property that x lies in the image of $F(A/A_0) \to F(A)$.

We now prove the key uniqueness statement.

Lemma. If (A', x') is minimal, then any morphism $u : A \to A'$ with $x = F(u)(x')$ is unique.

Proof of the lemma. If $v : A \to A'$ is another morphism with $x = F(v)(x')$, then $F(u - v)(x') = 0$. Since F is contravariant and left exact, the epimorphism $u - v : A \to \mathrm{Im}(u - v)$ induces an injection

$$F(\mathrm{Im}(u - v)) \hookrightarrow F(A).$$

Let $i : \mathrm{Im}(u - v) \hookrightarrow A'$ be the canonical injection, and set $B := \mathrm{Coker}(u - v)$. The short exact sequence

$$0 \longrightarrow \mathrm{Im}(u - v) \xrightarrow{i} A' \longrightarrow B \longrightarrow 0$$

gives, after applying F, that $F(B)$ is the kernel of $F(i)$. Hence $x' \in F(B)$. By minimality of (A', x'), this forces $B = A'$, so $u - v$ is surjective and therefore $u = v$.

The minimal pairs form a filtered system: if (A_1, x_1) and (A_2, x_2) are minimal, then they are both dominated by $(A_1 \oplus A_2, (x_1, x_2))$, which in turn is dominated by a minimal pair. Let $(I_i, x_i)_{i \in I}$ be a filtered system of minimal pairs obtained in this way.

Set $I := \varinjlim_i I_i$. The elements x_i define a canonical element $x \in F(I)$, and for every object $A \in \mathscr{C}$ and every morphism $f \in \mathrm{Hom}_{\mathscr{C}}(A, I)$, define

$$\Phi_A(f) := F(f)(x) \in F(A).$$

This gives a natural transformation

$$\Phi : \mathrm{Hom}_{\mathscr{C}}(A, I) \longrightarrow F(A).$$

We claim that Φ_A is an isomorphism for every A.

It is injective: if $f : A \to I_i$ satisfies $F(f)(x_i) = 0$, then the minimality of (I_i, x_i) implies $f = 0$.

It is surjective: given $y \in F(A)$, choose a minimal pair (I_i, x_i) dominating (A, y); then by definition, there exists a morphism $f : A \to I_i$ with $y = F(f)(x_i)$, hence $y = \Phi_A(f)$.

Therefore $F \simeq \varinjlim_i \mathrm{Hom}_{\mathscr{C}}(-, I_i)$, as required. □

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Notation Index

This file collects the main notation used throughout the notes. It is meant as a running index rather than a list of new definitions.

Groups and Cohomology. —

G, H, U, N : Usually profinite groups; U is often an open subgroup and N a closed normal subgroup.

G_F, G_K : The absolute Galois group $\mathrm{Gal}(F^{\mathrm{sep}}/F)$ or $\mathrm{Gal}(K^{\mathrm{sep}}/K)$.

Γ_F, Γ_K : Alternative notation for the absolute Galois group of a local field.

I_K : The inertia subgroup $\mathrm{Gal}(K^{\mathrm{sep}}/K^{\mathrm{ur}})$ of G_K.

P_K : The wild inertia subgroup of I_K.

P_F **:** For a local field F of residue characteristic p, the wild inertia group $\mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{tr}})$; it is a pro-$p$ group.

I_K^{t} **:** The tame inertia quotient I_K/P_K.

G^{ab} **:** The abelianization of a group G.

G_K^{ab}**,** Γ_K^{ab} **:** The profinite abelianization of $G_K = \Gamma_K$, namely the quotient by the closure of the commutator subgroup; canonically $G_K^{\mathrm{ab}} \simeq \mathrm{Gal}(K^{\mathrm{ab}}/K)$.

$\Gamma/n\Gamma$ **:** For a profinite abelian group Γ, the quotient by the closed subgroup $n\Gamma$; in Lecture 30, $\Gamma = \mathrm{Gal}(K^{\mathrm{ab}}/K)$.

$\widehat{\mathbf{Z}}$ **:** The profinite completion of $\mathbf{Z}$, also $\prod_p \mathbf{Z}_p$.

$\widehat{\mathbf{Z}}^{(p)}$ **:** The prime-to-p profinite completion $\prod_{\ell \neq p} \mathbf{Z}_\ell$.

$\mathscr{C}_G$ **:** The category of discrete G-modules.

A^G **:** The subgroup of G-invariants of a G-module A.

$C^q(G, A)$ **:** The group of continuous q-cochains of G with values in A.

$H^q(G, A)$ **:** Continuous group cohomology for profinite G and discrete G-module A.

$\widehat{H}^q(G, A)$ **:** Tate cohomology of a finite group G with coefficients in A.

Res**,** Inf**,** Cor **:** Restriction, inflation, and corestriction maps in group cohomology.

$\mathrm{Ind}_H^G A$ **:** The induced module from a closed subgroup H to G, in the continuous setting.

$cd_p(G)$**,** $cd(G)$ **:** p-cohomological dimension and cohomological dimension.

$scd_p(G)$**,** $scd(G)$ **:** Strict p-cohomological dimension and strict cohomological dimension.

$A^\vee$ **:** The Tate dual $\mathrm{Hom}(A, \mu_{p^\infty})$ for finite p-primary Galois modules.

M' **:** The Cartier dual of a finite Galois module M, namely $\mathrm{Hom}_{\mathbf{Z}}(M, \overline{K}^\times) = \mathrm{Hom}(M, \mu)$.

A^* **:** The Pontryagin dual $\mathrm{Hom}(A, \mathbf{Q}/\mathbf{Z})$ when A is a torsion abelian group.

δ **:** A connecting homomorphism in a long exact cohomology sequence.

I_G **:** The augmentation ideal $\ker(\mathbf{Z}[G] \to \mathbf{Z})$.

N_G **:** The norm element $\sum_{g \in G} g$, or the induced norm map on a G-module.

$\bar{s} \in \widehat{H}^{-2}(G, \mathbf{Z})$ **:** The Tate cohomology class corresponding to $s \in G^{\mathrm{ab}}$.

$\bar{a}_0$**,** $\bar{a}^0$ **:** For a finite group G acting on A, the class of an element of norm zero in $\widehat{H}^{-1}(G, A)$, respectively the class of an invariant element in $\widehat{H}^0(G, A)$.

Fields and Local Fields. —

L/K **:** The default notation for a finite extension of local fields; when Galois, $G = \mathrm{Gal}(L/K)$.

K^{sep} **:** A separable closure of K.

$\overline{K}$ **:** An algebraic closure of K; in Galois cohomology it is usually chosen to contain K^{sep}.

K^{ur} **:** The maximal unramified extension of a local field K.

K^{tr} **:** The maximal tamely ramified extension of K.
K^{ab} **:** The maximal abelian extension of K.
$\widehat{K^\times}$ **:** The profinite completion of the multiplicative group $K^\times$.
$\mathscr{O}_K$ **:** The ring of integers of a local field K.
$\mathfrak{m}_K$ **:** The maximal ideal of $\mathscr{O}_K$.
k **:** Often the residue field of a local field; context may also use k as a general base field.
π **:** A uniformizer of a discretely valued local field.
v_K **:** The normalized valuation on K, with $v_K(K^\times) = \mathbf{Z}$.
$\|\cdot\|_K$ **:** The normalized absolute value on K.
U_K **:** The unit group $\mathscr{O}_K^\times$.
U_K^n **:** The higher unit subgroup $1 + \mathfrak{m}_K^n$ when this filtration is used.
$N_{L/K}$ **:** The norm map $L^\times \to K^\times$ for a finite extension L/K.
$N_{L/K}L^\times$ **:** The norm subgroup of $K^\times$ attached to a finite extension L/K; in local class field theory, finite abelian extensions correspond to these subgroups.
M_m **:** In Lecture 30, the subgroup $\{a \in K^\times : v_K(a) \equiv 0 \pmod{m}\}$, equal to the norm group of the unramified extension of degree m.
$\widetilde{K}$ **:** The inverse limit $\varprojlim_L K^\times/N_{L/K}L^\times$ over finite abelian extensions L/K, i.e. the completion of $K^\times$ for the topology defined by norm subgroups.
i, $\widetilde{\theta}$ **:** In Lecture 30, the factorization maps $K^\times \xrightarrow{i} \widetilde{K} \xrightarrow{\widetilde{\theta}} G_K^{\mathrm{ab}}$ of the reciprocity map.
L_M **:** For a norm subgroup $M \subset K^\times$, the finite abelian extension corresponding to M under local class field theory.
$e(L/K)$, $f(L/K)$ **:** The ramification index and residue degree of a finite extension of local fields.
Frob, F **:** In local class field theory, the arithmetic Frobenius in an unramified Galois group; it acts on the residue field by $x \mapsto x^{\#k_K}$, and the convention is $(\pi_K, L/K) = F$ for finite unramified L/K.
F^{tr} **:** The maximal tamely ramified extension of a non-archimedean local field F.
G_i **:** The lower ramification groups of a finite Galois extension of local fields.
U_E^i **:** The higher unit groups $1 + \mathfrak{p}_E^i$ in a finite extension E/F.

Brauer Groups and Central Simple Algebras. —

$\mathrm{Br}(K)$ **:** The Brauer group of the field K, equivalently $H^2(G_K, (K^{\mathrm{sep}})^\times)$.
$\mathrm{Br}(L/K)$ **:** The relative Brauer group $\ker(\mathrm{Br}(K) \to \mathrm{Br}(L))$.
A, D **:** A usually denotes a central simple algebra and D a central division algebra.
$Z_A(B)$ **:** The centralizer in A of a subalgebra or subset B.
A^{op} **:** The opposite algebra of A.
$M_n(K)$ **:** The algebra of $n \times n$ matrices over K.
Nred **:** The reduced norm of a central simple algebra.

Tred **:** The reduced trace of a central simple algebra.
$SL_1(A)$ **:** The group of elements of reduced norm 1 in $A^\times$.
$a_{\sigma,\tau}$ **:** A factor set or 2-cocycle attached to a crossed product algebra.
$A(a)$ **:** The crossed product algebra associated with a 2-cocycle $a \in Z^2(G, L^\times)$.
$u_{L/K}$ **:** The fundamental class of a finite Galois extension L/K of local fields.
inv_K **:** The local invariant map $\mathrm{Br}(K) \to \mathbf{Q}/\mathbf{Z}$.

Roots of Unity, Characters, and Symbols. —

μ_n **:** The group of n-th roots of unity in a separable or algebraic closure.
μ_{p^∞} **:** The group $\bigcup_r \mu_{p^r}$.
$\mathbf{Q}/\mathbf{Z}(1)$ **:** The Tate twist of $\mathbf{Q}/\mathbf{Z}$, identified with all roots of unity.
ω **:** The mod-p cyclotomic character $G_K \to \mathbf{F}_p^\times$, defined by $g(\zeta) = \zeta^{\omega(g)}$ for $\zeta \in \mu_p$.
$\mathbf{F}_\chi$ **:** A one-dimensional vector space over a finite field $\mathbf{F}$ with G_K-action through χ.
χ_a **:** The Kummer character associated with $a \in k^\times/k^{\times n}$ once a primitive n-th root of unity is fixed.
d_χ **:** For a character $\chi \in H^1(G, \mathbf{Q}/\mathbf{Z})$, its image in $H^2(G, \mathbf{Z})$ under the connecting homomorphism from $0 \to \mathbf{Z} \to \mathbf{Q} \to \mathbf{Q}/\mathbf{Z} \to 0$.
$(a, b)_n$ **:** The n-th Hilbert or local symbol, defined by cup product using χ_a.
$(\alpha, L/K)$ **:** The image of $\alpha \in K^\times$ under the finite-level local reciprocity map for L/K.
$\theta_{L/K}$ **:** The finite-level reciprocity map $K^\times/N_{L/K}L^\times \to \mathrm{Gal}(L/K)^{\mathrm{ab}}$, defined as the inverse of cup product with the fundamental class.
θ_K, ω_K **:** The local reciprocity, or Artin, map $K^\times \to \mathrm{Gal}(K^{\mathrm{ab}}/K)$ obtained by passing to the limit over finite abelian extensions.
θ_n **:** In the proof of the existence theorem, the map $K^\times \to \Gamma/n\Gamma$ induced by θ_K, where $\Gamma = \mathrm{Gal}(K^{\mathrm{ab}}/K)$.

Homological Algebra. —

R^iF **:** The i-th right derived functor of a left exact functor F.
$\mathrm{Ext}^i_R(A, B)$ **:** The Ext groups over a ring R.
$I^\bullet$ **:** Usually an injective or acyclic resolution.
$\mathscr{F}(G, A)$ **:** The set of all functions from G to A.
$C(G, A)$ **:** The group of continuous functions from G to the discrete module A.
$\varprojlim$ **:** Projective limit.
$\varinjlim$ **:** Inductive limit.

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